

CONDITIONAL LOCAL TYPE I NAVIER–STOKES REDUCTION: CONCENTRATION, SEREGIN LIMITS, AND LIOUVILLE-CLASS CLOSURES

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ABSTRACT. We prove the concentration, compactness, and Liouville reductions needed for a conditional exclusion of local pointwise Type I Navier–Stokes singularities. Starting from a finite-energy local pointwise Type I singular point, we obtain a raw extracted bounded centered Seregin ancient profile with positive compact local L^3 velocity mass, then pass to the raw generated Seregin state space with invariant local nonvanishing. We define the small-amplitude, stationary L^3 , uniformly L^3 -tight, and structure and decay classes, and prove their exclusion by perturbative, stationary, endpoint L^3 , and axisymmetric or weighted-vorticity Liouville theorems. The exclusion of the uniformly L^3 -tight class rests on the Albritton–Barker endpoint ancient Liouville theorem after a physical pullback on truncated ancient intervals. The only conditional ingredient is the residual closure hypothesis [Hypothesis 16.1](#). The final Type I exclusion is conditional on residual closure for the raw generated Seregin state space $\mathcal{S}_{\text{raw-gen}}(u, p, z_*, R_0, \eta_*)$ defined below.

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INTRODUCTION AND STRUCTURE OF THE ARGUMENT

The conditional local Type I reduction reduces a local pointwise Type I singularity to a raw extracted bounded centered ancient solution, then to a raw generated Seregin state with invariant local nonvanishing, and excludes the non-residual Liouville classes obtained from standard local compactness and explicit Liouville theorems. The residual class is handled only by the residual-class hypothesis [Hypothesis 16.1](#); no unconditional residual exclusion is claimed in this paper.

The generated state space used in the final assembly is a raw local Seregin state space. The projected mild formulation is not imposed on every raw generated state. Instead, mildness is treated as a branch condition in those Liouville classes whose exclusion theorems require it. Generated states not known to satisfy the full hypotheses of one of the explicit Liouville branches belong to the remaining class $\mathcal{R}(\mathcal{S})$, and are removed only by the residual closure hypothesis.

The reduction has four components. First, the local concentration theorem shows that a singular point carries positive Caffarelli–Kohn–Nirenberg critical mass at arbitrarily small scales and gives the local Type I/local Type II dichotomy. It also identifies the endpoint weak-Serrin estimate used to obtain terminal velocity no-escape for local pointwise Type I rescalings. Second, the Type I blow-up analysis gives the admissible Seregin extraction: an admissible Type I extraction produces a raw smooth bounded ancient solution of the centered self-similar equation with positive compact local L^3 velocity mass. The compact mass is converted into invariant local nonvanishing before forming the raw generated descendant hull. The paper then defines the Liouville classes and the residual class on that raw hull.

Third, the uniformly tight analysis excludes nonzero uniformly L^3 -tight generated raw Seregin states that lie in the mild stratum. Boundedness plus uniform L^3 -tightness gives a backward sequence of finite critical norm after physical pullback, so the Albritton–Barker endpoint ancient Liouville theorem applies. Fourth, the structured part gives the axisymmetric and perturbative weighted-vorticity exclusions used in the Liouville-class analysis. These results turn the corresponding structured hypotheses into the zero ancient solution, contradicting the invariant local nonvanishing retained by the raw generated Seregin state space.

Consequently, a generated raw Seregin state in $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ must either belong to one of the excluded Liouville classes or to the residual class. The Type I contradiction is therefore conditional on the residual-class hypothesis [Hypothesis 16.1](#). No unconditional Type I regularity theorem is claimed in this paper; the final regularity conclusion is conditional on the residual closure hypothesis for the raw generated state space.

Dependency table.

Step	Status/source	Role
Velocity epsilon criterion	imported from Gustafson–Kang–Tsai	singularity gives L^3 concentration
Weak-Serrin Type I bridge	imported from Albritton–Barker, Lemma 2.5; used here	local Type I gives A, C, D, E bounds
Terminal no-escape	proved here using Albritton–Barker	prevents mass from escaping into $t = T$
Seregin extraction	proved here	produces a raw extracted centered ancient profile
Raw generated closure	defined here	gives the ambient state space for residual closure
Mild stratum	defined here	supplies the branch condition for mild Liouville theorems
Tight L^3 branch	proved here using Albritton–Barker, Theorem 1.2	excludes a non-residual class
Structured Liouville branches	imported theorem by theorem plus Type I constant removal	exclude the structured classes
Residual closure	conditional input from the companion residual paper	removes $\mathcal{R}(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*))$
Contradiction	conditional on residual closure	rules out local pointwise Type I singularities

How to read this paper. Readers interested only in the construction of the ancient limit should read Parts I and II together with the final assembly [Theorem 33.1](#). Readers interested in the non-residual Liouville exclusions should read Parts III and IV. The residual closure hypothesis [Hypothesis 16.1](#) is stated as a conditional analytic assumption.

Pressure notation. Lowercase p denotes the source pressure of a suitable weak solution. Q denotes a physical ancient pressure obtained as a local distributional limit after blow-up. Π denotes a centered pressure obtained from Q by the self-similar change of variables. All pressures are considered modulo functions of time only; this gauge ambiguity is invisible to the equations and to the local energy inequality ([Lemmas 10.9 and 11.4](#) and [Remarks 11.5 and 13.1](#)).

PART I. LOCAL CONCENTRATION AND THE TYPE I/TYPE II DICHOTOMY

Overview. This part proves a local analytic reduction for possible finite-time singularities of the three-dimensional Navier–Stokes equations. For a suitable weak solution, every singular point at time T has arbitrarily small backward parabolic cylinders on which both the standard mean-subtracted Caffarelli–Kohn–Nirenberg scale-invariant quantity $C + D$ and the scale-invariant velocity quantity C are bounded below by positive constants. If these local scale-invariant quantities tend to zero at each point of the singular set, the solution is regular at time T by the ε -regularity theorem. Along any such positive concentration sequence, the solution either satisfies a local Type I scale-invariant bound or belongs to the local Type II alternative. For local pointwise Type I terminal singularities of suitable weak solutions, the endpoint weak-Serrin argument gives the velocity no-escape condition used in the Type I blow-up analysis.

1. INTRODUCTION

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \quad \nabla \cdot u = 0.$$

The viscosity is normalized to one. The weak solution theory goes back to Leray [6]. Suitable weak solutions and their partial regularity theory were introduced and developed by Scheffer [8] and by Caffarelli, Kohn, and Nirenberg [2]; see also Lin [7]. The local reduction below connects a possible singularity at time T with the Type I/Type II blow-up rate dichotomy.

Scale-invariant regularity and blow-up criteria provide the standard background for this formulation; see Serrin [10], Giga [4], Escauriaza–Seregin–Šverák [3], and Seregin [9]. The principal regularity theorem is the Caffarelli–Kohn–Nirenberg ε -regularity theorem.

The proof has four components. First, the local scale-invariant quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg ε -regularity theorem implies that vanishing of $C + D$ at a putative singular point gives local regularity, while the velocity ε -regularity criterion gives positive concentration in the scale-invariant $L^3_{x,t}$ quantity. Third, a finite-time singularity therefore produces sequences of shrinking cylinders on which $C + D$, and separately C , are bounded below by positive constants. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or belongs to the local Type II alternative.

The concentration and Type I/Type II dichotomy proofs are local in spacetime. Apart from the standard Caffarelli–Kohn–Nirenberg ε -regularity theorem, all quantities and alternatives used in that local analysis are defined in this part. The local Type I reduction section shows how the endpoint weak-Serrin estimate upgrades the local pointwise Type I envelope to terminal velocity no-escape through the local endpoint weak-Serrin Type I estimate.

2. SUITABLE WEAK SOLUTIONS AND SINGULAR POINTS

Definition 2.1 (Suitable weak solution). Let $I \subset \mathbb{R}$ be an open interval. A pair (u, p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L_t^\infty L_{x,\text{loc}}^2 \cap L_t^2 H_{x,\text{loc}}^1, \quad p \in L_{\text{loc}}^{3/2}, \quad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned} & \int_{\mathbb{R}^3} \frac{|u(x, t_2)|^2}{2} \phi(x, t_2) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi dx dt \\ & \leq \int_{\mathbb{R}^3} \frac{|u(x, t_1)|^2}{2} \phi(x, t_1) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \phi + \Delta \phi) dx dt \\ & \quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \phi dx dt. \end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

Also write $Q_r(x_0, t_0)$ for $Q_r((x_0, t_0))$.

Definition 2.2 (Singular set at a fixed time). For a time T , define

$$\Sigma(T) = \{x \in \mathbb{R}^3 : u \text{ is not locally bounded in any } Q_r(x, T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x, T))} \leq M$.

If a finite-time singularity occurs at time T , then $\Sigma(T) \neq \emptyset$ in this local sense.

Temporal cutoff convention.

Lemma 2.3 (Temporal cutoff convention). *For a suitable weak solution, every local energy inequality used below with endpoint times or compact time slabs is obtained by applying the standard distributional local energy inequality to a test function of the form $\phi(x, t)\chi_\varepsilon(t)$, where χ_ε is a smooth nonincreasing temporal cutoff with $\chi_\varepsilon = 1$ on a closed sub-interval and supported in a slightly larger open interval, and then letting $\varepsilon \downarrow 0$ along an admissible set of times where the L^2 trace exists.*

We invoke this convention silently in subsequent proofs.

3. PARABOLIC RESCALING AND SCALE-INVARIANT QUANTITIES

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y, s) = r u(x_0 + ry, T + r^2 s), \quad p^{(r)}(y, s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y, s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 3.1 (Invariance under parabolic rescaling). *If (u, p) is a suitable weak solution in a cylinder containing (x_0, T) , then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. The local energy inequality is verified to fix the normalization.

Let $\psi \in C_c^\infty$ be nonnegative and compactly supported in the rescaled domain. Define

$$\phi(x, t) = \psi \left(\frac{x - x_0}{r}, \frac{t - T}{r^2} \right).$$

If $y = (x - x_0)/r$ and $s = (t - T)/r^2$, then

$$\partial_t \phi = r^{-2}(\partial_s \psi)(y, s), \quad \nabla_x \phi = r^{-1}(\nabla_y \psi)(y, s), \quad \Delta_x \phi = r^{-2}(\Delta_y \psi)(y, s).$$

For almost every $s_1 < s_2$, the corresponding times $T + r^2 s_1 < T + r^2 s_2$ are admissible in the original local energy inequality. Apply that inequality for (u, p) to ϕ on $[T + r^2 s_1, T + r^2 s_2]$. After the change of variables $x = x_0 + ry$, $t = T + r^2 s$, using

$$u(x_0 + ry, T + r^2 s) = r^{-1}u^{(r)}(y, s), \quad p(x_0 + ry, T + r^2 s) = r^{-2}p^{(r)}(y, s),$$

each term contains the same common factor r . Dividing by this factor gives the rescaled inequality

$$\begin{aligned} & \int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_2)|^2}{2} \psi(y, s_2) dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} |\nabla_y u^{(r)}|^2 \psi dy ds \\ & \leq \int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_1)|^2}{2} \psi(y, s_1) dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \frac{|u^{(r)}|^2}{2} (\partial_s \psi + \Delta_y \psi) dy ds \\ & \quad + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\frac{|u^{(r)}|^2}{2} + p^{(r)} \right) u^{(r)} \cdot \nabla_y \psi dy ds. \end{aligned}$$

Thus $(u^{(r)}, p^{(r)})$ is suitable on the rescaled domain. The pressure transforms according to the Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s . $\square$

Definition 3.2 (Scale-invariant local quantities). For $z_0 = (x_0, t_0)$ and $r > 0$, set

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt$$

and

$$D(z_0, r) = r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot, t)$ on $B_r(x_0)$.

The subtraction of the spatial mean removes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 3.3 (Local finiteness and scaling). *Let (u, p) be a suitable weak solution in a cylinder containing $z_0 = (x_0, T)$. Then $C(z_0, r)$ and $D(z_0, r)$ are finite for every sufficiently small r . Moreover, for every fixed $R < \infty$,*

$$\begin{aligned} & \int_{-R^2}^0 \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy ds \\ & = R^2 (C(z_0, rR) + D(z_0, rR)). \end{aligned}$$

Proof. Local finiteness of the velocity term follows from

$$u \in L_t^\infty L_{x,\text{loc}}^2 \cap L_t^2 H_{x,\text{loc}}^1 \subset L_{\text{loc}}^3$$

on finite cylinders. Indeed, if $B \Subset \mathbb{R}^3$ and $(a, b) \Subset I$, Sobolev and interpolation give

$$\int_a^b \|u(t)\|_{L^3(B)}^3 dt \leq C \left(\operatorname{ess\,sup}_{a < t < b} \|u(t)\|_{L^2(B)} \right)^{3/2} \left(\int_a^b \|u(t)\|_{H^1(B)}^2 dt \right)^{3/4} (b - a)^{1/4}.$$

Thus $u \in L_{\text{loc}}^3$. The pressure term is finite because $p \in L_{\text{loc}}^{3/2}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability. More explicitly, for a ball B ,

$$|p - (p)_B(t)|^{3/2} \leq C \left(|p|^{3/2} + |(p)_B(t)|^{3/2} \right),$$

while Jensen's inequality gives

$$|(p)_B(t)|^{3/2} \leq \frac{1}{|B|} \int_B |p(x, t)|^{3/2} dx.$$

After integration over B and over time, the mean-subtracted pressure term is controlled by the local $L^{3/2}$ norm of p .

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2s$. The velocity term transforms as

$$\int_{-R^2}^0 \int_{B_R} |u^{(r)}|^3 dy ds = r^{-2} \int_{T-(rR)^2}^T \int_{B_{rR}(x_0)} |u|^3 dx dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2(p)_{B_{rR}(x_0)}(T + r^2s),$$

because

$$\frac{1}{|B_R|} \int_{B_R} r^2 p(x_0 + ry, T + r^2s) dy = \frac{r^2}{|B_{rR}|} \int_{B_{rR}(x_0)} p(x, T + r^2s) dx.$$

Therefore

$$\begin{aligned} \int_{-R^2}^0 \int_{B_R} |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} dy ds \\ = r^{-2} \int_{T-(rR)^2}^T \int_{B_{rR}(x_0)} |p - (p)_{B_{rR}(x_0)}(t)|^{3/2} dx dt. \end{aligned}$$

Adding the velocity and pressure identities gives the displayed formula. $\square$

4. ε -REGULARITY AND THE VANISHING CASE

Theorem 4.1 (Caffarelli–Kohn–Nirenberg ε -regularity). *There exists a universal constant $\varepsilon_{\text{CKN}} > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_{\text{CKN}},$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. We use the local ε -regularity theorem of Caffarelli–Kohn–Nirenberg [2]; see also Lin [7]. The statement above is written with the spatial mean subtraction used in D . Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. $\square$

Theorem 4.2 (Velocity ε -regularity). *There exists a universal constant $\varepsilon_v > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) < \varepsilon_v,$$

then u is locally bounded in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. We use the one-scale velocity criterion of Gustafson–Kang–Tsai [5] in the case $p = q = 3$. In their notation the scaled $L_{x,t}^{p,q}$ velocity norm is small under the range $3/p + 2/q \leq 2$; taking $p = q = 3$ gives the scale-invariant quantity $C(z_0, r)$ used here, up to the harmless normalization convention. Thus a singular point must satisfy the concentration consequence

$$\limsup_{r \downarrow 0} C(z_0, r) \geq \varepsilon_v.$$

The statement is scale invariant because C is invariant under the parabolic Navier–Stokes scaling. $\square$

Definition 4.3 (Vanishing local scale-invariant quantities at time T). We say that the local scale-invariant quantities vanish at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0.$$

If $\Sigma(T) = \emptyset$, there is no condition to impose.

Proposition 4.4 (Equivalent rescaled formulation). *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r \downarrow 0} \int_{-R^2}^0 \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy ds = 0.$$

Proof. By Proposition 3.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2 (C((x_0, T), rR) + D((x_0, T), rR)).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. $\square$

Theorem 4.5 (Regularity under vanishing local quantities). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. If the local scale-invariant quantities vanish at time T , then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that

$$C((x_0, T), r) + D((x_0, T), r) < \varepsilon_{\text{CKN}}.$$

Theorem 4.1 implies that u is bounded in $Q_{r/2}(x_0, T)$, hence $x_0 \notin \Sigma(T)$. This contradicts the choice of x_0 . Hence $\Sigma(T) = \emptyset$. $\square$

Corollary 4.6 (Vanishing local quantities exclude a terminal singularity). *Let (u, p) be suitable near (x_0, T) . If*

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0,$$

then u is bounded in some backward cylinder $Q_\rho(x_0, T)$, and hence $x_0 \notin \Sigma(T)$. Equivalently, membership in $\Sigma(T)$ requires that $C + D$ have a positive lower bound along a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C + D$ is below the universal threshold in Theorem 4.1. The ε -regularity theorem gives local boundedness in a backward cylinder ending at T , so $x_0 \notin \Sigma(T)$. $\square$

5. POSITIVE LOCAL CONCENTRATION

Lemma 5.1 (Failure of vanishing gives a positive concentration sequence). *Suppose the local scale-invariant quantities do not vanish at time T . Then there exist $x_0 \in \Sigma(T)$, a constant $\eta > 0$, and radii $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Equivalently, the rescaled sequence around (x_0, T) has a positive lower bound for the corresponding local scale-invariant quantities on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of

$$C((x_0, T), r) + D((x_0, T), r)$$

as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 3.3. $\square$

Definition 5.2 (Positive local concentration sequence). Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0, T) if there is $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

The corresponding parabolic rescalings are those defined by

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 5.3 (Persistence of positive concentration). *Let (u_n, p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2, 0)$. Suppose that*

$$\int_{-R^2}^0 \int_{B_R} \left(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2} \right) dy ds \geq \eta > 0$$

for all n . Suppose further that, after passing to a subsequence,

$$u_n \rightarrow U \quad \text{in } L^3(B_R \times (-R^2, 0))$$

and

$$p_n - (p_n)_{B_R}(s) \rightarrow \Pi \quad \text{in } L^{3/2}(B_R \times (-R^2, 0)).$$

Then

$$\int_{-R^2}^0 \int_{B_R} \left(|U|^3 + |\Pi|^{3/2} \right) dy ds \geq \eta.$$

In particular, the limiting pair retains the same positive lower bound for the mean-subtracted local scale-invariant quantity. If $\Pi = P - (P)_{B_R}(s)$, this is the corresponding Caffarelli–Kohn–Nirenberg quantity of (U, P) on $B_R \times (-R^2, 0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^0 \int_{B_R} |u_n|^3 dy ds \longrightarrow \int_{-R^2}^0 \int_{B_R} |U|^3 dy ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the mean-subtracted pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. $\square$

Remark 5.4. [Proposition 5.3](#) is an auxiliary compactness observation, independent of the proof of the dichotomy below. The Seregin extraction in Part II uses only the velocity-only version below; strong $L^{3/2}$ -pressure convergence is not needed for nontriviality.

Proposition 5.5 (Persistence of velocity concentration). *If $u_n \rightarrow U$ strongly in $L^3(B_R \times (-R^2, 0))$ and*

$$\int_{-R^2}^0 \int_{B_R} |u_n|^3 dy ds \geq \eta,$$

then

$$\int_{-R^2}^0 \int_{B_R} |U|^3 dy ds \geq \eta.$$

Proof. Immediate from continuity of $\|\cdot\|_{L^3}^3$ under strong L^3 -convergence. $\square$

Proposition 5.6 (Local alternative). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. Exactly one of the following alternatives holds.*

(i) For every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) = 0.$$

(ii) *There exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set, and the second is its failure expressed by a sequence. If the first alternative fails, then for some $x_0 \in \Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a sequence along which the lower bound holds gives the second alternative. Conversely, the second alternative contradicts the first. Hence precisely one alternative holds. $\square$

Theorem 5.7 (Finite-time singularity yields positive local scale-invariant concentration). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Proof. If the first alternative in Proposition 5.6 held, then Theorem 4.5 would imply $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore the second alternative in Proposition 5.6 holds. $\square$

Corollary 5.8 (Singular points carry positive velocity concentration). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $x_0 \in \Sigma(T)$, then there are $\eta_v > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) \geq \eta_v \quad \text{for all } n.$$

Proof. If the conclusion failed, then

$$\limsup_{r \downarrow 0} C((x_0, T), r) = 0.$$

Choose $r > 0$ so small that

$$C((x_0, T), r) < \varepsilon_v,$$

where ε_v is the constant in Theorem 4.2. The velocity ε -regularity theorem gives boundedness of u in $Q_{r/2}(x_0, T)$, contradicting $x_0 \in \Sigma(T)$. Hence the limsup is positive; taking any η_v below it and passing to a sequence with $C((x_0, T), r_n) \geq \eta_v$ gives the result. $\square$

Proposition 5.9 (Concentration consequences at a terminal singular point). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ and let $x_* \in \Sigma(T)$. Then:*

- (1) $\limsup_{r \downarrow 0} (C + D)((x_*, T), r) > 0$ by Theorem 4.5;
- (2) $\limsup_{r \downarrow 0} C((x_*, T), r) > 0$ by Corollary 5.8;
- (3) *if the local pointwise Type I bound (8.1) holds at $z_* = (x_*, T)$ (or the global pointwise Type I bound holds), then Theorem 8.4 (respectively Theorem 6.5) supplies an admissible local Type I concentration sequence in the sense of Definition 8.1 suitable for the Seregin extraction.*

Proof. Items (1) and (2) are restatements of the cited results at the singular point x_* . Item (3) combines (2) with the no-escape conclusion in Theorem 8.4 or Theorem 6.5, together with Corollary 8.5 for the local entry and Corollary 6.6 for the global entry. $\square$

6. NO-ESCAPE FOR FINITE-ENERGY TYPE I SINGULARITIES

The preceding concentration result gives the positive velocity concentration sequence used in the Type I blow-up analysis. In the Seregin extraction theorem Theorem 9.1, the additional compactness hypothesis is a no-escape condition excluding the possibility that all of the selected scale-invariant velocity mass collapses into the final rescaled time slice. The next theorem shows that this no-escape condition is forced by the finite-energy Type I hypotheses.

For this section we use, in addition to C and D , the standard local kinetic-energy and dissipation quantities

$$A(z_0, r) = r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx,$$

and

$$E(z_0, r) = r^{-1} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |\nabla u(x, t)|^2 dx dt.$$

These quantities are invariant under the Navier–Stokes scaling.

Definition 6.1 (Finite-energy Type I terminal singularity). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. We say that $z_* = (x_*, T)$ is a finite-energy Type I terminal singularity if $x_* \in \Sigma(T)$,

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)),$$

and there are $T_0 < T$ and $M < \infty$ such that

$$(6.1) \quad \|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T-t}}, \quad T_0 < t < T.$$

The main estimate is a uniformly local energy propagation bound. It is stated at unit scale because the application is obtained by parabolic rescaling from (x_*, T) . The estimate is a local differential inequality: the unknown is the uniformly-local kinetic energy, and the Type I envelope turns the nonlinear pressure contribution into an integrable linear coefficient. No bound below depends on the size of the global L^2 energy or the global dissipation; finite energy is used to place the pressure in the standard whole-space Leray gauge and to provide fixed-scale terminal L^3 absolute continuity for each fixed rescaling.

The standard Leray pressure gauge is used throughout this section. At almost every terminal time on which the Type I bound holds, $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$. The whole-space pressure

$$p^L(t) = \mathcal{R}_i \mathcal{R}_j (u_i(t)u_j(t))$$

is then defined by the Calderon–Zygmund theory. The distributional pressure equation gives $-\Delta p = \partial_i \partial_j (u_i u_j) = -\Delta p^L$, and the whole-space de Rham argument applied to the momentum equation shows that $p - p^L$ is a function of time. Thus replacing p by p^L is only a pressure-gauge change and does not alter suitability. This gauge choice uses the finite-energy setting to define the representative; the uniform constants below, in particular B_{uloc} and B_A , do not contain $\|u\|_{L^\infty(0, T; L^2(\mathbb{R}^3))}$ or $\int_0^T \|\nabla u(t)\|_{L^2(\mathbb{R}^3)}^2 dt$.

Lemma 6.2 (Uniformly-local energy propagation under a Type I envelope). *Let (v, π) be a suitable weak solution on $\mathbb{R}^3 \times (-4, 0)$ with $v(s) \in L^2(\mathbb{R}^3)$ for a.e. s , and take π to be the whole-space Leray pressure representative, modulo functions of time. Assume that*

$$(6.2) \quad |v(x, s)| \leq M(-s)^{-1/2}, \quad \text{for a.e. } (x, s) \in \mathbb{R}^3 \times (-4, 0),$$

and that for some admissible initial time $s_0 \in (-4, -3)$,

$$(6.3) \quad \alpha_0 := \sup_{a \in \mathbb{R}^3} \int_{B_2(a)} |v(x, s_0)|^2 dx \leq K.$$

Then there is a constant $B_{\text{uloc}} = B_{\text{uloc}}(M, K) < \infty$ such that

$$(6.4) \quad \operatorname{ess\,sup}_{-1 < s < 0} \sup_{a \in \mathbb{R}^3} \int_{B_1(a)} |v(x, s)|^2 dx + \sup_{a \in \mathbb{R}^3} \int_{-1}^0 \int_{B_1(a)} |\nabla v|^2 dx ds \leq B_{\text{uloc}}.$$

Proof. The estimate supplies the terminal no-escape bound for [Theorem 6.5](#). Let

$$\alpha(s) = \sup_{a \in \mathbb{R}^3} \int_{B_1(a)} |v(x, s)|^2 dx.$$

By [\(6.2\)](#),

$$(6.5) \quad \alpha(s)^{1/2} \leq CM(-s)^{-1/2}, \quad -4 < s < 0.$$

Fix $a \in \mathbb{R}^3$ and choose $\phi_a \in C_c^\infty(B_2(a))$ with $0 \leq \phi_a \leq 1$, $\phi_a \equiv 1$ on $B_1(a)$, and $|\nabla \phi_a| + |\Delta \phi_a| \leq C$, uniformly in a . Applying the local energy inequality on (s_0, t) , using the standard monotone approximation of temporal cutoffs and with the pressure gauge chosen below, gives for a.e. $t \in (s_0, 0)$

$$(6.6) \quad \begin{aligned} & \int |v(t)|^2 \phi_a dx + 2 \int_{s_0}^t \int |\nabla v|^2 \phi_a dx ds \\ & \leq \int |v(s_0)|^2 \phi_a dx + C \int_{s_0}^t \int_{B_2(a)} |v|^2 dx ds + C \int_{s_0}^t \int_{B_2(a)} |v|^3 dx ds \\ & + C \int_{s_0}^t \int_{B_2(a)} |\pi - c_a(s)| |v| dx ds. \end{aligned}$$

Here $c_a(s)$ is an arbitrary function of time; adding or subtracting such a function does not change the local energy inequality, since $\nabla \cdot v = 0$ and ϕ_a is compactly supported.

The velocity terms are controlled by the uniformly local kinetic energy. Since $B_2(a)$ is covered by finitely many unit balls,

$$\int_{B_2(a)} |v|^2 dx \leq C\alpha(s),$$

and by [\(6.2\)](#),

$$(6.7) \quad \int_{B_2(a)} |v|^3 dx \leq M(-s)^{-1/2} \int_{B_2(a)} |v|^2 dx \leq CM(-s)^{-1/2} \alpha(s).$$

It remains to estimate the pressure term uniformly in a .

Use the whole-space pressure representative $\pi = \mathcal{R}_i \mathcal{R}_j (v_i v_j)$, modulo functions of time. Let $\chi_a \in C_c^\infty(B_8(a))$, $\chi_a \equiv 1$ on $B_4(a)$, and write inside $B_4(a)$

$$\pi = q_a + h_a, \quad q_a = \mathcal{R}_i \mathcal{R}_j (\chi_a v_i v_j).$$

Then h_a is harmonic in $B_4(a)$. Calderon–Zygmund boundedness [\[11\]](#) and [\(6.2\)](#) give, for a.e. s ,

$$\begin{aligned} \|q_a(s)\|_{L^2(B_4(a))} & \leq C \|\chi_a v(s) \otimes v(s)\|_{L^2(\mathbb{R}^3)} \\ & \leq CM(-s)^{-1/2} \|v(s)\|_{L^2(B_8(a))} \\ & \leq CM(-s)^{-1/2} \alpha(s)^{1/2}, \end{aligned}$$

where the last step again uses a finite covering by unit balls. Hence

$$(6.8) \quad \int_{B_2(a)} |q_a| |v| dx \leq CM(-s)^{-1/2} \alpha(s).$$

For the harmonic part, set $c_a(s) = h_a(a, s)$. The kernel estimate

$$|K_{ij}(x - z) - K_{ij}(a - z)| \leq C|x - a| |z - a|^{-4}, \quad x \in B_2(a), \quad |z - a| > 4,$$

where K_{ij} is the kernel of $\mathcal{R}_i \mathcal{R}_j$, gives

$$|h_a(x, s) - c_a(s)| \leq C \sum_{m=2}^{\infty} 2^{-4m} \int_{2^m < |z-a| < 2^{m+1}} |v(z, s)|^2 dz.$$

The annulus $\{2^m < |z - a| < 2^{m+1}\}$ is covered by at most $C2^{3m}$ unit balls. Therefore

$$(6.9) \quad \|h_a(s) - c_a(s)\|_{L^\infty(B_2(a))} \leq C \sum_{m=2}^{\infty} 2^{-m} \alpha(s) \leq C\alpha(s).$$

Consequently,

$$(6.10) \quad \begin{aligned} \int_{B_2(a)} |h_a - c_a| |v| dx &\leq C\alpha(s) \int_{B_2(a)} |v| dx \\ &\leq C\alpha(s)^{3/2} \\ &\leq CM(-s)^{-1/2} \alpha(s), \end{aligned}$$

where the last inequality is precisely the Type I absorbing estimate (6.5). Combining (6.8) and (6.10),

$$(6.11) \quad \int_{B_2(a)} |\pi - c_a(s)| |v| dx \leq CM(-s)^{-1/2} \alpha(s).$$

Insert (6.7) and (6.11) into (6.6), then take the supremum over a . Since $(-s)^{-1/2} \in L^1(-4, 0)$, Gronwall's inequality yields

$$\operatorname{ess\,sup}_{s_0 < s < 0} \alpha(s) \leq CK \exp \left(C \int_{s_0}^0 (1 + M(-s)^{-1/2}) ds \right) \leq B_1(M, K).$$

Returning to (6.6), using this bound, and then letting $t \uparrow 0$ through admissible times gives the corresponding uniformly-local dissipation estimate on $(-1, 0)$. This proves (6.4). $\square$

Remark 6.3 (Local energy estimate as an absorbing bound). The preceding proof can be read as a three-term estimate for the local quantity $\alpha(s)$. The diffusion and transport terms are linear in α , with coefficient $1 + M(-s)^{-1/2}$. The local pressure term is also linear after Calderon-Zygmund and the Type I envelope. The remaining nonlinear term is the harmonic pressure flux, which is $O(\alpha^{3/2})$; the pointwise Type I envelope implies $\alpha^{1/2} \leq CM(-s)^{-1/2}$, reducing it to the same integrable linear coefficient. Thus large values of α cannot persist: the resulting differential inequality bounds α in terms of constants depending only on M and the initial uniformly local energy level.

Proposition 6.4 (Terminal kinetic tightness). *Let (u, p, z_*) , $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity in the sense of Definition 6.1. Then there exist $r_0 > 0$ and $B_A < \infty$, with B_A depending only on the Type I constant M , such that*

$$(6.12) \quad A(z_*, r) \leq B_A, \quad 0 < r < r_0.$$

Moreover,

$$(6.13) \quad r^{-1} \int_{T-r^2}^T \int_{B_r(x_*)} |\nabla u|^2 dx dt \leq B_A, \quad 0 < r < r_0.$$

Proof. Choose $r_0 > 0$ so small that $T - 4r^2 > T_0$ for $0 < r < r_0$. For such r , use the whole-space Leray pressure gauge fixed above and define the rescaled solution

$$v^{(r)}(y, s) = ru(x_* + ry, T + r^2s), \quad \pi^{(r)}(y, s) = r^2p(x_* + ry, T + r^2s),$$

on $\mathbb{R}^3 \times (-4, 0)$. The Type I bound (6.1) gives

$$|v^{(r)}(y, s)| \leq M(-s)^{-1/2}, \quad -4 < s < 0.$$

For any admissible $s_0 \in (-4, -3)$ at which the Type I bound holds, and any $a \in \mathbb{R}^3$,

$$\begin{aligned} \int_{B_2(a)} |v^{(r)}(y, s_0)|^2 dy &= r^{-1} \int_{B_{2r}(x_* + ra)} |u(x, T + r^2 s_0)|^2 dx \\ &\leq r^{-1} |B_{2r}| \frac{M^2}{r^2 |s_0|} \leq CM^2. \end{aligned}$$

Such times form a full-measure subset of $(-4, -3)$. The preceding estimate is uniform in this choice, so [Lemma 6.2](#) applies with $K = CM^2$, uniformly in r . In particular K depends only on M , so $B_A = B_{\text{uloc}}(M, CM^2)$ depends only on M . Hence

$$\text{ess sup}_{-1 < s < 0} \int_{B_1} |v^{(r)}(y, s)|^2 dy + \int_{-1}^0 \int_{B_1} |\nabla v^{(r)}|^2 dy ds \leq B_A.$$

Undoing the parabolic change of variables gives precisely [\(6.12\)](#) and [\(6.13\)](#). The constant B_A depends only on M ; no global energy norm of u has been used in the estimate. $\square$

Theorem 6.5 (Global pointwise Type I implies finite-energy velocity no-escape). *Let (u, p, z_*) , $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Let $\lambda_k \downarrow 0$ be any sequence of scales, and set*

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then

$$(6.14) \quad \limsup_{\sigma \downarrow 0} \sup_k \int_{-\sigma}^0 \int_{B_1} |u_k(x, t)|^3 dx dt = 0.$$

Proof. By [Proposition 6.4](#), after discarding at most finitely many indices,

$$\text{ess sup}_{-1 < t < 0} \int_{B_1} |u_k(x, t)|^2 dx \leq B_A.$$

The Type I bound gives, for the same tail of the sequence,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(-t)^{-1/2}, \quad -1 < t < 0.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt &\leq \int_{-\sigma}^0 \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x, t)|^2 dx dt \\ &\leq MB_A \int_{-\sigma}^0 (-t)^{-1/2} dt = 2MB_A \sigma^{1/2}. \end{aligned}$$

This bound is uniform over the tail. For each of the finitely many discarded indices k , choose $\sigma_k > 0$ so that $T + \lambda_k^2 t > T_0$ for $-\sigma_k < t < 0$. On this terminal subinterval the Type I envelope and the finite-energy assumption imply

$$\int_{-\sigma_k}^0 \int_{B_1} |u_k|^3 dx dt \leq \int_{-\sigma_k}^0 \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x, t)|^2 dx dt < \infty.$$

Indeed, on $(-\sigma_k, 0)$,

$$\|u_k(t)\|_{L^\infty(B_1)} \leq M(-t)^{-1/2}, \quad \int_{B_1} |u_k(x, t)|^2 dx \leq \lambda_k^{-1} \|u(T + \lambda_k^2 t)\|_{L^2(\mathbb{R}^3)}^2,$$

and the right-hand side is finite for this fixed k by finite energy. Thus each discarded index has absolute continuity of its L^3 -integral at the terminal slice. Taking the maximum over this finite set gives the same limit for the full supremum over k . This proves [\(6.14\)](#). $\square$

Corollary 6.6 (Admissibility of finite-energy Type I concentration sequences). *Let (u, p, z_*) , $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Then there exist $\eta_v > 0$ and $\lambda_k \downarrow 0$ such that*

$$C(z_*, \lambda_k) \geq \eta_v \quad \text{for every } k,$$

and the corresponding rescaled velocities satisfy the no-escape condition (6.14). Hence the concentration and no-escape admissibility condition used in the Type I blow-up limit theorem Theorem 9.1 follows for finite-energy Type I singularities.

Proof. Since $x_* \in \Sigma(T)$, Corollary 5.8 gives $\eta_v > 0$ and $\lambda_k \downarrow 0$ with $C(z_*, \lambda_k) \geq \eta_v$ for all k . The no-escape condition for this same sequence follows from Theorem 6.5. $\square$

7. THE TYPE I/TYPE II BLOW-UP RATE DICHOTOMY

Let $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ be given by Theorem 5.7. Define

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) are suitable weak solutions on expanding backward cylinders, and by Proposition 3.3 they retain a positive lower bound for the mean-subtracted local scale-invariant quantity on a fixed bounded cylinder.

Definition 7.1 (Type I concentration sequence). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T) . The sequence is called Type I if the concentration point (x_0, T) satisfies a local Type I scale-invariant bound: there are $\rho > 0$ and $M < \infty$, with $\rho^2 < \delta$, such that

$$\operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_0))} \leq M.$$

This is the Type I alternative.

Definition 7.2 (Local Type II alternative). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T) . The sequence belongs to the local Type II alternative if the concentration point (x_0, T) satisfies no local Type I scale-invariant bound. Equivalently, for every $\rho > 0$ with $\rho^2 < \delta$,

$$\operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_0))} = \infty.$$

Theorem 7.3 (Local blow-up rate dichotomy). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

The sequence r_n is a positive local concentration sequence. The corresponding concentration regime satisfies either the local Type I bound or the local Type II alternative. Consequently, once the vanishing case has been excluded by ε -regularity, every finite-time singular case belongs to precisely one of the two blow-up rate alternatives.

Proof. If the local scale-invariant quantities vanished at every point of $\Sigma(T)$, then Theorem 4.5 would give $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore Theorem 5.7 gives $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ with the stated lower bound. The rescaled sequence has a positive local scale-invariant lower bound by Proposition 3.3. By definition, either the Type I bound in Definition 7.1 holds, or the local Type II alternative in Definition 7.2 holds. $\square$

Remark 7.4. This theorem is local. Its conclusion is the existence of a positive mean-subtracted scale-invariant concentration sequence and the dichotomy between a local Type I bound and failure of every such bound at the concentration point.

8. LOCAL POINTWISE TYPE I REDUCTION TO SEREGIN EXTRACTION

The following consequence of the weak-Serrin endpoint argument is used. The local pointwise Type I alternative in [Definition 7.1](#) is strong enough, after rescaling at the singular point, to give compactness on every compact negative-time cylinder. The endpoint weak-Serrin implication reads: for suitable weak solutions the local pointwise Type I envelope belongs to $L_t^{2,\infty} L_x^\infty$, hence the Albritton–Barker weak Serrin local Type I estimate gives the scale-invariant A -bound on smaller cylinders and therefore terminal velocity no-escape. Thus positive local Type I concentration sequences are admissible for the local Seregin extraction in [Lemma 8.7](#).

Throughout this section $z_* = (x_*, T)$, and

$$Q_1(0, 0) = B_1(0) \times (-1, 0).$$

Definition 8.1 (Admissible local Type I concentration sequence). For a local pointwise Type I point $z_* = (x_*, T)$, a sequence $r_n \downarrow 0$ is an admissible local Type I concentration sequence if the rescaled velocities

$$u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t)$$

satisfy, for some $\eta_v > 0$,

$$\int_{Q_1(0,0)} |u_n|^3 dx dt \geq \eta_v \quad \text{for every } n,$$

and

$$\limsup_{\sigma \downarrow 0} \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dx dt = 0.$$

Proposition 8.2 (Local pointwise Type I rescalings have terminal compactness). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$ satisfying the finite-energy bounds*

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)).$$

Let $z_ = (x_*, T)$ be a singular point, and assume that z_* satisfies the local pointwise Type I bound: there exist $\rho > 0$ and $M < \infty$ such that*

$$(8.1) \quad \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

For any sequence $r_n \downarrow 0$, define

$$(8.2) \quad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t), \quad p_n(x, t) = r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Then the following hold.

(i) *If*

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \quad R, S < \infty, \quad \sigma > 0,$$

then, for all sufficiently large n ,

$$(8.3) \quad \|u_n\|_{L^\infty(K)} \leq M \sigma^{-1/2}.$$

(ii) *On each such compact cylinder K , after subtracting time-dependent local pressure gauges $a_{n,K}(t)$, the sequence satisfies uniform local Seregin bounds*

$$(8.4) \quad \|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \leq C_K.$$

Here C_K may depend on K, M, ρ , and on the finite Leray energy level of the original solution, but it does not depend on n and does not use a terminal no-escape assumption.

- (iii) *After passing to a subsequence, $(u_n, p_n - a_{n,K})$ converges on every compact $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ to an ancient suitable weak solution (U, Q) , with*

$$\begin{aligned} u_n &\rightarrow U \quad \text{strongly in } L^q(K), \quad 1 \leq q < \infty, \\ p_n - a_{n,K}(t) &\rightharpoonup Q \quad \text{weakly in } L^{3/2}(K), \end{aligned}$$

and

$$|U(x, t)| \leq M(-t)^{-1/2}, \quad t < 0.$$

Proof. Fix $K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$. For large n ,

$$r_n R < \rho/2, \quad r_n^2 S < \rho^2/2.$$

Thus $x_* + r_n x \in B_\rho(x_*)$ and $T + r_n^2 t \in (T - \rho^2, T)$ for a.e. $(x, t) \in K$. The local pointwise Type I bound (8.1) gives

$$|u_n(x, t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \leq r_n M(T - (T + r_n^2 t))^{-1/2} = M(-t)^{-1/2} \leq M\sigma^{-1/2}.$$

This proves (8.3). It also gives uniform local $L_t^\infty L_x^2$ and $L_{x,t}^3$ bounds on K .

It remains to explain the pressure and dissipation bounds. At this point the ambient solution is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, not merely a local suitable weak solution. Hence the whole-space Leray pressure representative may be chosen, modulo functions of time, as $p = \mathcal{R}_i \mathcal{R}_j (u_i u_j)$. All local pressure estimates below are made for this representative, followed by local time-dependent mean subtraction. Choose $\chi_R \in C_c^\infty(B_{8R})$, $\chi_R \equiv 1$ on B_{4R} , and write inside B_{4R}

$$p_n = q_n + h_n, \quad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

Calderon–Zygmund boundedness and the local L^3 -bound for u_n give a uniform $L^{3/2}$ -bound for q_n . For h_n , take the gauge $a_{n,K}(t) = h_n(0, t)$. Let

$$A_m = \{2^m R < |z| < 2^{m+1} R\}, \quad m \geq 2.$$

For $x \in B_{2R}$, the harmonic pressure kernel estimate gives

$$|h_n(x, t) - a_{n,K}(t)| \leq CR \sum_{m \geq 2} (2^m R)^{-4} \int_{A_m} |u_n(z, t)|^2 dz.$$

Choose m_n by

$$2^{m_n} R r_n \leq \rho < 2^{m_n+1} R r_n.$$

After discarding finitely many indices, assume $m_n \geq 2$. If $m \leq m_n - 1$, then the physical image of A_m is contained in $B_\rho(x_*)$. On the time slab $[-S, -\sigma]$, the local Type I bound therefore gives

$$\int_{A_m} |u_n(z, t)|^2 dz \leq CM^2 \sigma^{-1} (2^m R)^3.$$

Consequently the inner part of the harmonic tail is bounded by

$$\begin{aligned} CR \sum_{m \leq m_n - 1} (2^m R)^{-4} \int_{A_m} |u_n(z, t)|^2 dz &\leq CRM^2 \sigma^{-1} \sum_{m \leq m_n - 1} (2^m R)^{-1} \\ &\leq CM^2 \sigma^{-1}. \end{aligned}$$

For $m \geq m_n$, the finite Leray energy gives

$$\int_{A_m} |u_n(z, t)|^2 dz \leq \int_{\mathbb{R}^3} |u_n(z, t)|^2 dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 dy.$$

Since $2^{m_n}R \simeq \rho/r_n$, the outer part satisfies

$$\begin{aligned} CR \sum_{m \geq m_n} (2^m R)^{-4} \int_{A_m} |u_n(z, t)|^2 dz &\leq CR r_n^{-1} \|u\|_{L^\infty(0, T; L^2)}^2 \sum_{m \geq m_n} (2^m R)^{-4} \\ &\leq C_{\rho, R} r_n^3 \|u\|_{L^\infty(0, T; L^2)}^2. \end{aligned}$$

Thus $h_n - a_{n, K}$ is bounded in $L^\infty([-S, -\sigma]; L^\infty(B_{2R}))$, uniformly in n . Together with the $L^{3/2}$ -bound for q_n , this gives the $L^{3/2}(K)$ pressure estimate in (8.4).

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder. The local kinetic, cubic velocity, and pressure-flux terms are bounded by the preceding estimates, so $\|\nabla u_n\|_{L^2(K)} \leq C_K$. The Navier–Stokes equation then bounds $\partial_t u_n$ locally in a negative Sobolev space. Aubin–Lions compactness and the pressure bound give the asserted Seregin subsequence convergence. Passing to the limit in the equations and local energy inequality gives an ancient suitable weak solution, and the pointwise Type I bound passes to the limit. $\square$

Lemma 8.3 (Terminal A -bound from the local weak-Serrin estimate). *Let (u, p) be a suitable weak solution in a backward cylinder $Q_\rho(z_*)$, $z_* = (x_*, T)$, and assume the local pointwise Type I bound (8.1). Then, for every $0 < \theta < 1$, there is $K_\theta < \infty$ such that, for all sufficiently small r ,*

$$r^{-1} \operatorname{ess\,sup}_{T-r^2 < t < T} \int_{B_r(x_*)} |u(x, t)|^2 dx \leq K_\theta.$$

Proof. After translating and scaling, assume $z_* = (0, 0)$ and work in

$$Q_\rho = B_\rho \times (-\rho^2, 0).$$

Put

$$f(t) = \|u(t)\|_{L^\infty(B_\rho)}.$$

The pointwise Type I bound gives $f(t) \leq M(-t)^{-1/2}$. Hence, for every $\alpha > 0$,

$$|\{t \in (-\rho^2, 0) : f(t) > \alpha\}| \leq \min\{\rho^2, M^2 \alpha^{-2}\},$$

and therefore

$$\|f\|_{L^{2, \infty}(-\rho^2, 0)} \leq CM.$$

Thus

$$u \in L_t^{2, \infty} L_x^\infty(Q_\rho).$$

Fix $0 < \theta < 1$. Apply the Albritton–Barker weak-Serrin Type I lemma [1, Lemma 2.5] with $p = \infty$ and $q = 2$. In their notation,

$$I(\omega) = \sup_{Q' \in \omega} (A(Q') + C(Q') + D(Q') + E(Q')),$$

and the endpoint weak-Serrin hypothesis above gives a finite bound

$$I(Q_{\theta\rho}) \leq K_\theta.$$

Let $r < (\theta\rho)/\sqrt{2}$ and $0 < \varepsilon < r^2$. Then

$$Q_r((0, -\varepsilon)) = B_r \times (-r^2 - \varepsilon, -\varepsilon) \Subset Q_{\theta\rho},$$

so the Albritton–Barker estimate gives

$$r^{-1} \operatorname{ess\,sup}_{-r^2 - \varepsilon < t < -\varepsilon} \int_{B_r} |u(x, t)|^2 dx \leq K_\theta.$$

Since $(-r^2, -\varepsilon) \subset (-r^2 - \varepsilon, -\varepsilon)$, it follows that

$$r^{-1} \operatorname{ess\,sup}_{-r^2 < t < -\varepsilon} \int_{B_r} |u(x, t)|^2 dx \leq K_\theta.$$

As $\varepsilon \downarrow 0$, the intervals $(-r^2, -\varepsilon)$ increase to $(-r^2, 0)$. By the definition of essential supremum over nested measurable sets,

$$r^{-1} \operatorname{ess\,sup}_{-r^2 < t < 0} \int_{B_r} |u(x, t)|^2 dx \leq K_\theta.$$

Undoing the translation and scaling gives the stated estimate at z_* . $\square$

Theorem 8.4 (Local endpoint weak-Serrin Type I estimate). *Let (u, p) be a suitable weak solution in a backward cylinder $Q_\rho(z_*)$, $z_* = (x_*, T)$, and assume the local pointwise Type I bound (8.1)*

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

Then, after possibly shrinking ρ , the rescaled velocities $u_n(y, s) = r_n u(x_ + r_n y, T + r_n^2 s)$ along every scale sequence $r_n \downarrow 0$ satisfy the terminal velocity no-escape condition*

$$\limsup_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n(y, s)|^3 dy ds = 0.$$

Proof. After translating and scaling, assume $z_* = (0, 0)$ and work in

$$Q_\rho = B_\rho \times (-\rho^2, 0).$$

Let $r_n \downarrow 0$ and

$$u_n(y, s) = r_n u(r_n y, r_n^2 s).$$

Fix $0 < \theta < 1$. By Lemma 8.3, the terminal A -quantity is bounded uniformly along the chosen scales. For all sufficiently large n ,

$$\operatorname{ess\,sup}_{-1 < s < 0} \int_{B_1} |u_n(y, s)|^2 dy = r_n^{-1} \operatorname{ess\,sup}_{-r_n^2 < t < 0} \int_{B_{r_n}} |u(x, t)|^2 dx \leq K_\theta.$$

Moreover,

$$\|u_n(s)\|_{L^\infty(B_1)} \leq M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dy ds &\leq \int_{-\sigma}^0 \|u_n(s)\|_{L^\infty(B_1)} \int_{B_1} |u_n(y, s)|^2 dy ds \\ &\leq M K_\theta \int_{-\sigma}^0 (-s)^{-1/2} ds = 2 M K_\theta \sigma^{1/2}. \end{aligned}$$

This proves the required terminal velocity no-escape for all sufficiently large n . The finitely many remaining indices are harmless by absolute continuity of the local L^3 integral, since suitable weak solutions are locally in L^3 on terminal subcylinders. Taking the supremum in n and then letting $\sigma \downarrow 0$ gives the claim. $\square$

Corollary 8.5 (Local pointwise Type I concentration sequences are admissible). *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be any sequence such that the rescaled velocities satisfy*

$$\int_{Q_1(0,0)} |u_n|^3 dx dt \geq \eta_v > 0 \quad \text{for all } n.$$

Then r_n is an admissible local Type I concentration sequence in the sense of Definition 8.1.

Proof. The displayed positive concentration lower bound is the first requirement in [Definition 8.1](#). The Albritton–Barker local weak-Serrin Type I estimate [\[1\]](#) gives terminal velocity no-escape from the local pointwise Type I envelope for every rescaled sequence; the precise statement used here is [Theorem 8.4](#). This gives the second requirement in [Definition 8.1](#). $\square$

Lemma 8.6 (Local Type I concentration enters the Seregin extraction). *Under the hypotheses of [Proposition 8.2](#), let $r_n \downarrow 0$ be a velocity concentration sequence satisfying*

$$\int_{Q_1(0,0)} |u_n|^3 dx dt \geq \eta_v > 0 \quad \text{for all } n.$$

Then r_n is admissible by [Corollary 8.5](#). There are $\sigma_0 \in (0, 1)$ and $\eta_1 > 0$ such that

$$(8.5) \quad \int_{-1}^{-\sigma_0} \int_{B_1} |u_n|^3 dx dt \geq \eta_1 \quad \text{for all } n.$$

Consequently, the rescaled sequence retains positive local L^3 mass on a compact negative-time cylinder. In particular the associated Seregin extraction has no loss of the local L^3 mass needed for nontriviality.

Proof. The no-escape condition in [Definition 8.1](#) gives $\sigma_0 > 0$ such that the terminal layer $B_1 \times (-\sigma_0, 0)$ carries at most half of the fixed Q_1 velocity mass. This proves [\(8.5\)](#). The compactness part follows from [Proposition 8.2](#) on the compact negative-time cylinder $B_1 \times [-1, -\sigma_0]$. Strong local L^3 convergence on this cylinder passes the positive mass to the ancient limit. Terminal no-escape is used only at this step.

Thus the local Type I concentration sequence is admissible for the Seregin extraction and retains positive localized L^3 -mass away from the terminal slice. $\square$

Lemma 8.7 (Local Seregin extraction from an admissible local Type I sequence). *Under the hypotheses of [Proposition 8.2](#), let $r_n \downarrow 0$ be an admissible local Type I concentration sequence in the sense of [Definition 8.1](#). Then, after passing to a subsequence, the rescaled velocities produce a smooth bounded centered solution V of [\(9.3\)](#) on $\mathbb{R}^3 \times \mathbb{R}$. The solution V satisfies the normalization used for Seregin ancient limits: there are a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$ such that*

$$\iint_K |V|^3 dy d\tau \geq \eta_0.$$

Proof. [Proposition 8.2](#) gives local Seregin compactness for the rescaled sequence on every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$. The local Type I bound passes to the ancient limit as

$$|U(x, t)| \leq M(-t)^{-1/2}, \quad t < 0.$$

Interior Serrin regularity then gives smoothness of U on $\mathbb{R}^3 \times (-\infty, 0)$ by the argument used in [Lemma 13.4](#). The centered change of variables

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t)$$

therefore gives a smooth bounded solution of the centered equation [\(9.3\)](#). The no-escape part of local admissibility and [Lemma 8.6](#) give positive compact L^3 -mass for U away from $t = 0$. On the compact image of that cylinder under $(x, t) = (e^{-\tau/2}y, -e^{-\tau})$, the factor $e^{-\tau}$ in

$$|U|^3 dx dt = e^{-\tau} |V|^3 dy d\tau$$

is bounded above and below by positive constants. Hence the positive lower bound for U becomes the displayed local L^3 nonvanishing for V . $\square$

Proposition 8.8 (Local Type I entry into the final assembly). *Assume the residual-class hypothesis [Hypothesis 16.1](#). Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$. Then every terminal point $z_* = (x_*, T)$ satisfying the local pointwise Type I bound [\(8.1\)](#) is regular. More precisely, if such a point were singular, it would generate a raw extracted normalized Seregin ancient limit with compact spacetime nonvanishing; [Lemma 13.12](#) would then give the invariant local nonvanishing used to form the associated generated descendant hull. That raw hull would be exhausted by the Liouville alternatives and the residual class removed by [Hypothesis 16.1](#).*

Proof. Assume [Hypothesis 16.1](#). Suppose, toward a contradiction, that $z_* = (x_*, T)$ is a singular point satisfying [\(8.1\)](#). Choose the positive Q_1 velocity concentration sequence supplied by the local concentration theorem. By [Corollary 8.5](#), this sequence satisfies terminal no-escape and is admissible. By [Lemma 8.6](#), the local Type I sequence has local compactness and positive compact L^3 -mass away from the terminal slice. By [Lemma 8.7](#), the sequence generates a raw extracted normalized Seregin ancient limit V . By [Lemma 13.12](#), V satisfies the invariant local nonvanishing condition for some data (R_0, η_*) . Form the raw generated descendant hull

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*).$$

By construction, $V \in \mathcal{S}$. By the class exhaustion in [Proposition 14.11](#), V lies either in one of the Liouville classes excluded by the stated Liouville theorems or in the remaining class $\mathcal{R}(\mathcal{S})$. The Liouville classes are excluded by the small-amplitude, stationary, uniformly tight, and structure-and-decay Liouville results; the last of these is routed through [Theorem 32.1](#). The residual alternative is removed by the residual-class hypothesis applied to $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. This contradicts the generated nonvanishing guaranteed by [Corollary 13.19](#). Hence z_* is regular. $\square$

Corollary 8.9 (Conditional exclusion of the local Type I alternative). *Assume the residual-class hypothesis [Hypothesis 16.1](#). Then the local Type I alternative is excluded for every finite-energy singular point. Hence every finite-energy singular point belongs to the local Type II alternative in the dichotomy above.*

Proof. This is [Proposition 8.8](#) restated in the terminology of the local Type I/local Type II dichotomy. $\square$

Part I dependency diagram.

singular point $z_* = (x_*, T)$
 $\downarrow$ [Corollary 5.8](#) (CKN)
 positive velocity concentration scales
 $\downarrow$ [Theorem 8.4](#) (local) or [Theorem 6.5](#) (global)
 no-escape away from terminal slice
 $\downarrow$ [Corollary 8.5](#) / [Corollary 6.6](#)
 admissible concentration sequence
 $\downarrow$ [Lemma 8.7](#) or [Theorem 9.1](#)
 raw extracted Seregin ancient limit, then raw generated state space and mild strata (Part II)
 $\downarrow$ class decomposition [Definition 14.7](#)
 non-residual classes (excluded here) $\cup$ residual class
 $\downarrow$ [Hypothesis 16.1](#) (conditional residual closure hypothesis)
 contradiction $\Rightarrow$ [Theorem 33.1](#).

This is a reduction theorem, not a self-contained residual proof; the residual closure assumption is isolated in [Hypothesis 16.1](#).

PART II. TYPE I BLOW-UP LIMITS AND THE RESIDUAL HYPOTHESIS

Overview. An admissible finite-energy Type I singularity for the three-dimensional Navier–Stokes equations is first reduced to a raw extracted nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation, and then to the raw generated state space used in the residual argument. The construction keeps track of pressure gauges, local compactness, stability of the local energy inequality, and the velocity concentration supplied by local Caffarelli–Kohn–Nirenberg theory. It yields a smooth bounded ancient solution for the centered equation on $\mathbb{R}^3 \times \mathbb{R}$, together with a quantitative local L^3 nonvanishing condition.

The possible generated raw Seregin states are organized by the analytic hypotheses used in the class decomposition: small amplitude, stationary L^3 , uniform L^3 -tightness, and structure or decay assumptions to which cited Liouville theorems apply. The small-amplitude case is excluded by the bounded ancient perturbative Liouville theorem in [Lemma 15.4](#), and the stationary L^3 case by the theorem of Nečas–Růžička–Šverák. The structure and decay class contains only hypotheses for which the cited Liouville theorems force $V \equiv 0$. The uniformly L^3 -tight class is excluded in [Theorem 25.8](#). The residual class is handled only through the residual-class hypothesis [Hypothesis 16.1](#). Consequently the Type I conclusion obtained from this extraction is conditional on that residual hypothesis within the raw generated finite-energy and local pointwise Type I extraction classes.

9. INTRODUCTION AND MAIN RESULTS

9.1. Type I blow-up limits. Let $u : \mathbb{R}^3 \times (0, T) \rightarrow \mathbb{R}^3$ solve the incompressible Navier–Stokes equations

$$(9.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \quad \operatorname{div} u = 0.$$

The local regularity theory of Caffarelli–Kohn–Nirenberg [\[2\]](#) detects singular points through scale-invariant concentration of velocity and pressure. A terminal point $z_* = (x_*, T)$ is of Type I if

$$(9.2) \quad \limsup_{t \uparrow T} \sqrt{T-t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

The blow-up extraction uses a selected velocity concentration sequence from the local concentration theorem and assumes only that this velocity concentration does not escape into the terminal slice after rescaling. The resulting admissibility condition is defined in [Definition 10.4](#). It replaces a stronger all-scale compactness assumption on scale-invariant energy, pressure, and dissipation quantities by the non-escape condition needed for nontriviality of the ancient limit.

9.2. Normalized Seregin limits. For scales $\lambda \downarrow 0$, the parabolic rescaling about z_* is

$$u^{(\lambda)}(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^{(\lambda)}(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

The rescaled time intervals exhaust $(-\infty, 0)$. The Type I bound becomes a uniform bound

$$\|u^{(\lambda)}(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(-t)^{-1/2}$$

on compact subintervals of $(-\infty, 0)$, and compactness produces an unrenormalized ancient solution (U, Q) . The associated centered variables are

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t).$$

They transform the ancient Navier–Stokes solution into the centered equation

$$(9.3) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0.$$

The selected velocity concentration at the singular point persists through the extraction and gives a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and a constant $\eta_0 > 0$ such that

$$(9.4) \quad \iint_K |V|^3 dy d\tau \geq \eta_0.$$

9.3. Liouville alternatives and the remaining case. No unconditional Liouville theorem is known for all bounded ancient solutions of (9.3). The reduction therefore uses a finite decomposition based on explicit Liouville theorems. A generated raw Seregin state is compared with the following analytic classes:

$$\mathcal{C}_{\text{small}}, \quad \mathcal{C}_{\text{stat}, L^3}, \quad \mathcal{C}_{L^3\text{-tight}}, \quad \mathcal{C}_{\text{sd}}, \quad \mathcal{R}(\mathcal{S}).$$

They denote, respectively, small amplitude, stationary L^3 , uniform L^3 -tightness, and structure or decay hypotheses to which the cited Liouville theorems apply (sd), and the complement of these four classes inside a raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. The last set carries no additional analytic hypothesis; it is the remaining case after the stated Liouville alternatives have been removed.

9.4. Main theorems.

Theorem 9.1 (Normalized Type I blow-up limit). *Let (u, p, z_*) be an admissible finite-energy Type I singularity. Then there exist scales $\lambda_k \downarrow 0$, pressure gauges $a_k(t)$, an ancient suitable weak solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0)$, and a centered solution (V, Π) on $\mathbb{R}^3 \times \mathbb{R}$, such that, after passing to a subsequence,*

$$u_k \rightarrow U \quad \text{strongly in } L_{\text{loc}}^q(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq q < \infty,$$

and

$$p_k - a_k(t) \rightharpoonup Q \quad \text{weakly in } L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (-\infty, 0)).$$

The centered velocity V is smooth, bounded, solves (9.3), and is a raw extracted normalized Seregin ancient limit with the compact spacetime L^3 nonvanishing condition (9.4).

For a raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$, the class definitions give, for each $V \in \mathcal{S}$, the exhaustive alternative

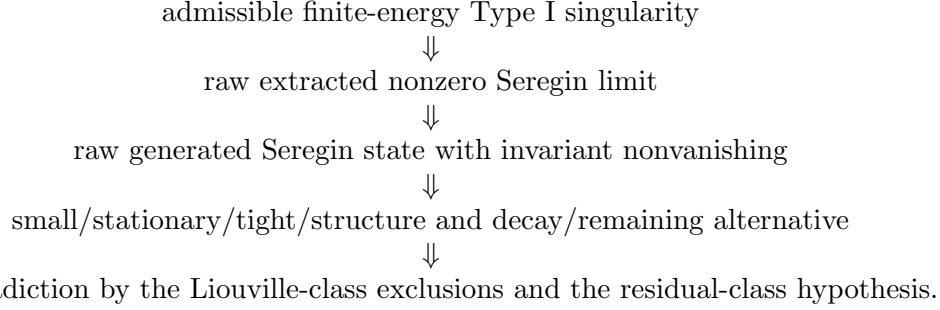
$$V \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}} \cup \mathcal{R}(\mathcal{S}).$$

This exhaustive alternative is recorded in [Proposition 14.11](#). It is a direct consequence of the definition of the remaining class, not an independent Liouville theorem.

Theorem 9.2 (Conditional Type I exclusion). *Assume the residual-class hypothesis [Hypothesis 16.1](#). Then no admissible finite-energy Type I singularity exists.*

The argument uses one closure theorem and one explicit residual-class hypothesis. The uniformly L^3 -tight class is excluded by [Theorem 25.8](#). The remaining class is excluded only under the residual-class hypothesis [Hypothesis 16.1](#). The structure and decay class is restricted to hypotheses to which cited Liouville theorems apply.

The resulting implication is



- Small amplitude: excluded in [Section 15](#).
- Stationary L^3 : excluded by [\[16\]](#).
- Uniformly L^3 -tight: proved in [Theorem 15.8](#), using [Theorem 25.8](#).
- Structure and decay: routed through [Theorem 32.1](#) and excluded in [Theorem 15.9](#).
- Remaining class: excluded under the residual-class hypothesis [Hypothesis 16.1](#).

9.5. Dependencies and conditional hypothesis. The proof of the extraction theorem uses the local velocity concentration result stated as [Proposition 10.7](#); no other result from the local concentration theory enters the blow-up construction. The extraction first produces a raw extracted limit carrying the local L^3 lower bound [\(9.4\)](#). The subsequent hull construction uses [Lemma 13.12](#) and [Lemma 13.16](#), which provide the invariant nonvanishing and the raw generated structure. Mildness is imposed only in the branch definitions that need it. The uniformly L^3 -tight Liouville theorem and the residual-class hypothesis are invoked only after this raw generated class is formed and the class exhaustion is applied. The statement [Theorem 15.8](#) follows from the endpoint L^3 closure in [Theorem 25.8](#), and [Hypothesis 16.1](#) is the residual-class hypothesis for $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. The structure and decay class is routed through [Theorem 32.1](#), whose hypotheses are recorded in the structured Liouville section. Thus the final Type I conclusion is conditional on the admissibility condition in [Corollary 6.6](#), the extraction theorem, the raw generated closure statements, the Liouville-class exclusions, and the residual-class hypothesis.

9.6. Two entry routes for the reduction. There are two routes to the same raw generated Seregin state space; they have different hypotheses and first produce the same type of raw extracted ancient solution.

- (1) *Local pointwise Type I.* The Part I local concentration and local endpoint weak-Serrin estimate [Theorem 8.4](#) produce an admissible local Type I concentration sequence at any singular point satisfying [\(8.1\)](#), and [Lemma 8.6](#) feeds the resulting rescalings into the Seregin extraction with positive compact L^3 velocity mass on a fixed negative-time cylinder. This is the entry point used by the final assembly theorem [Theorem 33.1](#).
- (2) *Global pointwise Type I.* The Part II extraction estimates, including the pressure decomposition in [Lemma 11.2](#), are written for an admissible Type I solution in the sense of [Definition 10.4](#), which encodes a global pointwise Type I bound on $\mathbb{R}^3$. This streamlines the harmonic-pressure tail estimates, which would otherwise require the local pressure-tail decomposition recorded in [Proposition 8.2](#).

In both cases the conclusion is first a raw extracted normalized Seregin ancient limit satisfying [\(9.4\)](#). The subsequent hull construction uses the invariant normalization [\(13.1\)](#). The local pointwise entry uses Part I to produce an admissible concentration sequence and pressure-tail control, while Part II then runs on the resulting compact cylinders. No analytic hypothesis beyond local pointwise Type I and finite Leray energy is added in this reduction.

Global route	global pointwise Type I	global no-escape $\Rightarrow$ finite-energy Seregin extraction
Local route	local pointwise Type I	AB endpoint weak-Serrin bounds $\Rightarrow$ terminal no-escape $\Rightarrow$ local Seregin extraction

9.7. Organization. [Section 10](#) introduces suitable solutions, scale-invariant quantities, the admissibility condition, scaling, pressure gauges, and local concentration. [Section 11](#) proves the compactness theorem for Type I rescalings away from the terminal time and constructs the ancient suitable weak limit. [Section 12](#) proves persistence of concentration and nontriviality. [Section 13](#) derives the centered equation and defines raw extracted limits, the raw generated state space, and its mild stratum. [Section 14](#) defines the Liouville classes and the remaining class. [Section 15](#) proves the class exclusions and the uniformly tight theorem. [Section 16](#) proves the conditional remaining-case criterion of [Theorem 9.2](#) and the residual closure hypothesis. [Appendix 17](#) gives the elementary symmetries of the centered equation.

10. SUITABLE SOLUTIONS, SCALE-INVARIANT QUANTITIES, AND LOCAL CONCENTRATION

For $z_0 = (x_0, t_0)$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

When $z_0 = (0, 0)$, write $Q_r = Q_r(0, 0)$. The spatial average of f over $B_r(x_0)$ is

$$(f)_{B_r(x_0)}(t) = \frac{1}{|B_r|} \int_{B_r(x_0)} f(x, t) dx.$$

Throughout this part, $\mathbb{P}$ denotes the Helmholtz–Leray projection onto divergence-free vector fields on $\mathbb{R}^3$.

Definition 10.1 (Terminal singular point). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. A terminal point $z_* = (x_*, T)$ is regular if there are $r > 0$ and $K < \infty$ such that

$$\operatorname{ess\,sup}_{Q_r(z_*)} |u| \leq K.$$

It is singular otherwise.

Definition 10.2 (Suitable weak solution). This definition repeats the Part I notion in the notation used here; the two definitions agree. Let $\Omega \subset \mathbb{R}^3 \times \mathbb{R}$ be open. A pair (u, p) is a suitable weak solution of (9.1) in Ω if

$$u \in L_t^\infty L_x^2(\Omega') \cap L_t^2 \dot{H}_x^1(\Omega'), \quad p \in L^{3/2}(\Omega')$$

for every compact cylinder $\Omega' \Subset \Omega$, if (u, p) solves (9.1) distributionally, $\operatorname{div} u = 0$, and if the local energy inequality

$$\begin{aligned} & \int_{\mathbb{R}^3} |u(x, t_2)|^2 \phi(x, t_2) dx + 2 \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi dx dt \\ & \leq \int_{\mathbb{R}^3} |u(x, t_1)|^2 \phi(x, t_1) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |u|^2 (\partial_t \phi + \Delta \phi) dx dt \\ & + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} (|u|^2 + 2p) u \cdot \nabla \phi dx dt \end{aligned} \tag{10.1}$$

holds for every non-negative $\phi \in C_c^\infty(\Omega)$ and almost every $t_1 < t_2$ for which the displayed terms are defined.

This is the Caffarelli–Kohn–Nirenberg notion of suitability [2]; the global finite-energy background is Leray’s Leray–Hopf class [6].

Definition 10.3 (Suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$). A suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, T)$ satisfying

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)), \quad u \in C_w([0, T]; L^2(\mathbb{R}^3)),$$

and the global Leray energy inequality. The pressure is taken in $L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T))$, modulo functions of time.

Throughout the global extraction and final assembly, the phrase *finite-energy suitable weak solution* on $\mathbb{R}^3 \times (0, T)$ means a suitable Leray–Hopf solution in the sense of Definition 10.3. Purely local statements use the phrase “suitable weak solution in a cylinder” and do not invoke the global energy or whole-space pressure representative unless explicitly stated.

Define

$$(10.2) \quad A(u; z_0, r) = r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx,$$

$$(10.3) \quad C(u; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |u|^3 dx dt, \quad D(p; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt,$$

$$(10.4) \quad E(u; z_0, r) = r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 dx dt,$$

Definition 10.4 (Admissible finite-energy Type I singularity). A triple (u, p, z_*) , with $z_* = (x_*, T)$, is an admissible finite-energy Type I singularity if:

- (i) (u, p) is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$;
- (ii) $z_* = (x_*, T)$ is a singular point;
- (iii) the Type I bound (9.2) holds;
- (iv) there are scales $\lambda_k \downarrow 0$ and a constant $\eta_v > 0$ such that

$$C(u; z_*, \lambda_k) \geq \eta_v \quad \text{for every } k;$$

- (v) for the corresponding rescaled velocities

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t),$$

the velocity no-escape condition holds:

$$(10.5) \quad \limsup_{\sigma \downarrow 0} \limsup_k \int_{-\sigma}^0 \int_{B_1} |u_k(x, t)|^3 dx dt = 0.$$

The sequence $\{\lambda_k\}$ is called an admissible concentration sequence at z_* .

Remark 10.5. Item (iv) is the velocity-only concentration supplied by the local concentration theorem; see Proposition 10.7. Thus, for a singular point, the additional admissibility requirement is the no-escape condition (10.5). It is the sole terminal-slice compactness assumption used to prove nontriviality of the ancient limit. No uniform bound on $A + C + D + E$ over all terminal scales is assumed.

Lemma 10.6 (Uniform kinetic energy implies no escape). *Let (u, p) satisfy the Type I bound (9.2), and let $\lambda_k \downarrow 0$. Suppose that*

$$\sup_k A(u; z_*, \lambda_k) \leq B < \infty.$$

Then the corresponding rescaled velocities satisfy the no-escape condition (10.5).

Proof. Choose $M < \infty$ and $t_0 < T$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(T-t)^{-1/2}, \quad t_0 < t < T.$$

For k sufficiently large, $T + \lambda_k^2 t > t_0$ whenever $-1 < t < 0$. Hence

$$\|u_k(t)\|_{L^\infty(B_1)} \leq M(-t)^{-1/2}, \quad -1 < t < 0.$$

By scale invariance of A ,

$$\operatorname{ess\,sup}_{-1 < t < 0} \int_{B_1} |u_k(x, t)|^2 dx = A(u; z_*, \lambda_k) \leq B.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt &\leq \int_{-\sigma}^0 \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x, t)|^2 dx dt \\ &\leq MB \int_{-\sigma}^0 (-t)^{-1/2} dt = 2MB\sigma^{1/2}. \end{aligned}$$

Taking the supremum over the tail $k \geq k_0$ and then letting $\sigma \downarrow 0$ gives (10.5) for the tail of the sequence. For each of the finitely many remaining indices $1 \leq k < k_0$, the rescaled velocity is defined on $B_1 \times (-\delta_k, 0)$ for some $\delta_k > 0$, and belongs to L^3 there. Therefore absolute continuity of the L^3 -integral gives

$$\lim_{\sigma \downarrow 0} \max_{1 \leq k < k_0} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt = 0.$$

Combining the finite initial segment with the tail proves the full supremum in (10.5). $\square$

Proposition 10.7 (Local velocity concentration). *Let (u, p) be suitable in a parabolic neighborhood of $z_* = (x_*, T)$. If z_* is singular, then there exist $\lambda_k \downarrow 0$ and $\eta_v > 0$ such that*

$$C(u; z_*, \lambda_k) \geq \eta_v \quad \text{for every } k.$$

Proof. This is the velocity-only concentration corollary of the local concentration result in Theorem 7.3. That result follows from the velocity ε -regularity criterion: if

$$\limsup_{r \downarrow 0} C(u; z_*, r) = 0,$$

then u is bounded in a backward parabolic cylinder ending at z_* , contradicting singularity of z_* . $\square$

Lemma 10.8 (Scaling). *Let (u, p) be suitable in a neighborhood of (x_*, T) . For $\lambda > 0$, set*

$$u^\lambda(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^\lambda(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

Then (u^λ, p^λ) is suitable on the rescaled domain. Moreover A, C, D, E are invariant under this scaling.

Proof. Suitability of the rescaled pair and preservation of pressure gauges follow by applying (10.1) to the test function

$$\phi^\lambda(X, S) = \phi\left(\frac{X - x_*}{\lambda}, \frac{S - T}{\lambda^2}\right)$$

and then changing variables $(X, S) = (x_* + \lambda x, T + \lambda^2 t)$. The differential identities

$$\partial_t u^\lambda = \lambda^3 (\partial_t u)(X, S), \quad (u^\lambda \cdot \nabla) u^\lambda = \lambda^3 ((u \cdot \nabla) u)(X, S),$$

$$\nabla p^\lambda = \lambda^3 (\nabla p)(X, S), \quad \Delta u^\lambda = \lambda^3 (\Delta u)(X, S)$$

give the rescaled equation and the divergence condition.

For the pressure average,

$$(p^\lambda)_{B_r}(t) = \lambda^2(p)_{B_{\lambda r}(x_*)}(S).$$

Consequently

$$\iint_{Q_r} |u^\lambda|^3 dx dt = \lambda^{-2} \iint_{Q_{\lambda r}(z_*)} |u|^3 dX dS$$

and

$$\iint_{Q_r} |p^\lambda - (p^\lambda)_{B_r}(t)|^{3/2} dx dt = \lambda^{-2} \iint_{Q_{\lambda r}(z_*)} |p - (p)_{B_{\lambda r}(x_*)}(S)|^{3/2} dX dS.$$

Multiplication by r^{-2} therefore gives invariance of C and D . The remaining two quantities have the same critical dimensional balance:

$$\begin{aligned} \int_{B_r} |u^\lambda(x, t)|^2 dx &= \lambda^{-1} \int_{B_{\lambda r}(x_*)} |u(X, S)|^2 dX, \\ \iint_{Q_r} |\nabla u^\lambda|^2 dx dt &= \lambda^{-1} \iint_{Q_{\lambda r}(z_*)} |\nabla u|^2 dX dS. \end{aligned}$$

Multiplication by r^{-1} on the rescaled cylinder is the same as multiplication by $(\lambda r)^{-1}$ on the original cylinder. Hence A and E are invariant as well. $\square$

Lemma 10.9 (Pressure gauges do not change the equations). *Let (v, q) be suitable in a cylinder Q , and let $\alpha(t) \in L_{\text{loc}}^{3/2}$ depend only on time. Then $(v, q - \alpha(t))$ satisfies the same distributional Navier–Stokes equation, the same divergence condition, and the same local energy inequality.*

Proof. In the distributional equation, replacing q by $q - \alpha(t)$ adds $-\nabla \alpha(t) = 0$. In the local energy inequality the only new term is

$$-2 \int \alpha(t) v(x, t) \cdot \nabla \phi(x, t) dx dt.$$

The product is integrable on the support of ϕ : by suitability, $v \in L_t^\infty L_x^2$ locally, and

$$\left| \int v(\cdot, t) \cdot \nabla \phi(\cdot, t) dx \right| \leq \|v(t)\|_{L^2(\text{supp } \phi(t))} \|\nabla \phi(t)\|_{L^2}.$$

For almost every t , $\text{div } v(\cdot, t) = 0$ in distributions and $\phi(\cdot, t)$ is compactly supported, so

$$\int v(x, t) \cdot \nabla \phi(x, t) dx = 0.$$

Thus the additional term vanishes. $\square$

11. THE TYPE I BLOW-UP SEQUENCE AND COMPACTNESS

11.1. The rescaled sequence. Choose $M < \infty$ so that, after decreasing $T_0 < T$ if necessary,

$$(11.1) \quad \|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T-t}}, \quad T_0 < t < T.$$

For $\lambda_k \downarrow 0$, define

$$(11.2) \quad u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \quad p_k(x, t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then u_k is defined on $\mathbb{R}^3 \times (-T/\lambda_k^2, 0)$, and these domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$.

11.2. Inherited Type I bounds.

Lemma 11.1 (Inherited Type I bound). *For every compact interval $I \Subset (-\infty, 0)$,*

$$\sup_k \|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} < \infty.$$

More precisely, for every fixed $t < 0$ and all sufficiently large k ,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}.$$

Proof. Since $T - (T + \lambda_k^2 t) = \lambda_k^2(-t)$, (11.1) gives, for all large k ,

$$\|u_k(t)\|_\infty = \lambda_k \|u(T + \lambda_k^2 t)\|_\infty \leq \lambda_k \frac{M}{\sqrt{\lambda_k^2(-t)}} = \frac{M}{\sqrt{-t}}.$$

Taking the supremum over $t \in I$ gives the interval statement. $\square$

11.3. Local energy, pressure, and time-derivative estimates.

Lemma 11.2 (Pressure gauges and local pressure bounds). *After fixing the whole-space Leray pressure representative below, set*

$$a_k(t) = (p_k)_{B_1}(t)$$

whenever the rescaled solution is defined. Then for every

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$$

one has

$$\sup_k \|p_k - a_k(t)\|_{L^{3/2}(Q)} < \infty.$$

The estimates below use the global pointwise Type I bound encoded in the admissible Type I assumption Definition 10.4 to control L^2 -mass on every annulus uniformly in k ; the corresponding local-Type-I version replaces the outer annular control by the Leray finite-energy tail estimate already recorded in Proposition 8.2.

Proof. For large k , the original times $T + \lambda_k^2 t$, $t \in [-S, -\sigma]$, lie in the interval where (11.1) holds. At the point where the whole-space Riesz pressure representation is used, the ambient solution is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$ in the sense of Definition 10.3; the pressure representative is therefore chosen, modulo functions of time, as the whole-space Leray pressure

$$p = \mathcal{R}_i \mathcal{R}_j (u_i u_j).$$

On those slices $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence

$$u_i(t) u_j(t) \in L^1(\mathbb{R}^3) \cap L^{3/2}(\mathbb{R}^3).$$

Use the whole-space Leray pressure representative

$$p^L = \mathcal{R}_i \mathcal{R}_j (u_i u_j),$$

modulo functions of time. The standard convention is

$$\widehat{\mathcal{R}_i f}(\xi) = i \xi_i |\xi|^{-1} \widehat{f}(\xi),$$

so that $\mathcal{R}_i \mathcal{R}_j$ has multiplier $-\xi_i \xi_j |\xi|^{-2}$ and $-\Delta(\mathcal{R}_i \mathcal{R}_j f) = \partial_i \partial_j f$. Thus this is the pressure reconstruction from $-\Delta p = \partial_i \partial_j (u_i u_j)$ and the Calderon–Zygmund theory of Riesz transforms [11, 18]. Since $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, interpolation gives $u(t) \in L^3(\mathbb{R}^3)$, hence $u_i(t) u_j(t) \in L^{3/2}(\mathbb{R}^3)$ and $p^L(t) \in L^{3/2}(\mathbb{R}^3)$ for a.e. t .

The given pressure and p^L differ by a function of time. Indeed, applying the Helmholtz projection to the whole-space weak momentum equation gives

$$\partial_t u - \Delta u + \mathbb{P} \operatorname{div}(u \otimes u) = 0$$

in solenoidal distributions. The complementary gradient part of $\operatorname{div}(u \otimes u)$ is precisely $-\nabla p^L$ with the convention above. Therefore the same velocity field solves the distributional equation with p^L in place of p . Subtracting the two formulations gives

$$\nabla(p - p^L) = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3)$$

for a.e. time. By the de Rham theorem on $\mathbb{R}^3$, $p - p^L$ is spatially constant for a.e. time. Replacing the pressure by p^L is therefore a time-dependent gauge change and does not affect suitability by [Lemma 10.9](#).

Fix $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R = 1$ on B_{4R} . For $t \in [-S, -\sigma]$, choose the representative so that, in B_{2R} ,

$$p_k = q_k + h_k, \quad q_k = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{k,i} u_{k,j}),$$

and $h_k(\cdot, t)$ is harmonic in B_{4R} for a.e. t . This choice is made before $a_k(t) = (p_k)_{B_1}(t)$ is evaluated. The harmonicity holds because $(1 - \chi_R)u_{k,i}u_{k,j}$ is supported outside B_{4R} , so the localized Riesz potential generated by this exterior part has zero Laplacian in B_{4R} . Calderon–Zygmund boundedness gives, for a.e. t ,

$$\|q_k(t)\|_{L^{3/2}(B_{2R})} \leq C \|u_k(t)\|_{L^3(B_{8R})}^2 \leq C(R, \sigma, M).$$

Integrating over $[-S, -\sigma]$ gives a uniform $L_{x,t}^{3/2}$ -bound for q_k . Let $c_{k,R}(t) = h_k(0, t)$, where $h_k(\cdot, t)$ is represented by its smooth harmonic representative. For $x \in B_R$, the kernel estimate

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq CR|z|^{-4}, \quad |z| > 4R,$$

implies

$$|h_k(x, t) - c_{k,R}(t)| \leq CR \sum_{m=2}^{\infty} (2^m R)^{-4} \int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 dz.$$

Using [Lemma 11.1](#),

$$\int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 dz \leq CM^2 \sigma^{-1} (2^m R)^3,$$

and therefore

$$|h_k(x, t) - c_{k,R}(t)| \leq CM^2 \sigma^{-1} \sum_{m=2}^{\infty} 2^{-m}, \quad x \in B_R,$$

which implies

$$\|h_k(t) - c_{k,R}(t)\|_{L^\infty(B_R)} \leq C(M, \sigma).$$

Thus

$$\sup_k \|p_k - c_{k,R}(t)\|_{L^{3/2}(B_R \times [-S, -\sigma])} < \infty.$$

If $R < 1$, enlarge R to 1. Otherwise $B_1 \subset B_R$. Jensen's inequality gives, for a.e. t ,

$$|a_k(t) - c_{k,R}(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_{k,R}(t)|^{3/2} dx.$$

Integrating in time and adding constants on B_R gives the desired bound for $p_k - a_k(t)$. $\square$

Lemma 11.3 (Compatibility of pressure gauges). *Let $B_1 \subset B_R$, $I \Subset (-\infty, 0)$, and suppose*

$$p_k - c_k(t) \text{ is bounded in } L^{3/2}(B_R \times I)$$

for some time-dependent gauges $c_k(t)$. If $a_k(t) = (p_k)_{B_1}(t)$, then $p_k - a_k(t)$ is bounded in $L^{3/2}(B_{R'} \times I)$ for every $R' \leq R$. Changing between such gauges changes the pressure only by a function of time and therefore does not affect the equation, the local energy inequality, or the centered equation.

Proof. For a.e. t , Jensen's inequality gives

$$|a_k(t) - c_k(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_k(t)|^{3/2} dx.$$

Integrating in t and using $B_{R'} \subset B_R$ yields

$$\|a_k - c_k\|_{L^{3/2}(I)}^{3/2} \leq |B_1|^{-1} \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}^{3/2}.$$

Thus

$$\|p_k - a_k(t)\|_{L^{3/2}(B_{R'} \times I)} \leq \|p_k - c_k(t)\|_{L^{3/2}(B_{R'} \times I)} + |B_{R'}|^{2/3} \|a_k - c_k\|_{L^{3/2}(I)}$$

and the right-hand side is bounded by

$$C(R') \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}.$$

This proves the asserted bound. The last assertion is [Lemma 10.9](#), and centering transforms the time-dependent gauge into another time-dependent gauge. $\square$

Lemma 11.4 (Pressure gauge atlas). *Let (u_n, p_n) be a rescaled sequence of suitable weak solutions satisfying the local compactness bounds on every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$. For every ball B_R and compact time interval $I \Subset (-\infty, 0)$ there are time-dependent functions $a_{n,R}(t)$ such that*

$$p_n - a_{n,R}(t) \text{ is bounded in } L^{3/2}(B_R \times I).$$

On nested balls, any two such gauges for the same n differ by functions of time only. After passing to a diagonal subsequence over R and I , the gauge-fixed pressures converge weakly in $L_{\text{loc}}^{3/2}$ to a pressure Q defined modulo functions of time. These gauge changes do not affect the distributional Navier-Stokes equations or the local energy inequality, and the same compatibility passes to pressure limits.

Proof. The local pressure bounds are those proved in [Lemma 11.2](#) for the global Type I extraction and in [Proposition 8.2](#) for the local Type I rescaling construction. On each fixed $B_R \times I$, choose any of the gauges provided there, for instance a spatial average on a unit ball contained in B_R after enlarging R if necessary. Compatibility on nested balls is [Lemma 11.3](#): subtracting two admissible representatives leaves the same pressure gradient, hence the difference is spatially constant for a.e. time. A countable exhaustion by balls and compact time intervals, together with Banach-Alaoglu in $L^{3/2}$, gives a diagonal subsequence and a compatible local pressure limit. Invariance of the equation and local energy inequality under time-only pressure changes is [Lemma 10.9](#). $\square$

Remark 11.5 (Pressure-gauge atlas). In the sequel, a pressure limit Q means a compatible family of local representatives modulo time-dependent gauges. On overlaps, the gradients agree; hence the representatives differ only by functions of time. This is the pressure atlas of [Lemma 11.4](#) for local suitable limits. When local compactness is run on a fixed compact cylinder K , we use the gauge $a_{k,K}(t) = (p_k)_{B_1}(t)$ supplied by [Lemma 11.2](#), and [Lemma 11.3](#) identifies a single pressure limit on overlaps modulo functions of time.

Lemma 11.6 (Local energy and dissipation). *Let*

$$Q = B_R \times [-S, -\sigma] \in \mathbb{R}^3 \times (-\infty, 0).$$

Then

$$\sup_k \|u_k\|_{L_t^\infty L_x^2(Q)} + \sup_k \|\nabla u_k\|_{L^2(Q)} < \infty.$$

Proof. The $L_t^\infty L_x^2$ estimate follows from

$$\|u_k(t)\|_{L^2(B_R)} \leq |B_R|^{1/2} M \sigma^{-1/2}, \quad t \in [-S, -\sigma].$$

Choose $\phi \in C_c^\infty(B_{2R} \times [-2S, -\sigma/2])$, $0 \leq \phi \leq 1$, with $\phi = 1$ on Q . Apply the local energy inequality to $(u_k, p_k - a_k(t))$, using Lemma 11.2 on the larger cylinder. For a.e. admissible times and then by monotone approximation of temporal cutoffs,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 |\partial_t \phi + \Delta \phi| + \iint (|u_k|^2 + 2|p_k - a_k(t)|) |u_k| |\nabla \phi|.$$

The first term is bounded because $\partial_t \phi + \Delta \phi$ is bounded and u_k is uniformly bounded in L^∞ on the finite-measure support of ϕ . The cubic velocity term satisfies

$$\iint |u_k|^3 |\nabla \phi| \leq \|\nabla \phi\|_\infty |\text{supp } \nabla \phi| \|u_k\|_{L^\infty(\text{supp } \nabla \phi)}^3 \leq C(Q, \phi, M, \sigma).$$

For the pressure term, Hölder's inequality on $\text{supp } \nabla \phi$ gives

$$\iint |p_k - a_k(t)| |u_k| |\nabla \phi| \leq \|p_k - a_k(t)\|_{L^{3/2}} \|u_k \nabla \phi\|_{L^3},$$

and both factors are uniformly bounded by Lemma 11.2 and Lemma 11.1. Since $\phi = 1$ on Q , the dissipation bound follows. $\square$

Lemma 11.7 (Time derivative bound). *For every compact cylinder $Q \in \mathbb{R}^3 \times (-\infty, 0)$,*

$$\partial_t u_k \text{ is bounded in } L_t^{3/2} W_x^{-1,3/2}(Q).$$

Proof. The equation is

$$\partial_t u_k = \Delta u_k - \text{div}(u_k \otimes u_k) - \nabla(p_k - a_k(t)).$$

Let $Q = B_R \times [-S, -\sigma]$. Throughout the proof we use the embedding $W_0^{1,3}(B_R) \hookrightarrow H_0^1(B_R)$ on the bounded domain B_R ; duality then gives $H^{-1}(B_R) \hookrightarrow W^{-1,3/2}(B_R)$, so that any $L_t^2 H_x^{-1}(Q)$ -bound becomes an $L_t^{3/2} W_x^{-1,3/2}(Q)$ -bound by Hölder in time on the finite interval $[-S, -\sigma]$. The term Δu_k is bounded in $L_t^2 H_x^{-1}(Q)$ by Lemma 11.6; since $W_0^{1,3}(B_R) \subset H_0^1(B_R)$ and the time interval is finite, this is also an $L_t^{3/2} W_x^{-1,3/2}$ -bound. Explicitly, for $\zeta \in W_0^{1,3}(B_R)$,

$$|\langle \Delta u_k, \zeta \rangle| = \left| \int_{B_R} \nabla u_k : \nabla \zeta \, dx \right| \leq C(R) \|\nabla u_k\|_{L^2(B_R)} \|\zeta\|_{W^{1,3}(B_R)}.$$

The L_t^2 -bound on ∇u_k , together with the finite length of $[-S, -\sigma]$, gives the claimed $L_t^{3/2}$ -bound.

For the nonlinear term, for $\zeta \in W_0^{1,3}(B_R)$,

$$\left| \int_{B_R} (u_k \otimes u_k) : \nabla \zeta \, dx \right| \leq \|u_k \otimes u_k\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

Moreover

$$\|u_k \otimes u_k\|_{L^{3/2}(Q)} \leq \|u_k\|_{L^\infty(Q)} \|u_k\|_{L^{3/2}(Q)} \leq C(Q).$$

Thus $\operatorname{div}(u_k \otimes u_k)$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. For the pressure term,

$$\left| \int_{B_R} (p_k - a_k(t)) \operatorname{div} \zeta \, dx \right| \leq \|p_k - a_k(t)\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

[Lemma 11.2](#) gives

$$p_k - a_k(t) \text{ bounded in } L^{3/2}(Q).$$

Hence $\nabla(p_k - a_k(t))$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. Summing the three terms gives the result. $\square$

11.4. Compactness away from the terminal time.

Proposition 11.8 (Compactness). *After passing to a subsequence,*

$$u_k \rightarrow U$$

strongly in $L_{\text{loc}}^q(\mathbb{R}^3 \times (-\infty, 0))$ for every finite q , and

$$p_k - a_k(t) \rightharpoonup Q \text{ weakly in } L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (-\infty, 0)).$$

Proof. Fix $Q_0 \in \mathbb{R}^3 \times (-\infty, 0)$. By [Lemma 11.6](#), u_k is bounded in $L_t^2 H_x^1(Q_0)$, and by [Lemma 11.7](#), $\partial_t u_k$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q_0)$. On bounded balls,

$$H^1 \Subset L^2 \hookrightarrow W^{-1,3/2}.$$

The Aubin–Lions–Simon theorem [[12](#), [15](#), [17](#)] gives relative compactness in $L^2(Q_0)$. Hence, after passing to a subsequence, $u_k \rightarrow U$ strongly in $L^2(Q_0)$ and a.e. on Q_0 .

The inherited Type I bound gives a common $L^\infty(Q_0)$ bound, and the limit belongs to the same L^∞ ball by a.e. convergence. For $q \geq 2$,

$$\|u_k - U\|_{L^q(Q_0)}^q \leq (2 \sup_j \|u_j\|_{L^\infty(Q_0)} + 2\|U\|_{L^\infty(Q_0)})^{q-2} \|u_k - U\|_{L^2(Q_0)}^2.$$

For $1 \leq q < 2$, Hölder's inequality on the finite-measure set Q_0 gives

$$\|u_k - U\|_{L^q(Q_0)} \leq |Q_0|^{1/q-1/2} \|u_k - U\|_{L^2(Q_0)} \rightarrow 0.$$

Choose an exhaustion $Q^{(m)} \Subset \mathbb{R}^3 \times (-\infty, 0)$ by compact cylinders. The preceding argument gives a subsequence converging on $Q^{(1)}$; extracting successively on $Q^{(m)}$ and taking the diagonal subsequence gives strong local convergence on every compact cylinder.

For pressure, [Lemma 11.2](#) gives boundedness of $p_k - a_k(t)$ in $L^{3/2}(Q^{(m)})$ for every m . Since $L^{3/2}$ is reflexive, each restriction has a weakly convergent subsequence. The same diagonal extraction gives a single limit Q in $L_{\text{loc}}^{3/2}$, because the weak limits agree on overlaps by uniqueness of distributional limits. $\square$

11.5. The ancient suitable weak limit.

Lemma 11.9 (Limit equation). *The limit (U, Q) solves*

$$\partial_t U + (U \cdot \nabla) U + \nabla Q = \Delta U, \quad \operatorname{div} U = 0$$

in distributions on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. For $\varphi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0); \mathbb{R}^3)$, the weak formulation for $(u_k, p_k - a_k(t))$ is

$$\iint [-u_k \cdot \partial_t \varphi - (u_k \otimes u_k) : \nabla \varphi + (p_k - a_k(t)) \operatorname{div} \varphi + \nabla u_k : \nabla \varphi] \, dx \, dt = 0.$$

The linear time term passes by strong L^2_{loc} -convergence of u_k . The pressure term passes by weak $L^{3/2}_{\text{loc}}$ -convergence of $p_k - a_k(t)$, since $\operatorname{div} \varphi \in L^3$. The viscous term passes by weak L^2_{loc} -convergence of ∇u_k , which follows from the uniform local dissipation bound. The nonlinear term passes because

$$\|u_k \otimes u_k - U \otimes U\|_{L^1(\operatorname{supp} \varphi)} \leq \|u_k - U\|_{L^2} (\|u_k\|_{L^2} + \|U\|_{L^2}) \rightarrow 0.$$

Testing with gradients of scalar functions gives $\operatorname{div} U = 0$. $\square$

Lemma 11.10 (Stability of suitability). *The pair (U, Q) is a suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$.*

Proof. Use the distributional form of the local energy inequality. For every non-negative $\phi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0))$,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 (\partial_t \phi + \Delta \phi) + \iint (|u_k|^2 + 2P_k) u_k \cdot \nabla \phi, \quad P_k = p_k - a_k(t).$$

This follows from (10.1) by applying (10.1) with temporal cutoffs which increase to 1 on the time projection of $\operatorname{supp} \phi$. The monotone convergence theorem passes the nonnegative dissipation term to the displayed distributional form.

The terms with $|u_k|^2$ converge strongly in L^1 because

$$\||u_k|^2 - |U|^2\|_{L^1(\operatorname{supp} \phi)} \leq \|u_k - U\|_{L^2} (\|u_k\|_{L^2} + \|U\|_{L^2}) \rightarrow 0.$$

The cubic flux converges strongly since $u_k \rightarrow U$ strongly in L^3_{loc} , hence

$$|u_k|^2 u_k \rightarrow |U|^2 U \quad \text{in } L^1_{\text{loc}}.$$

For the pressure flux, $u_k \cdot \nabla \phi \rightarrow U \cdot \nabla \phi$ strongly in L^3 , while $P_k \rightarrow Q$ weakly in $L^{3/2}$. Therefore

$$\iint P_k u_k \cdot \nabla \phi \rightarrow \iint Q U \cdot \nabla \phi.$$

Finally, $\nabla u_k \rightharpoonup \nabla U$ in L^2_{loc} , and

$$w \mapsto \iint |w|^2 \phi$$

is weakly lower semicontinuous because $\phi \geq 0$ and the expression equals $\|\sqrt{\phi} w\|_{L^2}^2$. Passing to the limit gives the local energy inequality for (U, Q) in distributional form. The endpoint form in Definition 10.2 is recovered by applying the distributional inequality to standard time cutoffs approximating characteristic functions of (t_1, t_2) ; weak L^2_{loc} -continuity of local energy-class solutions gives the endpoint traces for a.e. $t_1 < t_2$. $\square$

Lemma 11.11 (Inherited ancient Type I bound). *For every compact interval $I = [-S, -\sigma] \Subset (-\infty, 0)$,*

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M \sigma^{-1/2}.$$

Proof. Lemma 11.1 gives

$$\|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M \sigma^{-1/2}.$$

Fix $R < \infty$. Since $u_k \rightarrow U$ a.e. on $B_R \times I$, the essential supremum of U on $B_R \times I$ cannot exceed $M \sigma^{-1/2}$. If it did, $|u_k|$ would exceed this bound by a positive amount on a positive measure subset for large k . Letting $R \rightarrow \infty$ through integers gives the result on $\mathbb{R}^3 \times I$. $\square$

Proposition 11.12 (Ancient suitable Type I limit). *The pair (U, Q) is an ancient suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$ satisfying the interval Type I bound in Lemma 11.11.*

Proof. Combine Lemmas 11.9 to 11.11. $\square$

12. PERSISTENCE OF CONCENTRATION AND NONTRIVIALITY

12.1. Persistence of concentration away from the terminal time. The persistence of concentration is tracked through the velocity concentration sequence in [Definition 10.4](#), supplied by [Proposition 10.7](#). The no-escape condition says that this velocity mass does not concentrate entirely in the terminal layer $t = 0$ after rescaling.

12.2. Scale selection.

Lemma 12.1 (Scale selection from local concentration). *Let $\lambda_k \downarrow 0$ be an admissible concentration sequence in the sense of [Definition 10.4](#). Then there are $\sigma \in (0, 1)$, a compact cylinder*

$$K_0 = B_1 \times (-1, -\sigma) \Subset \mathbb{R}^3 \times (-\infty, 0),$$

and $\eta_1 > 0$ such that

$$(12.1) \quad \iint_{K_0} |u_k|^3 dx dt \geq \eta_1$$

for all k .

Proof. By scale invariance of C ,

$$\int_{-1}^0 \int_{B_1} |u_k|^3 dx dt = C(u; z_*, \lambda_k) \geq \eta_v.$$

The no-escape condition [\(10.5\)](#) gives $\sigma \in (0, 1)$ such that

$$\sup_k \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt \leq \frac{\eta_v}{2}.$$

Therefore

$$\int_{-1}^{-\sigma} \int_{B_1} |u_k|^3 dx dt \geq \frac{\eta_v}{2}$$

for every k . The assertion follows with $\eta_1 = \eta_v/2$. $\square$

12.3. Nonzero ancient limits.

Proposition 12.2 (Nonzero ancient limit). *For the scales chosen in [Lemma 12.1](#), the limit U in [Proposition 11.12](#) is nonzero. More precisely, there are a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$ and $\eta_1 > 0$ such that*

$$\iint_{K_0} |U|^3 dx dt \geq \eta_1.$$

Proof. Let $K_0 = B_1 \times (-1, -\sigma)$ and $\eta_1 > 0$ be given by [Lemma 12.1](#). Since $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, the compactness theorem gives, after passing to the subsequence used to define U ,

$$u_k \rightarrow U \quad \text{strongly in } L^3(K_0).$$

Hence

$$\int_{K_0} |U|^3 dx dt = \lim_{k \rightarrow \infty} \int_{K_0} |u_k|^3 dx dt \geq \eta_1.$$

$\square$

Thus concentration generated by the singular point propagates through the compactness limit. The velocity concentration and no-escape condition are the hypotheses that prevent the ancient limit from being identically zero.

13. CENTERED VARIABLES AND NORMALIZED SEREGIN LIMITS

13.1. **The centered self-similar equation.** Set

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t),$$

and

$$\Pi(y, \tau) = (-t)Q(y\sqrt{-t}, t),$$

up to addition of a function of τ .

Remark 13.1 (Pressure gauge transformation in centered variables). If Q is replaced by $Q + a(t)$ for a function a of t only, then Π is replaced by $\Pi + e^{-\tau}a(-e^{-\tau})$, which is a function of τ only. Thus the centered equation (9.3) is gauge-invariant: changing the pressure by a function of physical time amounts to changing the centered pressure by a function of centered time.

Lemma 13.2 (Centered pullback identities). *The centered and physical variables are related by*

$$t = -e^{-\tau}, \quad x = e^{-\tau/2}y, \quad U(x, t) = e^{\tau/2}V(y, \tau),$$

or equivalently

$$V(y, \tau) = e^{-\tau/2}U(e^{-\tau/2}y, -e^{-\tau}).$$

For every $\tau \in \mathbb{R}$, with $t = -e^{-\tau}$,

$$\int_{\mathbb{R}^3} |U(x, t)|^3 dx = \int_{\mathbb{R}^3} |V(y, \tau)|^3 dy.$$

Proof. The first identities are the definitions with $-t = e^{-\tau}$. Since

$$|U(x, t)|^3 = e^{3\tau/2}|V(y, \tau)|^3, \quad dx = e^{-3\tau/2} dy,$$

the critical L^3 identity follows by multiplication and integration. $\square$

Lemma 13.3 (Renormalized equation). *The pair (V, Π) solves (9.3).*

Proof. Write $s = -t = e^{-\tau}$, $x = s^{1/2}y$, and

$$U(x, t) = s^{-1/2}V(y, \tau).$$

The identities are first obtained in distributions by testing the equation for U against the inverse change of variables. After Lemma 13.4, the same identities hold classically for the smooth representative. Because $s_t = -1$, $\tau_t = s^{-1}$, and $y_t = (2s)^{-1}y$,

$$\partial_t U = s^{-3/2} \left(\partial_\tau V + \frac{1}{2}V + \frac{1}{2}y \cdot \nabla V \right),$$

$$(U \cdot \nabla_x)U = s^{-3/2}(V \cdot \nabla)V, \quad \Delta_x U = s^{-3/2}\Delta V, \quad \nabla_x Q = s^{-3/2}\nabla \Pi.$$

Substitution into the equation for U gives (9.3), and the divergence condition is preserved by the spatial scaling. $\square$

13.2. Boundedness, smoothness, and local energy.

Lemma 13.4 (Smoothness). *The ancient solution U is smooth on $\mathbb{R}^3 \times (-\infty, 0)$, and (V, Π) is smooth on $\mathbb{R}^3 \times \mathbb{R}$ after fixing a pressure gauge.*

Proof. On each compact cylinder $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, Lemma 11.11 gives $U \in L^\infty(Q_0)$. Hence $U \in L_t^s L_x^r(Q_0)$ for every finite r, s . Choose $r = s = 10$, so

$$\frac{2}{s} + \frac{3}{r} = \frac{1}{2} < 1.$$

Serrin's interior regularity criterion [10, 18] applies and gives local Hölder regularity of U on a slightly smaller cylinder. Since Q_0 was arbitrary, this removes every possible interior singular point in $\mathbb{R}^3 \times (-\infty, 0)$. The bootstrap starts from

$$\partial_t U - \Delta U + \nabla Q = -\operatorname{div}(U \otimes U), \quad \operatorname{div} U = 0,$$

with $U \otimes U$ locally bounded after Serrin regularity. Interior Stokes estimates on nested cylinders give $W_q^{2,1}$ -regularity for every finite q . Parabolic Sobolev embedding then improves the Hölder regularity of ∇U , and differentiating the equation and iterating gives

$$U \in C^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

The pressure is recovered locally from $-\Delta Q = \partial_i \partial_j (U_i U_j)$ plus a harmonic function and is smooth after fixing a time-dependent gauge. The centered change of variables is a smooth diffeomorphism for $t < 0$, so V and Π are smooth. $\square$

Corollary 13.5 (Classical Type I bound). *For the smooth representative from Lemma 13.4,*

$$\|U(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}, \quad t < 0.$$

Proof. Fix $t_0 < 0$. Choose $\delta_n \downarrow 0$ with

$$I_n = [t_0 - \delta_n, t_0 + \delta_n] \Subset (-\infty, 0).$$

Lemma 11.11 gives

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I_n)} \leq M(-t_0 - \delta_n)^{-1/2}.$$

Since the representative is continuous, the essential spacetime bound holds pointwise. Letting $n \rightarrow \infty$ gives the estimate at t_0 . $\square$

Lemma 13.6 (Renormalized boundedness and nontriviality). *The centered field satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M.$$

Moreover (9.4) holds for some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$.

Proof. The L^∞ bound follows from Corollary 13.5:

$$|V(y, \tau)| = \sqrt{-t} |U(y\sqrt{-t}, t)| \leq M.$$

For nontriviality, Proposition 12.2 gives a positive local L^3 lower bound on a compact cylinder in (x, t) . Under $(x, t) = (e^{-\tau/2}y, -e^{-\tau})$,

$$dx dt = e^{-5\tau/2} dy d\tau, \quad U = e^{\tau/2} V,$$

and hence

$$|U|^3 dx dt = e^{-\tau} |V|^3 dy d\tau.$$

On the compact image cylinder the factor $e^{-\tau}$ is bounded above and below by positive constants, so the positive lower bound becomes (9.4), with a possibly different η_0 . $\square$

Lemma 13.7 (Local energy bounds in centered variables). *Let V be obtained from an admissible finite-energy Type I singularity. For every $R < \infty$ and compact interval $I \subset \mathbb{R}$,*

$$\int_I \int_{B_R} |V|^2 dy d\tau + \int_I \int_{B_R} |\nabla V|^2 dy d\tau < \infty.$$

After subtracting a function of τ , the associated pressure satisfies

$$\Pi - b(\tau) \in L^{3/2}(B_R \times I).$$

Proof. The unrenormalized limit (U, Q) is suitable and has local energy, dissipation, and pressure bounds on compact negative-time cylinders by [Lemmas 11.2](#) and [11.6](#) and [Proposition 11.12](#); the bounds for the limit follow from the weak lower semicontinuity and weak compactness already used in [Lemma 11.10](#). On each compact (y, τ) -cylinder, the map

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2}y, -e^{-\tau})$$

has Jacobian $dx dt = e^{-5\tau/2} dy d\tau$, with

$$V = e^{-\tau/2}U, \quad \nabla_y V = e^{-\tau} \nabla_x U, \quad \Pi = e^{-\tau}Q.$$

If $I = [\tau_1, \tau_2]$, then $e^{-\tau}$ is bounded above and below by positive constants on I . Therefore

$$\int_I \int_{B_R} |V|^2 dy d\tau \leq C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |U|^2 dx dt,$$

and similarly

$$\int_I \int_{B_R} |\nabla_y V|^2 dy d\tau \leq C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |\nabla_x U|^2 dx dt.$$

For the pressure, subtract a spatial-average gauge $a(t)$ for Q on the corresponding x -ball and set $b(\tau) = e^{-\tau}a(-e^{-\tau})$. Then $\Pi - b = e^{-\tau}(Q - a)$, and the same bounded-weight change of variables gives $\Pi - b \in L^{3/2}(B_R \times I)$. $\square$

Remark 13.8. [Lemma 13.7](#) is local. It does not give uniform L^3 -tightness in y , nor a global $L_\tau^\infty L_y^3$ bound. This is the reason for retaining the remaining case.

13.3. Normalized Seregin ancient limits.

Definition 13.9 (Admissible Type I extraction). An admissible Type I extraction is one of the following two blow-up procedures:

- (i) the finite-energy Type I extraction in [Definition 10.4](#) and [Theorem 9.1](#);
- (ii) the local pointwise Type I extraction in [Definition 8.1](#) and [Lemma 8.7](#).

In both cases the conclusion is a smooth bounded solution of the centered equation [\(9.3\)](#) with compact local L^3 nonvanishing.

Definition 13.10 (Raw extracted normalized Seregin ancient limit). A smooth bounded solution (V, Π) of [\(9.3\)](#) is a *raw extracted normalized Seregin ancient limit* at z_* if it is produced by an admissible finite-energy Type I extraction in the sense of [Definition 13.9](#), or from an admissible local pointwise Type I concentration sequence at z_* in the sense of [Definition 8.1](#) and [Lemma 8.7](#), and satisfies the compact spacetime L^3 nonvanishing normalization [\(9.4\)](#).

Corollary 13.11 (Raw extracted limits are nonzero). *Every raw extracted normalized Seregin ancient limit satisfies the compact spacetime nonvanishing condition [\(9.4\)](#). In particular, it is not identically zero.*

Proof. This is part of the definition of a raw extracted normalized Seregin ancient limit. For the finite-energy extraction the condition is produced by [Lemmas 12.1 and 13.6](#) and [Proposition 12.2](#); for the local pointwise Type I concentration sequence it is produced by [Lemmas 8.6 and 8.7](#). $\square$

Lemma 13.12 (Time-translation invariant nonvanishing). *Let (V, Π) be a raw extracted normalized Seregin ancient limit satisfying (9.4) on the compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ with constant $\eta_{\text{raw}} = \eta_0 > 0$. Then there exist $R_0 > 0$ and $\eta_* > 0$, depending only on K and η_{raw} , such that*

$$(13.1) \quad \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_*.$$

Proof. Choose $R_0 > 0$ and an interval $[a, b] \subset \mathbb{R}$ with $K \subset B_{R_0} \times [a, b]$. Then

$$\eta_{\text{raw}} \leq \iint_K |V|^3 dy d\tau \leq \int_a^b \int_{B_{R_0}} |V(y, \tau)|^3 dy d\tau \leq (b - a) \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy.$$

Take $\eta_* = \eta_{\text{raw}} / (b - a)$. $\square$

Definition 13.13 (Raw generated Seregin state space). For a fixed triple (u, p, z_*) , fix invariant normalization data (R_0, η_*) obtained from a raw extracted profile by [Lemma 13.12](#). Let

$$\mathcal{S}_{\text{raw}}(u, p, z_*; R_0, \eta_*)$$

be the collection of raw extracted normalized Seregin ancient limits produced by admissible finite-energy Type I or admissible local pointwise Type I extractions at z_* which satisfy the invariant nonvanishing condition

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_*.$$

If different raw extractions yield different invariant normalization pairs, the construction below is carried out separately for each pair.

The *raw generated descendant hull*

$$\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

is, by definition, the smallest class containing $\mathcal{S}_{\text{raw}}(u, p, z_*; R_0, \eta_*)$ and closed under logarithmic time translations and locally smooth limits that retain:

- (i) the inherited centered L^∞ bound;
- (ii) smoothness and the centered local-pressure ancient formulation;
- (iii) the invariant local velocity nonvanishing condition (13.1) with the fixed parameters (R_0, η_*) .

The class is closed under locally smooth limits that still satisfy the invariant (R_0, η_*) -nonvanishing condition; no claim is made that this condition is lower semicontinuous under arbitrary uncentered locally smooth limits. An element of $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ is called a *generated raw Seregin state*. When the invariant normalization data are clear from the extracted profile under discussion, we write $\mathcal{S}_{\text{raw}}(u, p, z_*)$ for $\mathcal{S}_{\text{raw}}(u, p, z_*; R_0, \eta_*)$ and $\mathcal{S}_{\text{raw-gen}}(u, p, z_*)$ for $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. For notational economy, $\mathcal{S}(u, p, z_*)$ denotes this raw generated class.

Definition 13.14 (Mild stratum of the generated state space). The mild stratum of the raw generated state space is

$$\mathcal{S}_{\text{mild}}(u, p, z_*; R_0, \eta_*) \subset \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

consisting of those generated states V satisfying the finite-interval projected centered mild formulation, where

$$L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}, \quad S(s) = e^{sL}.$$

For every $\sigma < \tau$,

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_{\sigma}^{\tau} S(\tau - s)\mathbb{P}\nabla \cdot (V \otimes V)(s) ds$$

in the weak-* topology of $L^{\infty}(\mathbb{R}^3)$.

Remark 13.15 (Raw generated descendant hulls). No assertion is made here that an arbitrary locally smooth limit of translates is itself an extracted limit from the original physical sequence without a separate diagonal generation argument. Instead, the raw generated descendant hull is included in the definition of the class to which the residual closure hypothesis is applied. This is the class on which the residual set $\mathcal{R}(\mathcal{S})$ is formed in the final assembly. The projected mild formulation is imposed only on the mild stratum, not on the whole raw hull.

Lemma 13.16 (Raw generated closure preserves normalized ancient structure). *Every element of $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ is a smooth bounded centered ancient local-pressure solution, satisfies the inherited centered L^{∞} bound, and retains the time-translation invariant local L^3 nonvanishing normalization (13.1) with the fixed (R_0, η_*) .*

Proof. Raw extracted elements have these raw properties by [Theorem 9.1](#) and [Lemma 8.7](#) together with [Lemma 13.12](#), which converts the raw compact cylinder mass lower bound into the invariant condition (13.1). Logarithmic time translations preserve the autonomous centered equation, the centered L^{∞} bound, smoothness, and the local-pressure formulation; they also preserve the supremum on the left of (13.1), since the supremum is over all $\tau \in \mathbb{R}$. For a locally smooth limit of such translates, smoothness and the equation pass by local convergence; the L^{∞} bound passes by pointwise convergence. The raw generated hull is closed only under limits retaining the invariant nonvanishing condition with the fixed (R_0, η_*) , so (13.1) remains part of the structure. $\square$

Lemma 13.17 (Mild stratum stability). *The class $\mathcal{S}_{\text{mild}}(u, p, z_*; R_0, \eta_*)$ is invariant under logarithmic time translations. Moreover, locally smooth limits of elements of $\mathcal{S}_{\text{mild}}$ with a common L^{∞} bound remain in $\mathcal{S}_{\text{mild}}$, provided the limiting state belongs to the raw generated hull and retains the fixed invariant nonvanishing normalization.*

Proof. Time translation preserves the finite-interval projected mild identity because the centered equation and the centered Stokes semigroup are autonomous. Consider a locally smooth sequence $V_j \in \mathcal{S}_{\text{mild}}$ with a common L^{∞} bound, converging locally smoothly to V . For fixed $\sigma < \tau$, the linear term $S(\tau - \sigma)V_j(\sigma)$ converges to $S(\tau - \sigma)V(\sigma)$ in distributions and weak-* L^{∞}_{loc} . The Duhamel terms pass to the limit against compactly supported test fields by the Oseen-kernel estimate in [Lemma 15.1](#), the common L^{∞} bound, and local smooth convergence of $V_j \otimes V_j$. Hence the projected mild formula holds for V on $[\sigma, \tau]$. The convergence against compactly supported tests upgrades to weak-* L^{∞} convergence against arbitrary $L^1(\mathbb{R}^3)$ tests by approximating the L^1 test function by compactly supported functions and using the common L^{∞} bound on the sequence and on the limit. The remaining raw generated and nonvanishing requirements are part of the stated limiting hypotheses. $\square$

Remark 13.18. The raw compact spacetime L^3 nonvanishing condition is part of the normalization, not an additional hypothesis. It is inherited from the selected velocity concentration and the no-escape condition through [Lemma 12.1](#) and [Proposition 12.2](#) for the finite-energy extraction, and, for the local pointwise Type I extraction, through [Lemmas 8.6](#) and [8.7](#). The raw generated descendant hull uses the time-translation invariant consequence (13.1); this is the nonzero normalization used below for generated raw states.

Corollary 13.19 (Generated raw Seregin states are nonzero). *Every generated raw Seregin state in $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ satisfies the time-translation invariant local L^3 nonvanishing condition (13.1); in particular it is not identically zero.*

Proof. For raw finite-energy extractions this is [Lemmas 12.1 and 13.6](#) and [Proposition 12.2](#) together with [Lemma 13.12](#). For the raw local pointwise Type I concentration sequence it is [Lemmas 8.6 and 8.7](#) together with [Lemma 13.12](#). Generated closure elements retain (13.1) by [Lemma 13.16](#). $\square$

Proof of Theorem 9.1. Choose an admissible concentration sequence $\lambda_k \downarrow 0$ from [Definition 10.4](#), and form the rescaled sequence (11.2). The compact ancient limit is [Proposition 11.12](#); the strong and weak convergence are [Proposition 11.8](#). The no-escape argument in [Lemma 12.1](#), followed by strong L^3 -compactness away from $t = 0$, gives nontriviality through [Proposition 12.2](#). The pointwise Type I bound is [Corollary 13.5](#), and the centered equation, smoothness, boundedness, and raw compact L^3 nonvanishing are [Lemmas 13.3, 13.4 and 13.6](#). $\square$

14. LIOUVILLE CLASSES AND THE REMAINING CLASS

Fix a raw generated descendant hull

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*, R_0, \eta_*)$$

in the sense of [Definition 13.13](#). Complements below are taken relative to this raw hull. Write

$$\mathcal{S}_{\text{mild}} = \mathcal{S}_{\text{mild}}(u, p, z_*, R_0, \eta_*)$$

for its mild stratum. Structural conditions involving the unrenormalized ancient field refer to

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

Definition 14.1 (Small-amplitude class). Let $\varepsilon_{\text{small}} > 0$ be the small-data Liouville constant for bounded mild ancient solutions of (9.3). We say $V \in \mathcal{C}_{\text{small}}$ if

$$V \in \mathcal{S}_{\text{mild}} \quad \text{and} \quad \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_{\text{small}}.$$

Definition 14.2 (Stationary L^3 class). We say $V \in \mathcal{C}_{\text{stat}, L^3}$ if

$$V(y, \tau) = W(y) \quad \text{and} \quad W \in L^3(\mathbb{R}^3).$$

Definition 14.3 (Uniformly L^3 -tight class). We say $V \in \mathcal{C}_{L^3\text{-tight}}$ if

$$V \in \mathcal{S}_{\text{mild}}$$

and

$$(14.1) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Every $V \in \mathcal{S}$ is bounded. [Theorem 25.8](#) shows that, for mild states, the tail condition (14.1) gives the endpoint sequence- L^3 control needed after physical pullback; no separate $L_\tau^\infty L_y^3$ hypothesis is required for the tight-class exclusion.

Definition 14.4 (Structure and decay class). A generated raw Seregin state $V \in \mathcal{S}_{\text{raw-gen}}$ belongs to $\mathcal{C}_{\text{sd}}$ if $V \in \mathcal{S}_{\text{mild}}$ and its physical ancient representative U satisfies at least one of the following analytic hypotheses:

- (i) U is axisymmetric without swirl;
- (ii) U is axisymmetric and satisfies the pointwise scale-invariant bound

$$|U(x, t)| \leq \frac{C}{r}$$

for some $C < \infty$, where r is the cylindrical radius;

(iii) U is axisymmetric and

$$rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))$$

for some $1 \leq p < \infty$;

(iv) U is axisymmetric, periodic in the axial variable, and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0));$$

(v) the associated Galloway–Wayne forward rescaled vorticity $w = \nabla \times v$, with v recovered from w by the three-dimensional Biot–Savart law, satisfies the perturbative weighted-vorticity condition of [Definition 31.2](#).

Remark 14.5. Only hypotheses for which a cited Liouville theorem gives $U \equiv 0$ are included in $\mathcal{C}_{\text{sd}}$. Broader qualitative descriptions are not included unless they imply one of the listed hypotheses. In the weighted-vorticity alternative the Biot–Savart representative excludes a harmonic constant velocity mode, so zero vorticity gives zero velocity. Membership in $\mathcal{C}_{\text{sd}}$ is checked profile by profile. No closedness of the structural class under all generated limits is used in the final contradiction; whenever a generated limit satisfies one of the listed conditions and lies in the mild stratum, the corresponding Liouville theorem is applied directly to that profile.

Lemma 14.6 (Structural quantities under centering). *The centered change of variables preserves axisymmetry and absence of swirl. If (r, z, t) and (ρ, Z, τ) are related by*

$$\rho = \frac{r}{\sqrt{-t}}, \quad Z = \frac{z}{\sqrt{-t}}, \quad \tau = -\log(-t),$$

then the swirl variables satisfy

$$\rho V_\theta(\rho, Z, \tau) = rU_\theta(r, z, t).$$

Proof. The map $x \mapsto y = x/\sqrt{-t}$ is radial in the cylindrical variables and does not mix the e_r, e_θ, e_z frame. Since $V(y, \tau) = \sqrt{-t}U(x, t)$, the θ -components obey $V_\theta = \sqrt{-t}U_\theta$, and the displayed identities follow. $\square$

Definition 14.7 (Remaining class). The remaining class is

$$\mathcal{R}(\mathcal{S}) = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*) \setminus (\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}}).$$

Remark 14.8. The first four classes are not intended to be disjoint. The remaining class is the complement of their union, so overlaps among the Liouville classes cause no ambiguity. The remaining class is a set-theoretic complement, not a new structural hypothesis: it carries no analytic content beyond “not in any of the four named classes.” The residual closure hypothesis [Hypothesis 16.1](#) is the sole input used to remove this complement in the final contradiction.

Remark 14.9 (Non-mild states are residual unless otherwise classified). The ambient state space is the raw generated Seregin hull. The explicit Liouville classes include exactly the hypotheses needed to invoke their corresponding Liouville theorems. In particular, a generated state for which the projected mild formulation has not been established is not placed in the small, tight, or structure/decay Liouville classes merely because it satisfies the corresponding pointwise or tail condition. Such a state lies in the remaining class unless it belongs to the stationary L^3 branch.

Remark 14.10 (Single-profile versus collection-level tightness). The exclusion of a single generated raw Seregin state in $\mathcal{C}_{L^3\text{-tight}}$ uses only the single-profile tail condition together with mildness and boundedness, via [Theorem 25.8](#). The stronger collection-level tightness modulus appearing in [Definition 22.6](#) is used only in the auxiliary compactness and minimal-element material of Part III; no hidden common modulus is imposed on individual generated raw Seregin states.

Proposition 14.11 (Exhaustion by defined classes). *Let*

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be a raw generated descendant hull. Then every $V \in \mathcal{S}$ belongs either to one of

$$\mathcal{C}_{\text{small}}, \quad \mathcal{C}_{\text{stat}, L^3}, \quad \mathcal{C}_{L^3\text{-tight}}, \quad \mathcal{C}_{\text{sd}},$$

or to

$$\mathcal{R}(\mathcal{S}) = \mathcal{S} \setminus (\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}}).$$

Proof. Let

$$\mathcal{U} = \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}}.$$

For $V \in \mathcal{S}$, either $V \in \mathcal{U}$, in which case it lies in at least one Liouville class, or $V \in \mathcal{S} \setminus \mathcal{U} = \mathcal{R}(\mathcal{S})$. $\square$

15. LIOUVILLE HYPOTHESES AND CLASS EXCLUSIONS

Lemma 15.1 (Centered Stokes kernel estimates). *Let*

$$L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}, \quad S(s) = e^{sL}.$$

There is a universal constant C_0 such that, for all $s > 0$,

$$\|S(s)f\|_{L^\infty} \leq e^{-s/2}\|f\|_{L^\infty}$$

and

$$\|S(s)\mathbb{P}\nabla \cdot F\|_{L^\infty} \leq C_0 e^{-s/2}(1 - e^{-s})^{-1/2}\|F\|_{L^\infty}.$$

Proof. The first estimate follows from the explicit Ornstein–Uhlenbeck kernel:

$$S(s)f(y) = e^{-s/2} \int_{\mathbb{R}^3} G_{1-e^{-s}}(e^{-s/2}y - z)f(z) dz,$$

where G_a is the heat kernel with variance a . The kernel has total mass one. For the projected-divergence estimate, write the corresponding kernel as the Oseen projected-gradient kernel after the same change of variables. More explicitly, the semigroup for $\Delta - \frac{1}{2}y \cdot \nabla$ is

$$(e^{s(\Delta - \frac{1}{2}y \cdot \nabla)}g)(y) = (e^{(1-e^{-s})\Delta}g)(e^{-s/2}y).$$

Applying this to the distribution $g = \mathbb{P}\nabla \cdot F$ and using the projected-gradient Oseen kernel estimate

$$\|e^{\theta\Delta}\mathbb{P}\nabla \cdot F\|_\infty \leq C\theta^{-1/2}\|F\|_\infty, \quad \theta > 0,$$

gives the displayed estimate with $\theta = 1 - e^{-s}$, together with the extra factor $e^{-s/2}$ from the zeroth-order term in L . The projected-gradient Oseen estimate follows from the L^1 -bound

$$\|\nabla O(\cdot, \theta)\|_{L^1} \leq C\theta^{-1/2}$$

for the gradient of the Oseen kernel $O(\cdot, \theta)$. This is the bounded-data Stokes estimate used by Kato [13]; it does not require boundedness of the Leray projector itself on L^∞ . $\square$

Remark 15.2 (Oseen interpretation in L^∞). All occurrences of $S(t)\mathbb{P}\nabla \cdot F$ in L^∞ are understood through the projected heat-divergence, or Oseen, kernel representation. They do not use boundedness of the Leray projector as a Calderon–Zygmund operator on L^∞ .

Lemma 15.3 (Mild representation for generated mild states). *Every $V \in \mathcal{S}_{\text{mild}}(u, p, z_*; R_0, \eta_*)$ satisfies, for all $\Theta > 0$,*

$$V(\tau) = S(\Theta)V(\tau - \Theta) - \int_0^\Theta S(\Theta - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - \Theta + \sigma) d\sigma.$$

Proof. This is the finite-interval projected mild formulation in [Definition 13.14](#), applied with $\sigma = \tau - \Theta$. The centered Stokes estimates in [Lemma 15.1](#) ensure that the Duhamel term is well-defined in weak- * L^∞ . $\square$

Lemma 15.4 (Small bounded ancient Liouville theorem). *There is $\varepsilon_{\text{small}} > 0$ such that every bounded mild ancient solution of (9.3) with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_{\text{small}}$$

is identically zero.

Proof. Write the projected centered equation as

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V), \quad L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}.$$

Let $S(s) = e^{sL}$. By [Lemma 15.1](#), the nonlinear kernel is integrable on $(0, \infty)$; set

$$A = C_0 \int_0^\infty e^{-s/2} (1 - e^{-s})^{-1/2} ds < \infty.$$

For fixed τ and $T > 0$, the mild identity on $[\tau - T, \tau]$ gives

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(T - \sigma) \mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma) d\sigma.$$

This Duhamel identity is valid for every $T > 0$ by [Lemma 15.3](#). If $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$, then

$$\|V(\tau)\|_\infty \leq e^{-T/2} M + AM^2.$$

Letting $T \rightarrow \infty$ and taking the supremum in τ gives

$$M \leq AM^2.$$

If $0 < M < 1/A$, this is impossible. Choose $\varepsilon_{\text{small}} = 1/(2A)$; then $M \leq \varepsilon_{\text{small}}$ implies $M = 0$. $\square$

Lemma 15.5 (Stationary centered profiles). *If $V(y, \tau) = W(y)$ is a smooth solution of (9.3), then after subtracting a function of τ from the pressure, W solves*

$$(W \cdot \nabla)W + \nabla P = \Delta W - \frac{1}{2}(W + y \cdot \nabla W), \quad \operatorname{div} W = 0.$$

Proof. Since V is independent of τ , $\partial_\tau V = 0$. Set

$$F(y) = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W.$$

The centered equation gives $\nabla_y \Pi(\cdot, \tau) = F$ for a.e. τ . Fix one such time τ_0 and put $P(y) = \Pi(y, \tau_0)$. Then $\nabla_y (\Pi(\cdot, \tau) - P) = 0$ for a.e. τ , so $\Pi(y, \tau) - P(y) = b(\tau)$ is spatially constant. Subtracting this gauge from Π leaves the equation unchanged and yields the displayed stationary self-similar equation for W . $\square$

Lemma 15.6 (Spatially constant normalized profiles vanish). *Let U be the physical ancient representative of a generated raw Seregin state V in the mild stratum. If U is spatially constant in the physical Type I frame, then $U \equiv 0$ and hence $V \equiv 0$.*

Proof. Mildness excludes parasitic spatially constant solutions: if $U(x, t) = c(t)$, the mild formula has zero nonlinear term and the heat semigroup fixes constants, so $c(t) = c(s)$ for all $s < t < 0$. Thus $U(x, t) \equiv c$. The centered profile is then

$$V(y, \tau) = e^{-\tau/2} c.$$

Since generated raw Seregin states are bounded for all $\tau \in \mathbb{R}$, letting $\tau \rightarrow -\infty$ forces $c = 0$. $\square$

Theorem 15.7 (Small and stationary class exclusions). *Let V be a generated raw Seregin state. Then*

$$V \notin \mathcal{C}_{\text{small}}, \quad V \notin \mathcal{C}_{\text{stat}, L^3}.$$

Proof. If $V \in \mathcal{C}_{\text{small}}$, then by definition $V \in \mathcal{S}_{\text{mild}}$, and Lemmas 15.3 and 15.4 give $V \equiv 0$, contradicting the invariant nonvanishing normalization of Corollary 13.19.

If $V \in \mathcal{C}_{\text{stat}, L^3}$, then $V(y, \tau) = W(y)$ with $W \in L^3(\mathbb{R}^3)$. By Lemma 15.5, W satisfies the stationary self-similar profile equation after a pressure gauge change. The stationary Leray theorem Theorem 21.4 gives $W \equiv 0$. Again the invariant nonvanishing normalization of Corollary 13.19 gives the contradiction. $\square$

Theorem 15.8 (Uniformly tight L^3 closure). *No generated raw Seregin state in a raw generated descendant hull $\mathcal{S}_{\text{raw-gen}}(u, p, z_*, R_0, \eta_*)$ belongs to $\mathcal{C}_{L^3\text{-tight}}$.*

Proof. Let V be a generated raw Seregin state in $\mathcal{C}_{L^3\text{-tight}}$. By definition $V \in \mathcal{S}_{\text{mild}}$ and satisfies the uniform tail condition. Thus Theorem 25.8 gives $V \equiv 0$, contradicting Corollary 13.19. $\square$

Theorem 15.9 (Structure and decay exclusions from Liouville theorems). *No nonzero generated raw Seregin state belongs to $\mathcal{C}_{\text{sd}}$.*

Proof. Let $V \in \mathcal{C}_{\text{sd}}$. By definition of the structure/decay class, $V \in \mathcal{S}_{\text{mild}}$, and its physical representative

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t))$$

satisfies one of the listed structural hypotheses. The inherited centered L^∞ bound gives the physical Type I estimate

$$|U(x, t)| \leq \|V\|_\infty (-t)^{-1/2}.$$

By Lemma 25.4, for every $s < t < 0$ the standard projected mild formula holds on the interval $[s, t]$: choose $T = t$ and apply the finite-terminal restriction lemma. Thus U is a smooth mild Type I ancient solution in the sense of Definition 28.1. Therefore Theorem 32.1 gives $U \equiv 0$, and hence $V \equiv 0$, contradicting the invariant nonvanishing normalization of Corollary 13.19. $\square$

Proposition 15.10 (Exclusion of the Liouville classes). *No nonzero generated raw Seregin state can belong to*

$$\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}}.$$

Proof. If $V \in \mathcal{C}_{\text{small}}$, then by definition $V \in \mathcal{S}_{\text{mild}}$, and the small mild Liouville theorem applies. If $V \in \mathcal{C}_{\text{stat}, L^3}$, then the stationary self-similar profile equation and the stationary Leray theorem apply. These two branches are excluded in Theorem 15.7. If $V \in \mathcal{C}_{L^3\text{-tight}}$, then by definition $V \in \mathcal{S}_{\text{mild}}$, so Theorem 15.8 applies. If $V \in \mathcal{C}_{\text{sd}}$, then by definition $V \in \mathcal{S}_{\text{mild}}$ and the physical representative satisfies the relevant Type I mild structural hypothesis, so Theorem 15.9 applies. $\square$

16. RESIDUAL CLOSURE HYPOTHESIS

The preceding sections prove that every generated raw Seregin state in a raw generated descendant hull associated with an admissible finite-energy Type I extraction or an admissible local pointwise Type I concentration sequence is either in one of the four non-residual Liouville classes or in the complement $\mathcal{R}(\mathcal{S})$. The four non-residual classes ($\mathcal{C}_{\text{small}}$, $\mathcal{C}_{\text{stat}, L^3}$, $\mathcal{C}_{L^3\text{-tight}}$, $\mathcal{C}_{\text{sd}}$) are excluded in Proposition 15.10. The remaining assumption for the Type I contradiction is the residual closure statement below. No result in the preceding sections depends on the proof of residual closure; it is invoked only in Theorem 9.2 and in the final assembly Theorem 33.1.

Hypothesis 16.1 (Residual closure for raw generated Seregin state spaces). *For every finite-energy suitable weak solution (u, p) , every terminal point $z_* = (x_*, T)$ satisfying the local pointwise Type I bound, and every raw generated descendant hull produced by admissible local Type I concentration sequences at z_* with invariant normalization data generated by a raw extracted profile,*

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*),$$

one has

$$\mathcal{R}(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)) = \emptyset.$$

The same assertion is assumed for raw generated descendant hulls produced by admissible finite-energy Type I extractions in the finite-energy criterion [Theorem 9.2](#). The residual class is formed after removing only those states satisfying the full hypotheses of the explicit Liouville branches. Thus the residual closure hypothesis also removes any generated raw state for which projected mildness is not available and which is not covered by the stationary L^3 branch.

Remark 16.2 (Residual closure as a conditional assumption). [Hypothesis 16.1](#) is the only residual-class assumption not established from the preceding compactness and Liouville arguments in this paper. It concerns the raw generated descendant hull $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ after the ordered removal of the non-residual classes defined in [Section 14](#). The hypothesis is a conditional analytic statement about the same raw generated state space constructed here. When this hypothesis is assumed, [Theorem 9.2](#) and [Theorem 33.1](#) give the Type I exclusions within the stated finite-energy Type I extraction and local pointwise Type I concentration sequence classes. All uses of [Hypothesis 16.1](#) are therefore conditional on this residual input.

Remark 16.3 (Interface with the residual paper). The companion residual paper is intended to prove

$$\mathcal{R}(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)) = \emptyset$$

for the raw generated descendant hull defined in [Definition 13.13](#). The final assembly is formulated only for this raw generated state space.

Corollary 16.4 (Rigidity in the remaining case). *Assume [Hypothesis 16.1](#). For every raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ associated with admissible finite-energy Type I extractions or admissible local pointwise Type I concentration sequences, every element of $\mathcal{R}(\mathcal{S})$ is identically zero.*

Proof. The assertion follows directly from [Hypothesis 16.1](#). □

Proposition 16.5 (Equivalence on generated raw Seregin states). *For raw generated descendant hulls of generated raw Seregin states, the residual-class hypothesis [Hypothesis 16.1](#) implies [Corollary 16.4](#). Conversely, the corresponding rigidity assertion removes the remaining class from every raw generated descendant hull.*

Proof. If [Hypothesis 16.1](#) holds, the rigidity statement follows immediately. Conversely, assume the residual rigidity assertion for every raw generated descendant hull: every element of $\mathcal{R}(\mathcal{S})$ is identically zero. If some raw generated descendant hull had $V \in \mathcal{R}(\mathcal{S})$, the rigidity assertion would give $V \equiv 0$, directly contradicting the invariant nonvanishing normalization of [Corollary 13.19](#). Hence no nonzero generated raw Seregin state can remain in $\mathcal{R}(\mathcal{S})$. By [Corollary 13.19](#), every element of $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$, and hence every element of $\mathcal{R}(\mathcal{S})$, is nonzero. Therefore $\mathcal{R}(\mathcal{S}) = \emptyset$. □

Proof of [Theorem 9.2](#). Assume [Hypothesis 16.1](#). Suppose that (u, p, z_*) is an admissible finite-energy Type I singularity. By [Theorem 9.1](#), there exists a raw extracted normalized nonzero

Seregin ancient limit V . By Lemma 13.12, V satisfies the invariant local nonvanishing condition. Let

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be the raw generated descendant hull associated with z_* , the admissible extraction scales, and the corresponding fixed invariant normalization component. By construction, $V \in \mathcal{S}$.

By Proposition 14.11, either V belongs to one of the Liouville classes or $V \in \mathcal{R}(\mathcal{S})$. The first alternative is impossible by Proposition 15.10. The second alternative is impossible by Hypothesis 16.1, applied to $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. Hence the assumed admissible finite-energy Type I singularity cannot exist. $\square$

The preceding theorem is conditional on the residual-class hypothesis for the raw generated descendant hull associated with the finite-energy extraction. The compactness construction, pressure normalization, compact-to-invariant nonvanishing passage, raw generated closure stability, and class exhaustion are established above, with the local velocity concentration result recorded in Proposition 10.7. The structure and decay class is restricted to the assumptions in Definition 14.4 and is routed through Theorem 32.1; any other structured behavior lies in the remaining class and is addressed only by Hypothesis 16.1. No whole-space critical-norm estimate on the original solution is used in the conditional closure step.

17. ELEMENTARY SYMMETRIES OF THE CENTERED EQUATION

Lemma 17.1 (Logarithmic-time translation). *If V solves (9.3), then*

$$V_{\tau_0}(y, \tau) = V(y, \tau + \tau_0)$$

also solves (9.3). If

$$U^\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t),$$

then the renormalized field associated with U^λ is

$$V^\lambda(y, \tau) = V(y, \tau - 2 \log \lambda).$$

Proof. The first statement follows because (9.3) is autonomous in τ . For the second, put $s = -t = e^{-\tau}$. Then

$$\sqrt{s} U^\lambda(y\sqrt{s}, -s) = \lambda\sqrt{s} U(\lambda y\sqrt{s}, -\lambda^2 s) = V(y, -\log(\lambda^2 s)) = V(y, \tau - 2 \log \lambda).$$

$\square$

Proposition 17.2 (Dilation and translation are not symmetries). *The centered equation (9.3) is autonomous in τ , but it is not invariant under*

$$V(y, \tau) \mapsto \alpha V(\alpha y, \alpha^2 \tau)$$

unless $\alpha = 1$. It is also not invariant under constant translations

$$V(y, \tau) \mapsto V(y - a, \tau)$$

unless $a = 0$.

Proof. For scaling, set $Y = \alpha y$, $T = \alpha^2 \tau$, and $\tilde{V}(y, \tau) = \alpha V(Y, T)$, $\tilde{\Pi}(y, \tau) = \alpha^2 \Pi(Y, T)$. The non-drift part scales with factor α^3 , while

$$-\frac{1}{2}(\tilde{V} + y \cdot \nabla \tilde{V}) = -\frac{\alpha}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

Using the equation for V , the remaining term in the equation for $\tilde{V}$ is

$$\frac{\alpha - \alpha^3}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

If $\alpha \neq 1$, this forces $V + Y \cdot \nabla V = 0$. Along each ray,

$$\frac{d}{dr}(rV(r\omega, T)) = 0.$$

Boundedness and smoothness at the origin force the constant to be zero, hence $V \equiv 0$. Thus a nonzero centered solution has no such scaling symmetry.

For translations, all translation-invariant terms transform by composition, but the drift satisfies

$$y \cdot \nabla V(y - a, \tau) = (y - a) \cdot \nabla V(y - a, \tau) + a \cdot \nabla V(y - a, \tau).$$

The extra term is absent from the original equation. Thus spatial translation is not a symmetry of the fixed centered equation unless $a = 0$, apart from exceptional fixed solutions satisfying $a \cdot \nabla V \equiv 0$. $\square$

PART III. UNIFORMLY TIGHT ANCIENT DYNAMICS

Overview. Bounded ancient solutions of the three-dimensional Navier–Stokes equations are considered in backward self-similar variables under uniform L^3 -tightness. The compactness theory shows that bounded mild ancient solutions are smooth, locally compact under time translation, and that the common $L^\infty_\tau L^3_y$ and tightness bounds used in compactness statements are stable under locally smooth limits. A minimal-element reduction is then obtained for closed normalized uniformly tight collections: any nonempty such collection contains a nonzero solution of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ size.

For the time-translation hull of a bounded ancient solution, compact invariant hulls and invariant probability measures are constructed by Krylov–Bogolyubov averaging. These measures satisfy a stationary statistical form of the self-similar Navier–Stokes equation. Passing to the barycenter produces an explicit quadratic covariance term, isolating the obstruction to obtaining a single stationary self-similar profile. This compact-hull and invariant-measure material is auxiliary only; it is not used in the final Type I contradiction.

Finally, the uniformly L^3 -tight class is closed directly from its defining hypotheses. The tail condition, together with the boundedness of the ancient solution, gives the sequence- L^3 bound needed after the physical pullback. Since this pullback preserves the critical L^3 -norm on each time slice, the endpoint ancient Liouville theorem of Albritton–Barker then forces the pullback, and hence the centered solution, to vanish. Therefore the uniformly L^3 -tight class of the Type I ancient-limit reduction is excluded without any compact-hull Lyapunov hypothesis.

18. INTRODUCTION AND MAIN RESULTS

18.1. The uniformly tight ancient class. Consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The centered velocity V and pressure P solve

$$(18.1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$. In [Theorem 9.1](#), admissible Type I singularities are reduced to nonzero normalized ancient solutions of (18.1). The uniformly L^3 -tight class is the class in which

$$(18.2) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

The compactness statements also record the common bound

$$(18.3) \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty,$$

but the endpoint exclusion of the tight class uses only boundedness and the tail condition (18.2).

This condition on the centered limit is distinct from the original-solution condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3)).$$

For finite-energy suitable weak solutions, the original-solution assumption $u \in L^\infty_t L^3_x$ near a putative singular time is incompatible with singularity by the theorem of Escauriaza–Seregin–Šverák [3]. Uniform L^3 -tightness is imposed on centered ancient solutions after blow-up.

18.2. Minimal elements and compact hulls.

Remark 18.1 (Auxiliary diagnostic material). The compact-hull, minimal-element, and invariant-measure material in this part is auxiliary diagnostic material. It is not used in the proof of [Theorem 33.1](#); the final Type I contradiction uses only the uniform L^3 -tight Liouville theorem [Theorem 25.8](#) and its interface corollary [Corollary 26.3](#). It is recorded here because the covariance

obstruction it identifies is the precise reason the residual class cannot be removed by a stationary statistical reduction alone, and because the same constructions are reused in the residual-branch paper.

Remark 18.2 (Paper I/Paper III interface and normalization alignment). Raw extractions in Part II produce the local lower bound (9.4). By Lemma 13.12, every raw extracted profile also satisfies the invariant local normalization used in (22.1). The raw generated class $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ is closed under time translations and locally smooth limits only within this invariantly normalized class, so the closed normalized collections of Definition 22.1 include every raw generated descendant hull as a special case (with (R_0, η_0) given by the raw extraction parameters of Lemma 13.12).

The first reduction is variational:

$$\text{nonempty tight normalized class} \implies L^\infty\text{-minimal ancient element.}$$

The normalization is a fixed local velocity nonvanishing condition. It prevents minimizing sequences from losing all mass by time translation. Once a minimal element V_* is selected, its time-translation hull is

$$\mathcal{H}(V_*) = \overline{\{\Theta_s V_* : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}, \quad (\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The hull is compact because bounded mild ancient solutions are locally smooth with estimates depending only on their common L^∞ bound.

18.3. Invariant measures and covariance. Every compact invariant hull K carries invariant probability measures. Indeed, Krylov–Bogolyubov averages

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds$$

have weak subsequential limits. Such invariant measures satisfy a stationary statistical form of (18.1). If

$$\overline{W}(y) = \int_K W(y, 0) d\mu(W),$$

then the averaged nonlinearity decomposes as

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Thus the barycenter solves the stationary self-similar equation only up to the explicit covariance term C . This invariant-measure construction does not prove that a recurrent compact hull contains a single stationary profile.

18.4. Endpoint L^3 closure of the tight class. The compact-hull and invariant-measure constructions identify the covariance obstruction in a purely local topology, but they are not needed to close the uniformly L^3 -tight class. The tail condition already gives the sequence- L^3 information required by the endpoint ancient Liouville theorem, because the centered solution is bounded. Under the physical change of variables

$$u(x, t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

the L^3 -norm is invariant:

$$\|u(t)\|_{L_x^3} = \|V(-\log(-t))\|_{L_y^3}.$$

Consequently every tail-tight centered ancient solution produces, after restriction to any finite terminal time and translation of that terminal time to 0, a bounded physical mild ancient solution whose terminal datum lies in $L^3(\mathbb{R}^3)$ and whose L^3 -norm is bounded along a sequence $t_k \downarrow -\infty$.

The endpoint theorem of Albritton–Barker [1] then gives $u \equiv 0$, and the change of variables gives $V \equiv 0$.

The endpoint closure uses only the boundedness and tail tightness already present in the ancient solution class and introduces no additional estimate, recurrence assumption, or Lyapunov functional. The compactness and invariant-measure results establish the local smooth compactness theory, the existence of minimal elements and compact trajectory hulls, and the stationary statistical identity with its covariance term. The endpoint L^3 exclusion of the uniformly tight class is then obtained by the physical pullback and the Albritton–Barker ancient Liouville theorem.

Thus the uniformly L^3 -tight class is closed under local smooth limits by its defining hypotheses. The compact-hull construction describes recurrent dynamics and the covariance obstruction, but no compact-recurrence or Lyapunov assumption enters the tight-class exclusion in Theorem 25.8.

18.5. Main results. The first main theorem is the analytic compactness and closure statement.

Theorem 18.3 (Compactness and closure of uniformly tight ancient solutions). *Let $\{V_n\}$ be a sequence of bounded mild ancient solutions of (18.1). Assume*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

If in addition there are $L < \infty$ and a modulus $\omega : [1, \infty) \rightarrow [0, \infty)$, $\omega(R) \rightarrow 0$, such that

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R) \quad (R \geq 1),$$

then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L, \quad \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R),$$

and in particular V is uniformly L^3 -tight.

Compatible pressure representatives also converge locally weakly modulo time-dependent gauges under the additional $L^\infty_\tau L^3_y$ assumption; see Lemma 20.3.

The second main theorem is the minimal-element reduction.

Theorem 18.4 (Minimal element in a normalized uniformly tight class). *Let A be a nonempty collection of bounded mild ancient solutions satisfying a common L^∞ bound, a common $L^\infty_\tau L^3_y$ bound, a common uniform L^3 -tightness modulus, and a fixed local nonvanishing normalization*

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0.$$

Assume A is invariant under time translation and closed under locally smooth limits which retain the normalization. Then A contains an element V_ minimizing*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$$

among all $V \in A$. In particular $V_ \not\equiv 0$.*

The third main theorem is the endpoint L^3 tight-class Liouville criterion.

Theorem 18.5 (Uniformly tight Liouville criterion). *Let A be a collection of bounded mild ancient solutions of (18.1). Assume there is a modulus $\omega : [1, \infty) \rightarrow [0, \infty)$, $\omega(R) \rightarrow 0$, such that*

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R) \quad \text{for every } V \in A, R \geq 1,$$

and assume that A has a fixed local nonvanishing normalization:

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0 \quad \text{for every } V \in A.$$

Then

$$A = \emptyset.$$

The proof of the last theorem is summarized by the implications

uniformly L^3 -tight locally nonzero class A

$\Downarrow$ boundedness plus tail tightness

$$\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for a sequence } \tau_k \rightarrow -\infty$$

$\Downarrow$ physical pullback and critical L^3 -invariance

mild ancient Navier–Stokes solution u

$$\text{with } \sup_k \|u(t_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for some } t_k \downarrow -\infty$$

$\Downarrow$ Albritton–Barker endpoint theorem

$$u = 0$$

$\Downarrow$

contradiction with local velocity nonvanishing in A .

18.6. Relation to the Type I reduction. [Theorem 9.1](#) first reduces admissible Type I singularities to raw extracted Seregin ancient limits. The subsequent normalization step fixes the time-translation invariant lower bound, and the generated descendant hull is then organized into small, stationary L^3 , uniformly tight, structure and decay, and residual classes. The tight-class exclusion requires no additional compact-dynamical hypothesis: the uniformly L^3 -tight class contains no nonzero generated raw ancient state in the mild stratum.

18.7. Organization. [Section 19](#) fixes the centered equation, mild formulation, time translation, and local smooth topology. [Section 20](#) proves smoothing, pressure reconstruction, stability, and compactness. [Section 21](#) proves closure of uniform L^3 -tightness and recalls stationary L^3 rigidity. [Section 22](#) proves the minimal-element reduction. [Section 23](#) constructs compact trajectory hulls and minimal invariant subsets. [Section 24](#) constructs invariant measures and identifies the covariance obstruction. [Section 25](#) proves the physical pullback identities, applies the endpoint L^3 ancient Liouville theorem, and states the consequence used in the Type I reduction.

19. CENTERED ANCIENT SOLUTIONS AND THE LOCAL SMOOTH TOPOLOGY

19.1. The centered equation. Throughout this part V denotes a centered ancient velocity and P its pressure. An element of a compact hull is denoted by W , and a stationary profile by $w(y)$. Define

$$LV = \Delta V - \frac{1}{2}y \cdot \nabla V - \frac{1}{2}V.$$

After applying the Helmholtz–Leray projection $\mathbb{P}$, the centered equation becomes

$$(19.1) \quad \partial_\tau V = LV - \mathbb{P} \operatorname{div}(V \otimes V).$$

19.2. The Ornstein–Uhlenbeck–Stokes semigroup. Let $S(a) = e^{aL}$, $a > 0$. It is given by

$$(19.2) \quad (S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|z - e^{-a/2}y|^2}{4(1 - e^{-a})}\right)}{[4\pi(1 - e^{-a})]^{3/2}} f(z) dz.$$

The factor $e^{-a/2}$ represents the damping caused by the term $-V/2$.

Lemma 19.1 (Centered Stokes kernel estimate). *For $a > 0$ and $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(19.3) \quad \|(S(a)\mathbb{P} \operatorname{div})^* \phi\|_{L^1(\mathbb{R}^3)} \leq C e^{-a/2} (1 - e^{-a})^{-1/2} \|\phi\|_{L^1(\mathbb{R}^3)}.$$

Consequently the Duhamel term in (19.4) is well-defined by duality for bounded V .

Proof. Set $\lambda = e^{-a/2}$ and $\theta = 1 - e^{-a}$. From (19.2),

$$S(a)f(y) = e^{-a/2} (e^{\theta\Delta} f)(\lambda y).$$

Thus

$$S(a)^* \phi(z) = e^{-a/2} \lambda^{-3} (e^{\theta\Delta} \psi)(z), \quad \psi(x) = \phi(x/\lambda),$$

and $\|\psi\|_{L^1} = \lambda^3 \|\phi\|_{L^1}$. The Leray projection is a homogeneous Fourier multiplier of degree zero, so it commutes with this dilation. The classical heat–Stokes kernel estimate

$$\|(e^{\theta\Delta} \mathbb{P} \operatorname{div})^* \phi\|_{L^1} \leq C \theta^{-1/2} \|\phi\|_{L^1}, \quad \theta > 0,$$

follows from the Oseen kernel bound; see [20, 21, 18]. Substituting $\theta = 1 - e^{-a}$ and using

$$e^{-a/2} \lambda^{-3} \|\psi\|_{L^1} = e^{-a/2} \|\phi\|_{L^1}$$

gives (19.3). Since $(1 - e^{-a})^{-1/2}$ is integrable at $a = 0$, the duality pairing in (19.4) is integrable on finite time intervals for $V \in L^\infty$. $\square$

19.3. Riesz transform convention. The convention is that R_i has Fourier symbol $i\xi_i/|\xi|$. Thus $R_i R_j$ has symbol $-\xi_i \xi_j / |\xi|^2$, and

$$P = R_i R_j (F_{ij}) \iff -\Delta P = \partial_i \partial_j F_{ij}.$$

Pressures are determined only modulo functions of time unless a whole-space Leray gauge is explicitly chosen.

19.4. Bounded mild ancient solutions.

Definition 19.2 (Bounded mild ancient solution). A vector field $V : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$ is a bounded mild ancient solution of (18.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \quad \operatorname{div} V = 0$$

in distributions, and for every $\sigma < \tau$,

$$(19.4) \quad V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s) ds$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

The Duhamel term in (19.4) is interpreted by duality. For $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s), \phi \rangle = \langle V \otimes V(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle,$$

and Lemma 19.1 makes the scalar integrand absolutely integrable on $s \in (\sigma, \tau)$.

19.5. Time translation. For $s \in \mathbb{R}$ define

$$(\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The equation, the mild formulation, the L^∞ norm, and the $L_\tau^\infty L_y^3$ and tail bounds used below are invariant under Θ_s . Indeed, replacing (σ, τ) by $(\sigma + s, \tau + s)$ in (19.4) and changing variables in the time integral gives the mild formula for $\Theta_s V$.

19.6. The locally smooth topology.

Definition 19.3 (Local smooth topology). The locally smooth topology on $C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R}; \mathbb{R}^3)$ is generated by the seminorms

$$\|V\|_{m,N} = \max_{|\alpha|+j \leq m} \sup_{\substack{|y| \leq N \\ |\tau| \leq N}} |\partial_y^\alpha \partial_\tau^j V(y, \tau)|, \quad m, N \in \mathbb{N}.$$

One compatible metric is

$$d(U, V) = \sum_{m,N=1}^{\infty} 2^{-(m+N)} \frac{\|U - V\|_{m,N}}{1 + \|U - V\|_{m,N}}.$$

Thus $V_n \rightarrow V$ locally smoothly if and only if the convergence is in $C^m(K)$ for every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and every $m \geq 0$.

20. SMOOTHING, PRESSURE RECONSTRUCTION, AND COMPACTNESS

20.1. Local smoothing.

Lemma 20.1 (Smoothing of bounded mild solutions). *Every bounded mild ancient solution is smooth on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies the projected equation (19.1) classically. Moreover, for each compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and each integer $m \geq 0$,*

$$\|V\|_{C^m(K)} \leq C(m, K, \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. Fix a compact time interval $J \Subset \mathbb{R}$ containing the time projection of K , and set $t = -e^{-\tau}$. On the corresponding compact interval $t(J) \Subset (-\infty, 0)$, define

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map $(x, t) \mapsto (x/\sqrt{-t}, -\log(-t))$ is a smooth diffeomorphism between $\mathbb{R}^3 \times t(J)$ and $\mathbb{R}^3 \times J$, with all derivatives bounded on compact subcylinders.

The centered mild formula (19.4) transforms, under this parabolic change of variables, into the usual projected Navier–Stokes mild formula on every compact subinterval of $t(J)$:

$$U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(U \otimes U)(r) dr, \quad s < t.$$

This transformation uses only the projected mild equation: the Leray projection is homogeneous of degree zero and commutes with the parabolic dilation, while the Ornstein–Uhlenbeck kernel in (19.2) becomes the heat kernel with time parameter $t - s$. No pressure formulation is used at this stage. The L^∞ norm of U on $\mathbb{R}^3 \times t(J)$ is bounded by a constant depending only on J and $\|V\|_{L^\infty}$.

The standard local smoothing theorem for bounded projected mild Navier–Stokes solutions [13, 21] gives first local derivative bounds away from the initial time in any mild representation. In the present ancient setting one may choose the initial time strictly before the compact subcylinder under consideration. Iterating the differentiated Duhamel formula and then applying the interior

parabolic bootstrap for the Stokes system [23, 18] gives uniform bounds for all derivatives of U on compact subcylinders of $\mathbb{R}^3 \times t(J)$. The mild formula may then be differentiated in time, so U solves the projected Navier-Stokes equation classically. Returning to (y, τ) gives the asserted estimates for V and the classical projected equation (19.1). The argument is uniform for families with a common L^∞ bound. $\square$

20.2. Pressure reconstruction.

Lemma 20.2 (Local pressure reconstruction). *Let V be a smooth bounded solution of the projected centered equation (19.1). Then for every ball B_R and compact interval $I \subseteq \mathbb{R}$ there exists*

$$P_R \in C^\infty(B_R \times I),$$

unique up to an additive function of time, such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P_R = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0,$$

on $B_R \times I$.

Proof. Set

$$F = \Delta V - \frac{1}{2}(V + y \cdot \nabla V) - \partial_\tau V - (V \cdot \nabla)V.$$

Since $(V \cdot \nabla)V = \operatorname{div}(V \otimes V)$ and V satisfies (19.1), self-adjointness of the Leray projection gives

$$\int_I \int_{B_R} F \cdot \Phi \, dy \, d\tau = 0$$

for every $\Phi \in C_c^\infty(B_R \times I; \mathbb{R}^3)$ satisfying $\operatorname{div}_y \Phi = 0$. The local de Rham theorem, applied in the spatial variables with τ as a parameter, gives a scalar distribution P_R on $B_R \times I$, unique up to a distribution depending only on time, such that $\nabla_y P_R = F$. Since F is smooth, all spatial derivatives of P_R of positive order are smooth. Fixing, for instance, the spatial mean of P_R over B_R to be zero for each τ selects a representative which is smooth on $B_R \times I$. This proves the local pressure formulation. $\square$

Lemma 20.3 (Leray pressure and local gauges). *Let V be a bounded mild ancient solution satisfying*

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then the whole-space Leray pressure

$$P = R_i R_j (V_i V_j)$$

belongs to $L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3))$. Moreover, if $V_n \rightarrow V$ locally smoothly, the V_n have a common $L_\tau^\infty L_y^3$ bound, and

$$P_n = R_i R_j (V_{n,i} V_{n,j}),$$

then on every compact cylinder $B_R \times [\tau_1, \tau_2]$ there are bounded measurable gauges $a_n(\tau)$ and $a(\tau)$ such that, after passing to a subsequence,

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

Equivalently, after replacing the limiting pressure by an equivalent representative modulo a function of time, the weak limit may be denoted by P .

Proof. The first assertion follows from the Calderon–Zygmund boundedness of Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ [11]:

$$\|P(\tau)\|_{L^{3/2}} \leq C\|V_i(\tau)V_j(\tau)\|_{L^{3/2}} \leq C\|V(\tau)\|_{L^3}^2.$$

For the local stability statement, choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R} . Decompose on each time slice

$$P_n = R_i R_j (\chi V_{n,i} V_{n,j}) + R_i R_j ((1 - \chi) V_{n,i} V_{n,j}).$$

The local term converges strongly in $L^q(B_{2R} \times [\tau_1, \tau_2])$ for every finite q . This is because $V_n \rightarrow V$ smoothly on $\overline{B_{4R}} \times [\tau_1, \tau_2]$, so $\chi V_{n,i} V_{n,j} \rightarrow \chi V_i V_j$ strongly in every finite L^q , and the Riesz transforms are Calderon–Zygmund operators on L^q , $1 < q < \infty$.

It remains to control the exterior part modulo constants. Let K_{ij} be the convolution kernel of $R_i R_j$. For $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq C_R |z|^{-4}.$$

Define the exterior gauges

$$b_n(\tau) = \int_{\mathbb{R}^3} K_{ij}(-z)(1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz$$

and $b(\tau)$ by the same formula with V in place of V_n . The cutoff removes the singularity at the origin, and the integral is absolutely convergent at infinity because, on the dyadic annulus $A_m = \{2^m R < |z| < 2^{m+1} R\}$,

$$\int_{A_m} |z|^{-3} |V_n(z, \tau)|^2 dz \leq C(2^m R)^{-3} \|V_n(\tau)\|_{L^3(A_m)}^2 |A_m|^{1/3} \leq CL^2(2^m R)^{-2}.$$

The same dyadic estimate shows that b_n and b are bounded uniformly on $[\tau_1, \tau_2]$. The maps $\tau \mapsto b_n(\tau)$ and $\tau \mapsto b(\tau)$ are measurable, because they are limits of measurable truncated integrals. After subtracting $b_n(\tau)$, the exterior contribution on B_R is

$$E_n(x, \tau) = \int_{\mathbb{R}^3} [K_{ij}(x - z) - K_{ij}(-z)](1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz.$$

The same dyadic estimate, now using $|z|^{-4}$, gives

$$\|E_n(\cdot, \tau)\|_{L^\infty(B_R)} \leq C(R)L^2$$

uniformly in n and τ . Splitting the exterior integral into $\{4R < |z| < R'\}$ and $\{|z| > R'\}$, the first part converges by local smooth convergence, while the second part is bounded uniformly by CL^2/R' . Letting first $n \rightarrow \infty$ and then $R' \rightarrow \infty$ identifies the local limit of E_n with the corresponding exterior contribution for V . Therefore, with $a_n = b_n$ and $a = b$, the decompositions of $P_n - a_n$ and $P - a$ have local parts converging strongly and exterior parts converging distributionally with a uniform local $L^{3/2}$ bound. Hence

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

□

20.3. Stability under locally smooth limits.

Lemma 20.4 (Closure of the mild formulation). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

If $V_n \rightarrow V$ locally smoothly, then V is a bounded mild ancient solution of (18.1).

Proof. The common L^∞ bound passes to V by pointwise convergence on compact subsets. Divergence freeness passes distributionally. Fix $\sigma < \tau$ and test (19.4) for V_n against $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$. The endpoint terms are

$$\langle V_n(\tau), \phi \rangle, \quad \langle V_n(\sigma), S(\tau - \sigma)^* \phi \rangle.$$

The first converges by local smooth convergence. For the second, note that $S(\tau - \sigma)^* \phi \in L^1(\mathbb{R}^3)$. Approximate this L^1 function by a compactly supported smooth function and use the common L^∞ bound to control the L^1 approximation error uniformly in n .

For the Duhamel term, write

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V_n \otimes V_n)(s), \phi \rangle = \langle V_n(s) \otimes V_n(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle.$$

For fixed s , the adjoint test belongs to L^1 ; pairing convergence again follows by compact approximation in L^1 , local smooth convergence, and the common L^∞ bound on $V_n \otimes V_n$. The membership in L^1 follows from Lemma 19.1: the operator $(S(\tau - s) \mathbb{P} \operatorname{div})^*$ acts on the compactly supported smooth test field ϕ by integration against the (smooth) adjoint OU–Stokes kernel, which has finite L^1 norm in the spatial variable for every $\tau - s > 0$. The approximation error is uniform in n for this fixed s . Moreover Lemma 19.1 gives the integrable domination

$$C \sup_n \|V_n\|_{L^\infty}^2 e^{-(\tau-s)/2} (1 - e^{-(\tau-s)})^{-1/2} \|\phi\|_{L^1}, \quad s \in (\sigma, \tau).$$

Dominated convergence in s passes the Duhamel integral to the limit. Thus V satisfies (19.4). $\square$

20.4. Compactness of bounded families.

Proposition 20.5 (Compactness of bounded ancient families). *Let $\{V_n\}$ be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

Proof. By Lemma 20.1, the sequence is bounded in every C^m -norm on every compact cylinder. Arzela–Ascoli on a countable exhaustion $B_N \times [-N, N]$, followed by a diagonal extraction over (m, N) , gives a subsequence converging locally smoothly to a smooth limit. Lemma 20.4 shows that the limit is again a bounded mild ancient solution. $\square$

21. UNIFORM L^3 -TIGHTNESS AND STATIONARY L^3 RIGIDITY

21.1. Uniform L^3 -tightness.

Definition 21.1 (Uniform L^3 -tightness). A bounded ancient solution V is uniformly L^3 -tight if

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

For the compactness statements in this section we sometimes assume, in addition, a common $L_\tau^\infty L_y^3$ bound. The endpoint Liouville argument in Section 25 does not take this bound as a separate hypothesis; it obtains the required sequence estimate from boundedness and the tail condition above.

21.2. Closedness under locally smooth convergence.

Lemma 21.2 (Closure of uniform L^3 -tightness). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L < \infty$$

and assume that they have a common tightness modulus: for some $\omega : [1, \infty) \rightarrow [0, \infty)$ with $\omega(R) \rightarrow 0$,

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

If $V_n \rightarrow V$ locally smoothly, then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

In particular V is uniformly L^3 -tight.

Proof. Fix τ . On B_R , local smooth convergence gives

$$\int_{B_R} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{B_R} |V_n(y, \tau)|^3 dy \leq L^3.$$

Here L bounds the L^3 norm (so L^3 bounds the integral of the cube), while $\omega(R)$ below bounds the integral on the tail directly, not its cube. Letting $R \rightarrow \infty$ and using monotone convergence gives $\|V(\tau)\|_{L^3} \leq L$. Since τ was arbitrary, the same bound is uniform in time.

For the tail bound, fix $1 \leq R < R'$. On the bounded annulus $\{R < |y| < R'\}$, dominated convergence gives

$$\int_{R < |y| < R'} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{R < |y| < R'} |V_n(y, \tau)|^3 dy \leq \omega(R).$$

Letting $R' \rightarrow \infty$ and using monotone convergence gives

$$\int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R).$$

The estimate is independent of τ , so taking the supremum in time gives the asserted tail bound. Since $\omega(R) \rightarrow 0$, V is uniformly L^3 -tight. $\square$

Proof of Theorem 18.3. The compactness assertion is Proposition 20.5. The $L^\infty_\tau L^3_y$ and tightness closure assertions follow from Lemma 21.2 with the displayed constants L and ω . $\square$

21.3. Stationary time-translate limits. The stationary self-similar profile equation used below is

$$(21.1) \quad (w \cdot \nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \quad \operatorname{div} w = 0.$$

Theorem 21.3 (Stationary limits under uniform L^3 -tightness). *Let V be uniformly L^3 -tight. Let $\tau_n \in \mathbb{R}$, and suppose*

$$\Theta_{\tau_n} V \rightarrow W$$

locally smoothly. If W is stationary, then $W \equiv 0$.

Proof. The sequence $\Theta_{\tau_n} V$ has the same L^3 bound and tightness modulus as V . By Lemma 21.2, the limit W belongs to $L^\infty_\tau L^3_y$. If W is stationary, write $W(y, \tau) = w(y)$. Then $w \in L^3(\mathbb{R}^3)$.

By Lemma 20.4, W is a bounded mild ancient solution. The smoothness from Lemma 20.1 and the local pressure lemma Lemma 20.2 give smooth local pressures p_N on B_N at time 0. Since $\partial_\tau W = 0$, each p_N satisfies

$$(w \cdot \nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w) \quad \text{on } B_N.$$

On overlaps the gradients of the local pressures agree, so the pressures differ only by constants. Adjusting those constants and passing to the compatible union gives a scalar distribution p on $\mathbb{R}^3$ for which (21.1) holds in distributions. The stationary L^3 rigidity theorem Theorem 21.4 applies and gives $w = 0$. $\square$

21.4. Stationary self-similar rigidity.

Theorem 21.4 (Stationary self-similar rigidity). *Let $w \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ solve (21.1) in distributions for some scalar distribution p , with $\operatorname{div} w = 0$. Then $w \equiv 0$.*

Proof. We use the theorem of Nečas, Růžička, and Šverák [16], applied to the stationary Leray self-similar system. The smooth distributional formulation stated here is a special case of their weak $L^3(\mathbb{R}^3)$ theorem. $\square$

22. NORMALIZED COLLECTIONS AND MINIMAL ANCIENT SOLUTIONS

22.1. Closed normalized collections.

Definition 22.1 (Closed normalized collection). A collection A of bounded mild ancient solutions is called a closed normalized collection if:

(i) there is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M \quad \text{for all } V \in A;$$

(ii) there are $R_0 > 0$ and $\eta_0 > 0$ such that

$$(22.1) \quad \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 \quad \text{for all } V \in A;$$

(iii) A is invariant under time translation:

$$V \in A, s \in \mathbb{R} \implies \Theta_s V \in A;$$

(iv) if $V_n \in A$, $V_n \rightarrow V$ locally smoothly, and V satisfies the same (R_0, η_0) -nonvanishing condition (22.1), then $V \in A$.

22.2. Time-centering.

Lemma 22.2 (Time-centering). *Let A be a closed normalized collection. If $V \in A$, then for every $\delta > 0$ there exists $s \in \mathbb{R}$ such that*

$$\int_{B_{R_0}} |\Theta_s V(y, 0)|^3 dy \geq \eta_0 - \delta.$$

Proof. By (22.1), the supremum in time of the local L^3 mass is at least η_0 . Choose s within δ of this supremum. Since $(\Theta_s V)(y, 0) = V(y, s)$, the displayed estimate follows. Time-translation invariance keeps $\Theta_s V$ in A . $\square$

22.3. Stability of local nonvanishing.

Lemma 22.3 (Stability of local nonvanishing). *If $V_n \rightarrow V$ locally smoothly and*

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - o(1),$$

then

$$\int_{B_{R_0}} |V(y, 0)|^3 dy \geq \eta_0.$$

Proof. Local smooth convergence implies uniform convergence on $\overline{B}_{R_0} \times \{0\}$, hence convergence in $L^3(B_{R_0})$. The map $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$, and the claim follows by passing to the limit. $\square$

22.4. Lower semicontinuity of size.

Lemma 22.4 (Lower semicontinuity of L^∞ size). *If $V_n \rightarrow V$ locally smoothly and*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty,$$

then

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Let L_0 be the liminf on the right. If $|V(y_0, \tau_0)| > L_0 + \varepsilon$, choose a subsequence with $\|V_n\|_{L^\infty} \rightarrow L_0$. Local uniform convergence at (y_0, τ_0) gives $|V_n(y_0, \tau_0)| > L_0 + \varepsilon/2$ for all large n , contradicting $\|V_n\|_{L^\infty} \rightarrow L_0$. Hence $|V| \leq L_0$ pointwise. Since the smooth representative has pointwise supremum equal to its essential L^∞ norm, the result follows. $\square$

22.5. Existence of a minimal element.

Theorem 22.5 (Existence of a minimal ancient solution). *Let A be nonempty and closed normalized. Set*

$$m = \inf_{V \in A} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Then $0 < m < \infty$, and there exists $V_ \in A$ such that*

$$\|V_*\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = m.$$

In particular $V_ \not\equiv 0$.*

Proof. The common L^∞ bound gives $m < \infty$. The local nonvanishing gives a positive lower bound:

$$\eta_0 \leq \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \leq |B_{R_0}| \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}^3,$$

so

$$m \geq (\eta_0/|B_{R_0}|)^{1/3} > 0.$$

Choose $V_n \in A$ with

$$m \leq \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq m + \frac{1}{n}.$$

By [Lemma 22.2](#), after replacing V_n by a time translate, which stays in A and preserves the L^∞ norm, we may assume

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - \frac{1}{n}.$$

The common L^∞ bound and [Proposition 20.5](#) give a locally smooth subsequential limit V_* , which is a bounded mild ancient solution. By [Lemma 22.3](#),

$$\int_{B_{R_0}} |V_*(y, 0)|^3 dy \geq \eta_0,$$

so V_* retains the fixed (R_0, η_0) -normalization. Closedness of A then gives $V_* \in A$. Finally, [Lemma 22.4](#) gives

$$\|V_*\|_{L^\infty} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty} = m.$$

Since m is the infimum over A and $V_* \in A$, equality holds. The positive lower bound on m rules out $V_* \equiv 0$. $\square$

22.6. The uniformly tight normalized class.

Definition 22.6 (Uniformly tight normalized collection). Fix $M, L, R_0, \eta_0 > 0$ and a modulus

$$\omega : [1, \infty) \rightarrow [0, \infty), \quad \omega(R) \rightarrow 0.$$

A collection A of bounded mild ancient solutions is called a uniformly tight normalized collection with data $(M, L, \omega, R_0, \eta_0)$ if:

(1) for every $V \in A$,

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M;$$

(2) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L;$$

(3) for every $V \in A$ and every $R \geq 1$,

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R);$$

(4) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0;$$

(5) A is invariant under time translation;

(6) A is closed under locally smooth limits retaining item (4).

Proposition 22.7 (Uniformly tight normalized collections are closed normalized). *Every uniformly tight normalized collection is a closed normalized collection.*

Proof. The L^∞ bound, local nonvanishing, time-translation invariance, and closedness clause in [Definition 22.1](#) are part of [Definition 22.6](#). It remains to verify that the stated L^3 bound and fixed tightness modulus are preserved under locally smooth limits. This follows from [Lemma 21.2](#). $\square$

Proof of Theorem 18.4. By [Proposition 22.7](#), the collection is closed normalized. [Theorem 22.5](#) gives the asserted nonzero minimizer. $\square$

23. COMPACT TRAJECTORY HULLS

The material in [Sections 23](#) and [24](#) is auxiliary: it is included to clarify why a purely recurrent compactness argument does not by itself produce a stationary profile, since the barycenter carries a covariance defect. None of these results is used in the proof of [Theorem 25.8](#) or in the final assembly [Theorem 33.1](#).

On compact hulls we use the metric induced by the seminorms $\|\cdot\|_{C^m(B_N \times [-N, N])}$, $m, N \in \mathbb{N}$; this gives the locally smooth topology a complete separable metric structure on bounded sets, used by the probability-measure arguments in [Section 24](#).

23.1. Trajectory hulls.

Definition 23.1 (Trajectory hull). For a bounded mild ancient solution V , its trajectory hull is

$$\mathcal{H}(V) = \overline{\{\Theta_s V : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}.$$

23.2. Compactness of bounded hulls.

Theorem 23.2 (Compact trajectory hull). *If V is a bounded mild ancient solution, then $\mathcal{H}(V)$ is compact in the locally smooth topology and invariant under every time translation Θ_s . Every $W \in \mathcal{H}(V)$ is a bounded mild ancient solution with*

$$\|W\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Every translate $\Theta_s V$ is a bounded mild ancient solution with the same L^∞ bound as V . Hence every sequence of translates has a locally smoothly convergent subsequence by [Proposition 20.5](#). If $(W_n) \subset \mathcal{H}(V)$, choose translates $\Theta_{s_n} V$ with $d(W_n, \Theta_{s_n} V) < 1/n$. A subsequence of $\Theta_{s_n} V$ converges locally smoothly, and the corresponding subsequence of W_n has the same limit. Thus $\mathcal{H}(V)$ is sequentially compact. Since the locally smooth topology is metrizable, it is compact.

The L^∞ estimate and the mild equation pass to elements of the hull by local smooth convergence and [Lemma 20.4](#). For invariance, note that Θ_t maps the orbit of V onto itself. If $W_n \rightarrow W$ locally smoothly, then $\Theta_t W_n \rightarrow \Theta_t W$ locally smoothly because time translation only shifts the compact time interval on which the seminorms are evaluated. Hence $\Theta_t \mathcal{H}(V) = \mathcal{H}(V)$. $\square$

Corollary 23.3 (Uniform tightness on hulls). *If V is uniformly L^3 -tight with bound L and modulus ω , then every $W \in \mathcal{H}(V)$ satisfies*

$$\sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |W(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

The same estimates hold uniformly on every compact invariant subset $K \subset \mathcal{H}(V)$.

Proof. Every translate of V has the same L^3 bound and tightness modulus. Applying [Lemma 21.2](#) to any sequence of translates converging locally smoothly to $W \in \mathcal{H}(V)$ gives the displayed estimates for W . Taking the supremum over $W \in K \subset \mathcal{H}(V)$ gives the final assertion. $\square$

23.3. Compact invariant sets.

Definition 23.4 (Compact invariant set). A nonempty compact subset K of the locally smooth space of bounded mild ancient solutions is invariant if

$$\Theta_t K = K \quad \text{for every } t \in \mathbb{R}.$$

It is minimal if it contains no proper nonempty compact invariant subset.

Lemma 23.5 (Continuity of the translation action). *Let K be a compact invariant set. For each $t \in \mathbb{R}$, $\Theta_t : K \rightarrow K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. For fixed t , convergence $W_n \rightarrow W$ locally smoothly implies

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R \times [-T, T])} = \|W_n - W\|_{C^m(B_R \times [-T+t, T+t])} \rightarrow 0.$$

Thus Θ_t is continuous, and its inverse is Θ_{-t} . If $t_n \rightarrow t$ and $W_n \rightarrow W$, then on any fixed compact cylinder the difference

$$\Theta_{t_n} W_n - \Theta_t W$$

is bounded by a term involving $W_n - W$ on a slightly larger compact cylinder and a term involving the uniform continuity of finitely many derivatives of W on that larger cylinder. Both terms tend to zero, proving joint continuity. $\square$

23.4. Minimal compact invariant subsets.

Proposition 23.6 (Minimal compact invariant subsets). *Let K be a nonempty compact invariant set. Then K contains a nonempty compact minimal invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K , ordered by reverse inclusion. For a chain $\{K_\alpha\}$, the intersection $\bigcap_\alpha K_\alpha$ is nonempty by compactness and the finite intersection property, since the chain is totally ordered by ordinary inclusion. The intersection is compact and invariant: invariance follows from $\Theta_t K_\alpha = K_\alpha$ for all α and the fact that Θ_t is a homeomorphism. Thus every chain has an upper bound in the reverse-inclusion order. Zorn's lemma gives a maximal element for that order, equivalently a minimal compact invariant subset under ordinary inclusion. $\square$

23.5. The zero trajectory.

Lemma 23.7 (The zero trajectory). *The singleton $\{0\}$ is compact and invariant. If a minimal compact invariant set contains the zero trajectory, then it equals $\{0\}$.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for all t . Therefore $\{0\}$ is a nonempty compact invariant subset of any invariant set containing zero. Minimality forces equality. $\square$

24. INVARIANT MEASURES AND THE COVARIANCE OBSTRUCTION

24.1. Krylov–Bogolyubov averages. The following is the standard Krylov–Bogolyubov averaging argument [22, 19].

Theorem 24.1 (Krylov–Bogolyubov averages). *Let K be a compact invariant set and $W_0 \in K$. For $T > 0$, define*

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds.$$

Then every sequence $T_n \rightarrow \infty$ has a subsequence such that

$$\mu_{T_n} \rightharpoonup \mu$$

weakly as Borel probability measures on K , and every such limit μ is invariant:

$$(\Theta_t)_\# \mu = \mu \quad \text{for all } t \in \mathbb{R}.$$

If K is minimal, then every invariant Borel probability measure on K has support equal to K .

Proof. The orbit map $s \mapsto \Theta_s W_0$ is continuous by Lemma 23.5; thus μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$. Since K is compact and metrizable, the space of Borel probability measures on K is weakly sequentially compact [19]. Hence μ_{T_n} has a weakly convergent subsequence.

For $F \in C(K)$ and $t > 0$,

$$\int_K F(\Theta_t W) d\mu_T(W) - \int_K F(W) d\mu_T(W) = \frac{1}{T} \int_T^{T+t} F(\Theta_s W_0) ds - \frac{1}{T} \int_0^t F(\Theta_s W_0) ds.$$

The absolute value is at most $2t\|F\|_{C(K)}/T$. Letting $T = T_{n_k} \rightarrow \infty$ gives invariance for $t > 0$; negative times follow by applying the positive-time result to $F \circ \Theta_t$.

If K is minimal and μ is invariant, then $\text{supp } \mu$ is nonempty, closed, and invariant. Indeed, if U is a neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is a neighborhood of W , and invariance shows $\mu(U) = \mu(\Theta_{-t}U) > 0$ whenever $W \in \text{supp } \mu$. Hence $\Theta_t(\text{supp } \mu) \subset \text{supp } \mu$. Applying the same argument with $-t$ gives the reverse inclusion, so $\text{supp } \mu$ is invariant. It is nonempty because $\mu(K) = 1$, and it is compact because it is closed in compact K . Minimality forces $\text{supp } \mu = K$. $\square$

24.2. Support on minimal hulls.

Corollary 24.2 (Nonzero support). *Let K be a minimal compact invariant set with $K \neq \{0\}$. Then zero is not in K , and every invariant probability measure on K has support equal to K .*

Proof. If zero were in K , Lemma 23.7 would give $K = \{0\}$, contrary to the assumption. The support conclusion is the minimal-support part of Theorem 24.1. $\square$

24.3. Stationary statistical identity.

Theorem 24.3 (Stationary statistical form). *Let K be a compact invariant set and μ an invariant Borel probability measure on K . Then for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(24.1) \quad 0 = \int_K \left[\int_{\mathbb{R}^3} W(y, 0) \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) dy + \int_{\mathbb{R}^3} W_i(y, 0) W_j(y, 0) \partial_j \phi_i(y) dy \right] d\mu(W).$$

Proof. Define

$$F(W) = \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) dy.$$

Local smooth convergence makes F continuous on K : if $W_n \rightarrow W$, then $W_n(\cdot, 0) \rightarrow W(\cdot, 0)$ uniformly on the compact support of ϕ . Invariance gives

$$\int_K \frac{F(\Theta_h W) - F(W)}{h} d\mu(W) = 0 \quad (0 < |h| \leq 1).$$

Choose R with $\text{supp } \phi \subset B_R$. On $\overline{B}_R \times [-1, 1]$, compactness of K in the locally smooth topology gives

$$M_\phi = \sup_{W \in K} \|\partial_\tau W\|_{L^\infty(\overline{B}_R \times [-1, 1])} < \infty.$$

For $0 < |h| \leq 1$,

$$\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3} \left(\frac{1}{h} \int_0^h \partial_\tau W(y, s) ds \right) \cdot \phi(y) dy,$$

and the absolute value is bounded by $|B_R| \|\phi\|_{L^\infty} M_\phi$, uniformly in W and h . For each fixed W , smoothness gives uniform convergence of the inner average to $\partial_\tau W(y, 0)$ on $\text{supp } \phi$. Dominated convergence on K yields

$$\int_K \int_{\mathbb{R}^3} \partial_\tau W(y, 0) \cdot \phi(y) dy d\mu(W) = 0.$$

For each W , the smooth local equation (18.1) at time zero, tested against solenoidal ϕ , gives

$$\begin{aligned} \int_{\mathbb{R}^3} \partial_\tau W \cdot \phi \, dy &= \int_{\mathbb{R}^3} W \cdot \Delta \phi \, dy + \int_{\mathbb{R}^3} W \cdot \phi \, dy + \frac{1}{2} \int_{\mathbb{R}^3} W \cdot (y \cdot \nabla \phi) \, dy \\ &\quad + \int_{\mathbb{R}^3} W_i W_j \partial_j \phi_i \, dy. \end{aligned}$$

The pressure term vanishes because $\operatorname{div} \phi = 0$; the drift identity uses

$$\int y_j \partial_j W_i \phi_i \, dy = -3 \int W \cdot \phi \, dy - \int W \cdot (y \cdot \nabla \phi) \, dy.$$

The right-hand side is a continuous bounded function of $W \in K$. Averaging over K gives (24.1). $\square$

24.4. Barycenter and covariance.

Theorem 24.4 (Barycenter and covariance). *Let K and μ be as in Theorem 24.3. Define*

$$\overline{W}(y) = \int_K W(y, 0) \, d\mu(W)$$

and

$$C_{ij}(y) = \int_K W_i(y, 0) W_j(y, 0) \, d\mu(W) - \overline{W}_i(y) \overline{W}_j(y).$$

Then $\overline{W}$ and C are C_{loc}^∞ , $\overline{W}$ is solenoidal, and for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$(24.2) \quad \int_{\mathbb{R}^3} \overline{W} \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) \, dy + \int_{\mathbb{R}^3} \overline{W}_i \overline{W}_j \partial_j \phi_i \, dy + \int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i \, dy = 0.$$

In particular, $\overline{W}$ is a weak stationary self-similar solution against solenoidal test fields if and only if

$$\int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i \, dy = 0$$

for every solenoidal test field ϕ .

Proof. Compactness of K in the locally smooth topology gives, for every compact ball and every derivative order, a uniform bound on $\partial_y^\alpha W(y, 0)$ for $W \in K$. Hence one may differentiate $\overline{W}$ under the integral sign. More explicitly, for every multiindex α and every y in a fixed compact ball,

$$\partial_y^\alpha \overline{W}(y) = \int_K \partial_y^\alpha W(y, 0) \, d\mu(W),$$

because the difference quotients are dominated by the next derivative seminorm, uniformly on K . The same argument applies to the second moment

$$M_{ij}(y) = \int_K W_i(y, 0) W_j(y, 0) \, d\mu(W),$$

using Leibniz' rule and the uniform bounds on derivatives of W . This gives C_{loc}^∞ regularity of $\overline{W}$, M , and $C = M - \overline{W} \otimes \overline{W}$. Since every W is divergence free, Fubini against scalar test functions gives $\operatorname{div} \overline{W} = 0$: for $\eta \in C_c^\infty(\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \nabla \eta \, dy = \int_K \int_{\mathbb{R}^3} W(y, 0) \cdot \nabla \eta(y) \, dy \, d\mu(W) = 0.$$

For each compactly supported test field ϕ , the maps

$$W \mapsto \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) \, dy, \quad W \mapsto \int_{\mathbb{R}^3} W_i(y, 0) W_j(y, 0) \partial_j \phi_i(y) \, dy$$

are continuous and bounded on K , again by compactness in C_{loc}^∞ . Thus all integrals below are finite, and Fubini's theorem applies.

Applying [Theorem 24.3](#) and passing the linear term through the K -integral gives the first integral in [\(24.2\)](#). Passing the nonlinear term through the integral gives

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Substitution yields [\(24.2\)](#). Comparing this identity with the pressure-free stationary weak formulation for $\overline{W}$ gives the stated equivalence with vanishing covariance contribution. $\square$

25. ENDPOINT L^3 CLOSURE OF THE UNIFORMLY TIGHT CLASS

25.1. Physical pullback. The pullback below may be unbounded as $t \uparrow 0$; it is bounded only on every truncated ancient interval $(-\infty, T]$ with $T < 0$ by the factor $(-T)^{-1/2}$ below. For this reason the Albritton–Barker endpoint Liouville theorem is applied after translating such a truncated interval to terminal time 0, not directly to the unbounded pullback on $(-\infty, 0)$.

Let V be a smooth centered ancient velocity and let P be a compatible local pressure. For $t < 0$ define

$$(25.1) \quad u(x, t) = (-t)^{-1/2} V \left(\frac{x}{\sqrt{-t}}, -\log(-t) \right),$$

and

$$(25.2) \quad p(x, t) = (-t)^{-1} P \left(\frac{x}{\sqrt{-t}}, -\log(-t) \right).$$

Equivalently,

$$V(y, \tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}), \quad P(y, \tau) = e^{-\tau} p(e^{-\tau/2} y, -e^{-\tau}).$$

Lemma 25.1 (Pullback solves physical Navier–Stokes). *Let V be a bounded mild ancient solution of [\(18.1\)](#). Then the pullback u in [\(25.1\)](#) is a smooth divergence-free solution of*

$$(25.3) \quad \partial_t u + (u \cdot \nabla) u + \nabla p = \Delta u, \quad \operatorname{div} u = 0,$$

on $\mathbb{R}^3 \times (-\infty, 0)$, with pressure p locally given by [\(25.2\)](#). Moreover, for every $T < 0$,

$$\|u\|_{L^\infty(\mathbb{R}^3 \times (-\infty, T])} \leq (-T)^{-1/2} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. By [Lemma 20.1](#), V is smooth. By [Lemma 20.2](#), on every compact parabolic cylinder there is a smooth pressure representative P , unique up to a function of τ , for which [\(18.1\)](#) holds classically. Adding a function of τ to P only adds a function of t to p , and hence does not affect ∇p .

Set

$$\lambda(t) = (-t)^{-1/2}, \quad y = \lambda(t)x, \quad \tau = -\log(-t).$$

Then $\dot{\tau} = \lambda^2$ and

$$\partial_t y = \frac{1}{2} \lambda^2 y, \quad \dot{\lambda} = \frac{1}{2} \lambda^3.$$

Using $u = \lambda V(y, \tau)$, one obtains

$$\partial_t u = \lambda^3 \left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V \right),$$

while

$$(u \cdot \nabla_x) u = \lambda^3 (V \cdot \nabla_y) V, \quad \Delta_x u = \lambda^3 \Delta_y V, \quad \nabla_x p = \lambda^3 \nabla_y P.$$

Substituting these identities into (18.1) gives (25.3). The divergence identity follows similarly:

$$\operatorname{div}_x u = \lambda^2 \operatorname{div}_y V = 0.$$

Finally, if $t \leq T < 0$, then $(-t)^{-1/2} \leq (-T)^{-1/2}$, and the stated L^∞ bound follows directly from (25.1). $\square$

25.2. Critical L^3 invariance.

Lemma 25.2 (Invariance of the critical norm). *Let u be the physical pullback of V . Then, for every $t < 0$,*

$$(25.4) \quad \|u(t)\|_{L^3(\mathbb{R}^3)} = \|V(-\log(-t))\|_{L^3(\mathbb{R}^3)}.$$

Proof. With $y = x/\sqrt{-t}$, (25.1) gives

$$|u(x, t)|^3 = (-t)^{-3/2} |V(y, -\log(-t))|^3, \quad dx = (-t)^{3/2} dy.$$

Multiplying these two identities and integrating over $\mathbb{R}^3$ gives (25.4). $\square$

Lemma 25.3 (Tail tightness gives an endpoint sequence). *Let V be a bounded function on $\mathbb{R}^3 \times \mathbb{R}$ satisfying*

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Then, for every sequence $\tau_k \in \mathbb{R}$,

$$\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. Let $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$. Choose $R \geq 1$ such that

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq 1.$$

Then, for every k ,

$$\int_{\mathbb{R}^3} |V(y, \tau_k)|^3 dy \leq M^3 |B_R| + \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq M^3 |B_R| + 1.$$

This gives the claimed endpoint sequence bound. $\square$

25.3. Mildness after finite terminal-time restriction.

Lemma 25.4 (Physical pullbacks are mild ancient solutions on finite intervals). *Let V be a bounded mild ancient solution of (18.1), and let u be its physical pullback. For every $T < 0$, the restriction of u to $\mathbb{R}^3 \times (-\infty, T]$ is a bounded mild ancient solution of the physical Navier-Stokes equations after translating the terminal time T to 0.*

Proof. Let $s < t \leq T < 0$, and set

$$\sigma = -\log(-s), \quad \tau = -\log(-t).$$

Then $\sigma < \tau$. Apply the centered mild formula for V on $[\sigma, \tau]$:

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - \alpha) \mathbb{P} \nabla \cdot (V \otimes V)(\alpha) d\alpha.$$

Put

$$a = \tau - \sigma, \quad \lambda = e^{-a/2} = \sqrt{\frac{-t}{-s}}, \quad \theta = 1 - e^{-a} = \frac{t-s}{-s}.$$

For the linear term, use the explicit Ornstein–Uhlenbeck semigroup formula (19.2). Writing $x = \sqrt{-t}y$ and changing variables $z = \zeta/\sqrt{-s}$, we obtain

$$\begin{aligned} (-t)^{-1/2}(S(a)V(\sigma))(y) &= (-t)^{-1/2}e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|e^{-a/2}y-z|^2}{4(1-e^{-a})}\right)}{[4\pi(1-e^{-a})]^{3/2}} V(z, \sigma) dz \\ &= \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|x-\zeta|^2}{4(t-s)}\right)}{[4\pi(t-s)]^{3/2}} u(\zeta, s) d\zeta \\ &= (e^{(t-s)\Delta}u(s))(x). \end{aligned}$$

For the Duhamel term, write $r = -e^{-\alpha}$ for $\alpha \in [\sigma, \tau]$. Then $dr = (-r) d\alpha$,

$$V(\xi, \alpha) \otimes V(\xi, \alpha) = (-r)(u \otimes u)(\sqrt{-r}\xi, r),$$

and the homogeneity of $\mathbb{P}\nabla \cdot$ together with the same scaling calculation gives

$$\begin{aligned} (-t)^{-1/2}(S(\tau - \alpha)\mathbb{P}\nabla \cdot (V \otimes V)(\alpha))(y) \\ = (-r)(e^{(t-r)\Delta}\mathbb{P}\nabla \cdot (u \otimes u)(r))(x). \end{aligned}$$

Indeed, if $F_\alpha(\xi) = V(\xi, \alpha) \otimes V(\xi, \alpha)$ and $H_r(x) = u(x, r) \otimes u(x, r)$, then

$$F_\alpha(\xi) = (-r)H_r(\sqrt{-r}\xi),$$

so

$$\mathbb{P}\nabla_\xi \cdot F_\alpha = (-r)^{3/2}(\mathbb{P}\nabla_x \cdot H_r)(\sqrt{-r}\xi),$$

and using $S(b)f(\eta) = e^{-b/2}(e^{(1-e^{-b})\Delta}f)(e^{-b/2}\eta)$ with $b = \tau - \alpha$ yields the displayed identity because $e^{-b/2}\sqrt{-r} = \sqrt{-t}$ and $(-r)(1 - e^{-b}) = t - r$. Therefore, after the change of variables $r = -e^{-\alpha}$,

$$\begin{aligned} (-t)^{-1/2} \int_\sigma^\tau S(\tau - \alpha)\mathbb{P}\nabla \cdot (V \otimes V)(\alpha) d\alpha \\ = \int_s^t e^{(t-r)\Delta}\mathbb{P}\nabla \cdot (u \otimes u)(r) dr. \end{aligned}$$

Under the parabolic change of variables

$$u(x, t) = (-t)^{-1/2}V(x/\sqrt{-t}, -\log(-t)),$$

the centered mild formula therefore becomes

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-r)\Delta}\mathbb{P}\operatorname{div}(u \otimes u)(r) dr$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$. Since $t \leq T < 0$, the physical pullback is bounded on $\mathbb{R}^3 \times (-\infty, T]$, and the Oseen kernel estimate makes the Duhamel term finite on every finite interval. Hence the restriction of u to $(-\infty, T]$, after translating T to 0, is a bounded mild ancient solution. $\square$

25.4. Endpoint L^3 ancient Liouville theorem.

Theorem 25.5 (Albritton–Barker endpoint ancient Liouville theorem). *Let u be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0]$. Suppose that there is a sequence $s_k \downarrow -\infty$ such that*

$$\sup_k \|u(s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

and suppose $u(0) \in L^3(\mathbb{R}^3)$. Then $u \equiv 0$. The statement is written after translating the terminal time to 0.

Proof. We use the L^3 -sequence formulation of Albritton–Barker’s ancient Liouville theorem [1, Theorem 1.2], after translating the terminal time to 0. $\square$

Remark 25.6 (Quantitative Lorentz/Besov form). Albritton–Barker also give a quantitative endpoint statement [1, Theorem 4.1] with additional quantitative endpoint hypotheses. Those hypotheses are not used in this paper. The only imported statement needed below is the L^3 -sequence Liouville theorem in Theorem 25.5.

Remark 25.7 (Notation match with Albritton–Barker). In the tight-class application below, the terminal datum is $u_T(0) = u(T)$. Tail tightness and the critical L^3 identity give $u(T) \in L^3(\mathbb{R}^3)$, so the L^3 terminal hypothesis of Theorem 25.5 is automatic. The backward sequence bound is obtained in L^3 , which is the form used below.

Theorem 25.8 (Tail-tight centered ancient solutions vanish). *Let V be a bounded mild ancient solution of (18.1). Assume*

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Then $V \equiv 0$.

Proof. Let u be the physical pullback of V . Fix $T < 0$. By Lemma 25.4, after translating the terminal time T to 0, the restriction of u to $(-\infty, T]$ is a bounded mild ancient solution. Moreover $u(T) \in L^3(\mathbb{R}^3)$: indeed, if $\tau_T = -\log(-T)$, then $\|u(T)\|_{L^3} = \|V(\tau_T)\|_{L^3}$, and the latter is finite by boundedness on B_R and tail tightness outside B_R . Choose $t_k = -e^k$. For all sufficiently large k , $t_k < T$, and $\tau_k = -\log(-t_k) = -k$. By Lemma 25.3,

$$\sup_k \|V(-k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Hence (25.4) gives

$$\|u(t_k)\|_{L^3(\mathbb{R}^3)} = \|V(-k)\|_{L^3(\mathbb{R}^3)}$$

uniformly along the same sequence, after discarding the finitely many indices for which $t_k \geq T$. Set $u_T(x, \sigma) = u(x, \sigma + T)$ for $\sigma \leq 0$ and $s_k = t_k - T$. Then $s_k \downarrow -\infty$ and

$$\|u_T(s_k)\|_{L^3(\mathbb{R}^3)} = \|u(t_k)\|_{L^3(\mathbb{R}^3)}$$

is uniformly bounded. Applying Theorem 25.5 to u_T gives $u_T \equiv 0$ on $\mathbb{R}^3 \times (-\infty, 0]$, equivalently $u = 0$ on $\mathbb{R}^3 \times (-\infty, T]$. Since $T < 0$ was arbitrary, $u = 0$ on $\mathbb{R}^3 \times (-\infty, 0)$. The inverse change of variables then gives

$$V(y, \tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}) = 0$$

for every (y, τ) . $\square$

26. TIGHT-CLASS LIOUVILLE CRITERION AND TYPE I REDUCTION

26.1. Normalized tight collections.

Theorem 26.1 (No nonempty normalized uniformly tight collection). *No uniformly tight normalized collection in the sense of Definition 22.6 is nonempty.*

Proof. Suppose A is such a collection and choose $V \in A$. By the tightness modulus in item (3) of Definition 22.6, V satisfies the tail condition in Theorem 25.8. Hence Theorem 25.8 gives $V \equiv 0$. This contradicts the local velocity normalization in the same definition:

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0.$$

Hence A is empty. $\square$

Proof of Theorem 18.5. If A were nonempty, choose $V \in A$. The common tail modulus gives the tail condition in Theorem 25.8, so $V \equiv 0$. This contradicts the fixed local nonvanishing normalization in Theorem 18.5. $\square$

26.2. Use in the Type I reduction.

Definition 26.2 (Tight-class Liouville consequence). The tight-class Liouville consequence is the assertion that no generated raw Seregin state satisfying the invariant nonvanishing normalization of Corollary 13.19 belongs to $\mathcal{C}_{L^3\text{-tight}}$.

Corollary 26.3 (Tight-class consequence for the Type I reduction). *Definition 26.2 holds.*

Proof. Suppose, contrary to Definition 26.2, that a generated raw Seregin state V belongs to $\mathcal{C}_{L^3\text{-tight}}$. By definition of $\mathcal{C}_{L^3\text{-tight}}$, $V \in \mathcal{S}_{\text{mild}}$ and satisfies the tail condition

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Theorem 25.8 applies and gives $V \equiv 0$, contradicting Corollary 13.19. Hence no generated raw Seregin state can belong to $\mathcal{C}_{L^3\text{-tight}}$. $\square$

The tight-class consequence used in the Type I reduction is therefore proved without the residual-class hypothesis. The proof relies only on boundedness, the tail-tightness condition of the uniformly L^3 -tight class, the local-in-time physical pullback, and the endpoint ancient Liouville theorem of Albritton–Barker.

PART IV. STRUCTURED ANCIENT SOLUTION EXCLUSIONS

Overview. The final part concerns bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. Explicit PDE hypotheses are stated under which such solutions are spatially constant; in the Type I class, these constants vanish. The perturbative weighted-vorticity Lyapunov rigidity used by the conditional Type I reduction is also proved.

For the unforced equations, the cited axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L_t^\infty L_x^p$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ . When one of these hypotheses holds for a nonzero Type I blow-up limit, the cited theorems reduce the solution to a spatially constant state in the physical Type I frame, and that constant is then forced to vanish, contradicting the nontriviality of the limit.

The perturbative part uses the Lyapunov rigidity argument coming from Gallay–Wayne’s small-data decay theorem for the three-dimensional rescaled vorticity equation. It yields a coercive Lyapunov functional forcing any ancient trajectory that stays in the perturbative weighted-vorticity ball to be zero. The section concludes with a summary of the Liouville hypotheses and cited PDE results.

27. INTRODUCTION

27.1. Ancient solutions from Type I blow-up. The Type I blow-up argument in [Theorem 9.1](#) produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$(27.1) \quad \partial_t U + (U \cdot \nabla)U + \nabla P = \Delta U, \quad \nabla \cdot U = 0, \quad t < 0,$$

satisfying the Type I ancient bound

$$(27.2) \quad \sup_{t < 0} \sqrt{-t} \|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (27.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \implies U \equiv 0 \implies \text{contradiction with nonzero blow-up limit.}$$

The hypotheses treated in this part are precisely those to which a cited Liouville theorem or the perturbative argument in [Section 31](#) applies:

$$\text{axisymmetric Liouville hypotheses,} \quad \text{small weighted rescaled vorticity.}$$

No conclusion is asserted for decaying or symmetry-restricted ancient solutions outside the stated hypotheses. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

For the Type I application in [Theorem 9.1](#), the exclusions used are precisely the cases in which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$. Ancient solutions outside these hypotheses are treated separately.

27.2. Axisymmetric Liouville hypotheses. Several unforced axisymmetric cases fall under cited Liouville theorems. Koch–Nadirashvili–Seregin–Šverák treat the no-swirl case and the pointwise scale-invariant condition $|u| \leq C/r$. The swirl-bearing finite- L^p condition

$$\Gamma = ru_\theta \in L_t^\infty L_x^p, \quad 1 \leq p < \infty,$$

is treated by Lei–Zhang–Zhao. The z -periodic case with bounded Γ is treated by Lei–Ren–Zhang.

For Type I ancient limits, a constant conclusion after the relevant Galilean reduction must be converted into zero in the blow-up frame. The proof time-shifts away from the terminal

time $t = 0$, applies the bounded ancient Liouville theorem, patches the resulting constants on backward intervals, and uses (27.2): if $U(x, t) \equiv c$, then

$$\sup_{t < 0} \sqrt{-t} \|U(t)\|_\infty = \sup_{t < 0} \sqrt{-t} |c|$$

is finite only when $c = 0$.

27.3. Lyapunov hypotheses. The Lyapunov result uses a forward Lyapunov functional in the Galloway–Wayne perturbative weighted-vorticity regime. If w solves the rescaled three-dimensional vorticity equation and remains in the Galloway–Wayne small ball of $L_\sigma^2(m)$, $m > 7/2$, then the functional

$$(27.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2$$

is coercive and contracts along the forward semiflow. Sending the starting time to $-\infty$ then forces every ancient trajectory in the small ball to be zero.

27.4. Main results. The first result states the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 27.1 (Axisymmetric Liouville consequences for Type I ancient solutions). *Let U be a smooth mild Type I ancient solution of (27.1) in the sense of Definition 28.1. If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

- (i) *axisymmetric without swirl;*
- (ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x, t)| \leq \frac{C}{r};$$

- (iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) *axisymmetric, periodic in the physical axial variable z , and with*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 27.2 (Perturbative weighted-vorticity Lyapunov rigidity). *Let $m > \frac{7}{2}$. We say that $w(\tau)$ on $\mathbb{R}^3 \times \mathbb{R}$ is an ancient mild solution of the Galloway–Wayne rescaled vorticity equation if every forward time-shift of w coincides with the Galloway–Wayne mild solution launched from its value at the starting time. Let $w(\tau)$ be such an ancient mild solution of Galloway–Wayne rescaled vorticity equation*

$$(27.4) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \quad \nabla \cdot w = 0,$$

with v recovered by the three-dimensional Biot–Savart law. Assume that w satisfies the perturbative weighted-vorticity condition of Definition 31.2. Then

$$w \equiv 0.$$

The weighted-vorticity condition is only used when $w = \nabla \times v$ for the rescaled velocity associated with the ancient flow and v is the Biot–Savart velocity in the function class of Galloway–Wayne, so that the weighted-vorticity exclusion can be transferred to the velocity Seregin profile.

Theorem 27.3 (Consequences for Type I blow-up limits). *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in [Theorem 27.1](#); nor can its corresponding rescaled vorticity satisfy the hypotheses of [Theorem 27.2](#).*

Only the explicit hypotheses in the theorem are used.

27.5. Relation to the preceding parts. [Theorem 9.1](#) reduces Type I singularities to nonzero ancient limits satisfying [\(27.2\)](#). [Theorem 25.8](#) treats uniformly L^3 -tight ancient solutions. The axisymmetric and perturbative weighted-vorticity exclusions used by the Type I argument are proved in [Sections 30](#) and [31](#). Cited Liouville theorems rule out the axisymmetric cases, and the Lyapunov argument handles the perturbative weighted-vorticity case. [Theorem 32.1](#) does not depend on [Theorem 25.8](#).

27.6. Summary of hypotheses.

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS
pointwise $ u \leq C/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$, $1 \leq p < \infty$	$U \equiv 0$	LZZ
z -periodic bounded Γ	$U \equiv 0$	LRZ
small weighted vorticity	$w \equiv 0$	Gallay–Wayne

27.7. Organization. [Section 28](#) introduces the Type I ancient setting, self-similar variables, the invariance facts, and the time-shift lemma. [Section 29](#) fixes axisymmetric notation and the cited Liouville theorems. [Section 30](#) proves the unforced axisymmetric Liouville consequences. [Section 31](#) proves the Gallay–Wayne Lyapunov rigidity, and [Section 32](#) summarizes the consequences.

28. ANCIENT TYPE I LIMITS AND SELF-SIMILAR VARIABLES

28.1. Type I ancient solutions.

Definition 28.1 (Type I ancient solution). A smooth mild solution U of [\(27.1\)](#) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if [\(27.2\)](#) holds. Mild is understood in the standard velocity formulation [\[13, 18\]](#): for every $-\infty < s < t < 0$,

$$(28.1) \quad U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (U(\sigma) \otimes U(\sigma)) d\sigma$$

in distributions, where $\mathbb{P}$ is the Leray projector.

28.2. Nonzero Type I blow-up limits.

Definition 28.2 (Nonzero Type I blow-up limit). A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 28.3 (Nontriviality of blow-up limits). *A nonzero Type I blow-up limit is not identically zero.*

Proof. The definition includes the condition $U \not\equiv 0$. □

28.3. Backward self-similar variables. For $t < 0$, define

$$(28.2) \quad V(y, \tau) = \sqrt{-t} U(x, t), \quad Q(y, \tau) = (-t)P(x, t), \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t).$$

Equivalently,

$$x = e^{-\tau/2} y, \quad t = -e^{-\tau}.$$

Lemma 28.4 (Self-similar equation). *Let (U, P) be a smooth solution of (27.1), and define (V, Q) by (28.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$(28.3) \quad \partial_\tau V + (V \cdot \nabla) V + \nabla Q = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0.$$

Moreover (27.2) is equivalent to

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t \tau = \frac{1}{-t} = \frac{1}{a^2}, \quad \partial_t y = \frac{y}{2a^2}, \quad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3} \left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V \right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2} \nabla_y V, \quad \Delta_x U = a^{-3} \Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3} \nabla_y Q.$$

Thus

$$(U \cdot \nabla_x) U = a^{-3} (V \cdot \nabla_y) V.$$

Multiplying (27.1) by a^3 yields (28.3); the divergence equation follows from $\nabla_x \cdot U = a^{-2} \nabla_y \cdot V$. The spatial map $y \mapsto x = e^{-\tau/2} y$ is a bijection for each τ , and

$$\|V(\tau)\|_\infty = \sqrt{-t} \|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau \in \mathbb{R}$, gives the last assertion. $\square$

28.4. Preservation of axisymmetry and swirl variables. For axisymmetric flows we write

$$x = (r \cos \theta, r \sin \theta, z), \quad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho \cos \theta, \rho \sin \theta, Z), \quad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r, z, t) = r U_\theta(r, z, t), \quad \Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau).$$

Lemma 28.5 (Preservation of axisymmetric structure). *Axisymmetry and absence of swirl are preserved under (28.2). If U is axisymmetric, then*

$$\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t), \quad (r, z) = \sqrt{-t}(\rho, Z).$$

Fixed physical z -periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ -dependent period in the Z -coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho, Z, \tau) = \sqrt{-t} U_\theta(r, z, t), \quad r = \sqrt{-t}\rho, \quad z = \sqrt{-t}Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho \sqrt{-t} U_\theta = r U_\theta = \Gamma_U.$$

If U has physical axial period L , then V has period $L/\sqrt{-t} = Le^{\tau/2}$ in Z , not a fixed period. $\square$

28.5. Time shifts away from the singular time.

Lemma 28.6 (Time shifts away from the singular time). *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x, s) = U(x, T + s), \quad \Pi_T(x, s) = P(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution of (27.1) on $\mathbb{R}^3 \times (-\infty, 0)$. Moreover the axisymmetric, no-swirl, L^p -swirl, bounded- Γ , and physical z -periodic properties are preserved.

Proof. Time translation preserves the differential equation because $\partial_s W_T(x, s) = \partial_t U(x, T + s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (28.1) to U between the physical times $T + s < T + t < 0$:

$$U(T + t) = e^{(t-s)\Delta} U(T + s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta} \mathbb{P} \nabla \cdot (U(\eta) \otimes U(\eta)) d\eta.$$

With $\eta = T + \sigma$, this becomes the mild formula for W_T on the interval (s, t) . For $s < 0$, $T + s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T + s)\|_\infty \leq \frac{1}{\sqrt{-T}} \sup_{\eta < 0} \sqrt{-\eta} \|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r, z, s) = \Gamma_U(r, z, T + s),$$

so $L_s^\infty L_x^p$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. $\square$

28.6. PDE hypotheses. The arguments are stated directly in terms of hypotheses on ancient solutions. The axisymmetric hypotheses are:

- (i) axisymmetry and $U_\theta \equiv 0$;
- (ii) axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$ for $r > 0$;
- (iii) axisymmetry and

$$\Gamma = r U_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) axisymmetry, periodicity in the physical axial variable z , and $\Gamma \in L^\infty$.

The additional hypothesis used later is the Gallay–Wayne small weighted-vorticity condition.

29. AXISYMMETRIC NOTATION AND CITED LIOUVILLE THEOREMS

29.1. Cylindrical variables and the swirl scalar. For an axisymmetric solution u , write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \quad \Gamma = r u_\theta, \quad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r \cos \theta, r \sin \theta, z)$, independent of θ :

$$\|\Gamma(t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r, z, t)|^p r dr dz, \quad 1 \leq p < \infty.$$

29.2. Constant solutions in the Type I frame.

Definition 29.1 (Reduction to a spatial constant in the Type I frame). The cited Liouville theorems are used below only through the consequence that, after passing to the finite-terminal shift in the physical Type I frame, the solution is spatially constant. The Type I bound then forces that constant to be zero.

Lemma 29.2 (Spatially constant mild solutions). *Let W be a smooth mild solution of (27.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If $W(x, t) = b(t)$ for all x and t , then b is constant in time.*

Proof. Insert $W(x, t) = b(t)$ into (28.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. $\square$

Lemma 29.3 (Patching constants on backward intervals). *Let $U : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x, t) = c_T \quad (x \in \mathbb{R}^3, t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x, t) = c \quad (x \in \mathbb{R}^3, t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. $\square$

Lemma 29.4 (Constant Type I ancient solutions vanish). *Let U be a Type I ancient solution. If $U(x, t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t} |c| \leq \sup_{s < 0} \sqrt{-s} \|U(s)\|_\infty < \infty \quad (t < 0).$$

Letting $t \rightarrow -\infty$ gives $c = 0$. $\square$

29.3. Cited Liouville theorems. The following Liouville theorems are used in the stated forms.

Remark 29.5 (Match with the cited papers). The axisymmetric theorems below are imported from the cited references and are recorded with the hypotheses used here. In the Type I application they are applied to the finite-terminal time shifts

$$W_T(x, s) = U(x, T + s), \quad s < 0, \quad T < 0,$$

which are bounded ancient mild solutions by Lemma 28.6. The resulting constants are then patched in T and removed by the Type I bound. The notation matches the cited papers modulo the centered-versus-physical translation recorded in Lemma 28.5.

Theorem 29.6 (KNSS no-swirl Liouville theorem). *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is reduced, in the form used here, to a spatially constant axial drift in the physical Type I frame.*

Proof. Imported theorem. We use the axisymmetric no-swirl Liouville theorem of Koch, Naderashvili, Seregin, and Šverák [14, Theorem 5.2].

Required hypotheses. The theorem is stated for bounded ancient mild solutions of the unforced Navier–Stokes equations in the whole space which are axisymmetric and have no swirl. No drift-only or circulation-only hypothesis is being substituted for the no-swirl assumption in this branch.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 28.6. The no-swirl branch of $\mathcal{C}_{\text{sd}}$ requires axisymmetry together with $u_\theta \equiv 0$, and this structural hypothesis is preserved under the time shift. Therefore the cited theorem applies to W_T .

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives that W_T is a spatially constant axial drift in the frame used here. The resulting constants patch in T by Lemma 29.3, and Lemma 29.4 then forces the patched constant to vanish. $\square$

Theorem 29.7 (KNSS pointwise scale-invariant condition). *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0)$$

is constant up to a Galilean transformation. Only the spatially constant conclusion in the physical Type I frame is used below; if the source theorem is stated modulo Galilean transformations, this branch is invoked only after reducing to that constant representative. Zero is obtained only after the constant-removal step.

Proof. Imported theorem. We use the pointwise scale-invariant axisymmetric Liouville theorem of Koch–Naderashvili–Seregin–Šverák [14, Theorem 5.3].

Required hypotheses. The source theorem applies to bounded ancient mild axisymmetric whole-space solutions satisfying the full velocity condition

$$r|u(x, t)| \leq C.$$

Equivalently, away from the axis this is $|u(x, t)| \leq C/r$. This branch therefore uses the full velocity hypothesis. Drift-only assumptions or circulation-only assumptions are not included in this class.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 28.6. The pointwise branch of $\mathcal{C}_{\text{sd}}$ requires the full velocity bound $r|u| \leq C$, not merely a bound on the drift $u^r e_r + u^z e_z$ or on the circulation $\Gamma = ru_\theta$, and this structural hypothesis is preserved under the time shift. Therefore the cited theorem applies to W_T .

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives a spatially constant solution in the frame used here. The resulting constants patch in T by Lemma 29.3. In the Type I ancient frame, the patched constant is eliminated by Lemma 29.4; in the pointwise $1/r$ branch it is also excluded directly by letting $r \rightarrow \infty$ in the pointwise bound, as recorded in Remark 30.3. $\square$

Theorem 29.8 (Lei–Zhang–Zhao). *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = ru_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty,$$

then the no-swirl case $u_\theta \equiv 0$ is handled by Theorem 29.6, while in the nontrivial-swirl case the cited theorem yields that u is a constant vector field.

Proof. Imported theorem. We use the ancient axisymmetric circulation Liouville theorem of Lei–Zhang–Zhao [26, Theorem 1.3].

Required hypotheses. The theorem assumes a bounded ancient mild axisymmetric solution in $\mathbb{R}^3 \times (-\infty, 0)$, nontrivial swirl, and finite critical circulation integrability

$$\Gamma = ru_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty.$$

The no-swirl case $u_\theta \equiv 0$ is handled separately by Theorem 29.6; this theorem is invoked for the finite L^p -circulation branch.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 28.6. The finite- L^p -swirl branch requires axisymmetry together with $\Gamma = ru_\theta \in L_t^\infty L_x^p$, $1 \leq p < \infty$, and this structural hypothesis is preserved under the time shift. Therefore the cited theorem applies to W_T .

Conclusion in the Type I frame. In the nontrivial-swirl case, for each $T < 0$, the cited theorem gives that W_T is a constant vector field. The resulting constants patch in T by Lemma 29.3, and Lemma 29.4 then eliminates the patched constant in the Type I ancient frame. $\square$

Theorem 29.9 (Lei–Ren–Zhang). *Let u be a bounded ancient mild axisymmetric solution, periodic in z , with*

$$\Gamma = ru_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then u is a constant axial vector field.

Proof. Imported theorem. We use the periodic axisymmetric ancient Liouville theorem of Lei–Ren–Zhang [25, Theorem 1.1].

Required hypotheses. The theorem is stated on the quotient $\mathbb{R}^2 \times \mathbb{T}^1$ for bounded mild ancient axisymmetric solutions with bounded swirl circulation $\Gamma = ru_\theta$. The periodic branch in this paper is deliberately no larger than that source theorem: the physical representative is assumed z -periodic and satisfies the same bounded Γ condition.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 28.6. In the periodic branch the periodicity is imposed on the physical axial variable z , not on the centered field, and the branch also requires bounded circulation $\Gamma = ru_\theta$. These structural hypotheses are preserved under the time shift. Therefore the cited theorem applies to W_T after descending to $\mathbb{R}^2 \times \mathbb{T}^1$ and normalizing the period by a Navier–Stokes scaling, which does not change boundedness, mildness, axisymmetry, or boundedness of Γ .

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives a spatially constant solution in the frame used here. The resulting constants patch in T by Lemma 29.3, and Lemma 29.4 then eliminates the patched constant in the Type I ancient frame. $\square$

30. UNFORCED AXISYMMETRIC LIOUVILLE CONSEQUENCES

30.1. No swirl.

Proposition 30.1 (No-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$. By Lemma 28.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 29.6 gives that W_T is a spatially constant axial drift, say $W_T(x, s) = b_T(s)e_z$. By Lemma 29.2, $b_T(s)$ is constant in s . Thus $U(x, t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Lemma 29.3 gives $U(x, t) = c$ for all $t < 0$. Finally, Lemma 29.4 gives $c = 0$. $\square$

30.2. Pointwise scale-invariant control.

Proposition 30.2 (Pointwise $1/r$ Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*

$$|U(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 28.6, the shifted solution

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a bounded ancient mild axisymmetric solution and satisfies the same pointwise bound

$$|W_T(x, s)| \leq \frac{C}{r}.$$

By Theorem 29.7, W_T is spatially constant in the frame used here. Write $W_T \equiv c_T$. The pointwise bound gives

$$|c_T| \leq \frac{C}{r} \quad \text{for every } r > 0,$$

hence $c_T = 0$ by letting $r \rightarrow \infty$. Equivalently, one may patch the constants in T and apply Lemma 29.4. Thus $W_T \equiv 0$, so $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ was arbitrary, $U \equiv 0$. $\square$

Remark 30.3 (Pointwise bound rules out a constant directly). In the pointwise scale-invariant case, even before invoking the Type I constant-removal lemma, the pointwise bound $|c| \leq C/r$ for all $r > 0$ forces $c = 0$ by letting $r \rightarrow \infty$ when the Liouville theorem returns a constant. The Type I argument used in the proof above is consistent with this stronger conclusion.

 30.3. Finite L^p control of Γ .

Proposition 30.4 (Finite- L^p swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \leq p < \infty$,*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. If $U_\theta \equiv 0$, then the conclusion follows from Proposition 30.1. Hence assume $U_\theta \not\equiv 0$.

Fix $T < 0$. By Lemma 28.6,

$$W_T(x, s) = U(x, T + s), \quad s < 0,$$

is a bounded ancient mild axisymmetric solution. The finite- L^p circulation hypothesis is preserved by time shift:

$$\Gamma_{W_T} \in L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

The nontrivial-swirl alternative is the case covered by Theorem 29.8, so W_T is a constant vector field. Patch the constants in T using Lemma 29.3 and then apply Lemma 29.4. $\square$

30.4. Periodic bounded swirl. The periodic hypothesis below is always imposed on the physical ancient representative U , not on the centered profile V . Under the self-similar transform a fixed physical z -period becomes a τ -dependent period in Z , so it is not a fixed-period condition on V (Lemma 28.5).

Proposition 30.5 (Periodic bounded-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z , and satisfies*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 28.6, the shifted solution W_T is bounded, ancient, axisymmetric, physically z -periodic, and has bounded Γ . Theorem 29.9 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Lemma 29.4. $\square$

30.5. Proof of the axisymmetric Liouville theorem.

Proof of Theorem 27.1. The four cases are precisely the hypotheses of Propositions 30.1, 30.2, 30.4 and 30.5. Each proposition gives $U \equiv 0$. This contradicts Lemma 28.3 when U is a nonzero Type I blow-up limit. $\square$

31. PERTURBATIVE WEIGHTED-VORTICITY LYAPUNOV RIGIDITY

31.1. The rescaled vorticity equation. The Galloway–Wayne variable used here is the forward rescaled vorticity time; its generator Λ below has the opposite drift convention from the centered velocity generator used in Parts II–III. No result is transferred between the two generators without the explicit change of variables; the weighted-vorticity Lyapunov rigidity is stated and proved entirely in the Galloway–Wayne forward time.

In this section τ is the forward self-similar time used in the Galloway–Wayne vorticity formulation. The ancient trajectories are indexed by all $\tau \in \mathbb{R}$, and the forward semiflow acts by increasing τ .

Let $w = \nabla \times v$. The rescaled vorticity equation is

$$(31.1) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \quad \nabla \cdot w = 0,$$

where

$$(31.2) \quad \Lambda w = \Delta w + \frac{1}{2} \xi \cdot \nabla w + w,$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \geq 0$, set

$$L_\sigma^2(m) = \{f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 d\xi.$$

31.2. Galloway–Wayne small-data decay.

Theorem 31.1 (Galloway–Wayne small-data decay). *Let $0 < \mu \leq 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $r_m > 0$ and $C_m \geq 1$ such that if $\|w_0\|_m \leq r_m$, then (31.1) has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in $C([0, \infty); L_\sigma^2(m))$ and*

$$\|\Phi_\tau w_0\|_m \leq C_m e^{-\mu\tau} \|w_0\|_m, \quad \tau \geq 0.$$

In particular, if $m > \frac{7}{2}$, the choice $\mu = 1$ is admissible.

Proof. Imported theorem. We use Gallay–Wayne [24, Theorem 2.3] for the rescaled three-dimensional vorticity equation.

Required hypotheses. Their theorem is stated in the weighted divergence-free space $L_\sigma^2(m)$. It assumes $0 < \mu \leq 1$, $m > 2\mu + 1/2$, and small initial data $\|w_0\|_m \leq r_0(m, \mu)$. In this formulation no additional moment, orthogonality, spectral-projection, or codimension condition is used beyond membership in the stated small ball of $L_\sigma^2(m)$.

Verification for our profile. The weighted-vorticity branch of $\mathcal{C}_{\text{sd}}$ requires exactly the small-ball condition in $C(\mathbb{R}; L_\sigma^2(m))$, with $m > 7/2$, for the rescaled vorticity associated with the generated raw Seregin state. This places every forward time shift of the ancient trajectory in the small-data regime of the cited theorem.

Conclusion in the Type I frame. The theorem constructs a unique global mild solution of the integral rescaled vorticity equation and proves

$$\|w(\tau)\|_m \leq K_0 e^{-\mu\tau} \|w_0\|_m, \quad \tau \geq 0.$$

We denote the smallness radius and decay constant by r_m and C_m . For an ancient trajectory staying in the small ball for all $\tau \in \mathbb{R}$, the forward decay applied from arbitrarily negative times forces the rescaled vorticity to vanish. The Biot–Savart representative in this branch then gives $U \equiv 0$, hence $V \equiv 0$.

Variable convention. The equation (31.1) is the same forward self-similar vorticity equation after the notational change $\xi \leftrightarrow y$; the sign of the confining drift is the forward Gallay–Wayne convention used in this weighted-vorticity subsection. $\square$

31.3. The Lyapunov functional.

Definition 31.2 (Perturbative weighted-vorticity condition). Fix $m > \frac{7}{2}$, let $r_m > 0$ be the Gallay–Wayne small-data radius from Theorem 31.1, and fix $0 < \rho_m < r_m$. An ancient vorticity trajectory satisfies the perturbative weighted-vorticity condition with exponent m if it is a function

$$w \in C(\mathbb{R}; L_\sigma^2(m))$$

that solves (31.1) for all $\tau \in \mathbb{R}$ and satisfies

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_m \leq \rho_m.$$

Solving for all τ means that every forward time shift is the Gallay–Wayne forward mild solution launched from the value of w at the corresponding starting time; this is the semiflow compatibility used in the Lyapunov argument.

Proposition 31.3 (Semiflow Lyapunov functional). Fix $m > \frac{7}{2}$. For every initial datum w_0 in the Gallay–Wayne small-data ball $\{\|w\|_m \leq r_m\}$, define

$$(31.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2.$$

Then $\mathcal{L}_m(w_0) < \infty$, and

$$(31.4) \quad \|w_0\|_m^2 \leq \mathcal{L}_m(w_0) \leq C_m^2 \|w_0\|_m^2.$$

Moreover, whenever both w_0 and $\Phi_s w_0$ lie in the Gallay–Wayne small-data ball,

$$(31.5) \quad \mathcal{L}_m(\Phi_s w_0) \leq e^{-2s} \mathcal{L}_m(w_0).$$

In particular, this contraction applies along any ancient trajectory satisfying the perturbative weighted-vorticity condition.

Proof. The Gallay–Wayne estimate with $\mu = 1$ gives

$$e^{2\tau} \|\Phi_\tau w_0\|_m^2 \leq C_m^2 \|w_0\|_m^2,$$

which proves finiteness and the upper bound. The lower bound is the value at $\tau = 0$. By uniqueness,

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0.$$

If also $\|\Phi_s w_0\|_m \leq r_m$, then $\mathcal{L}_m(\Phi_s w_0)$ is defined. Hence

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_{\tau+s} w_0\|_m^2 = e^{-2s} \sup_{\sigma \geq s} e^{2\sigma} \|\Phi_\sigma w_0\|_m^2 \leq e^{-2s} \mathcal{L}_m(w_0).$$

□

31.4. Ancient rigidity in the small weighted ball.

Proof of Theorem 27.2. Let w be an ancient solution in the small ball and fix $\tau_0 \in \mathbb{R}$. For $s < \tau_0$, the shifted function

$$W_s(\theta) = w(s + \theta), \quad \theta \geq 0,$$

is the Gallay–Wayne mild solution with initial datum $w(s)$. The small-ball hypothesis gives $\|w(s)\|_m \leq \rho_m$, so the forward solution is in the domain of Φ_θ , and uniqueness gives

$$W_s(\theta) = \Phi_\theta w(s), \quad \theta \geq 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0-s} w(s).$$

Since $\|w(\tau_0)\|_m \leq \rho_m$, the Lyapunov functional is also defined at $w(\tau_0)$. Applying Proposition 31.3 with forward time $\tau_0 - s$,

$$\mathcal{L}_m(w(\tau_0)) \leq e^{-2(\tau_0-s)} \mathcal{L}_m(w(s)) \leq C_m^2 \rho_m^2 e^{-2(\tau_0-s)}.$$

Letting $s \rightarrow -\infty$ gives $\mathcal{L}_m(w(\tau_0)) = 0$. The coercive lower bound (31.4) gives $w(\tau_0) = 0$. Since τ_0 was arbitrary, $w \equiv 0$. □

32. SUMMARY OF PDE CONSEQUENCES

32.1. Zero conclusions.

Theorem 32.1 (Zero conclusions for structured Type I ancient solutions). *Let U be a smooth mild Type I ancient solution in the sense of Definition 28.1. If U satisfies one of the following hypotheses:*

- (i) *axisymmetry and $U_\theta \equiv 0$;*
- (ii) *axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$;*
- (iii) *axisymmetry and finite- L^p control of the swirl scalar,*

$$\Gamma \in L_t^\infty L_x^p, \quad 1 \leq p < \infty;$$

- (iv) *axisymmetry, z -periodicity, and bounded Γ ;*
- (v) *the associated Gallay–Wayne forward rescaled vorticity satisfies Definition 31.2, and the velocity is the Biot–Savart representative of that vorticity.*

then $U \equiv 0$.

Proof. Items (i)–(iv) are Theorem 27.1. Item (v) is Theorem 27.2; since the velocity in that case is recovered from w by the Biot–Savart law, $w \equiv 0$ gives the zero velocity. □

Corollary 32.2 (Structured hypotheses cannot hold for nonzero Type I blow-up limits). *A nonzero Type I blow-up limit cannot satisfy any of the hypotheses in Theorem 32.1.*

Proof. If a nonzero Type I blow-up limit satisfied one of the hypotheses, then [Theorem 32.1](#) would give $U \equiv 0$, contradicting [Lemma 28.3](#). $\square$

32.2. Consequences for Type I blow-up arguments.

Proof of [Theorem 27.3](#). The assertion is [Corollary 32.2](#). $\square$

Theorem 32.3 (Combined zero conclusion). *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in [Theorem 32.1](#), then $U \equiv 0$, a contradiction.*

Proof. The assertion is precisely the conclusion of [Theorem 32.1](#). $\square$

32.3. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$	$U \equiv 0$	LZZ plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
small weighted vorticity	$w \equiv 0$	Gallay–Wayne Lyapunov functional

The no-swirl, pointwise scale-invariant, finite- L^p -swirl, and periodic bounded-swirl cases follow from the cited axisymmetric Liouville theorems. The perturbative weighted-vorticity case follows from a Lyapunov functional derived from Gallay–Wayne decay. Thus every zero conclusion used in the Type I application rests on one of the explicit PDE hypotheses in the table.

PART V. FINAL ASSEMBLY: CONDITIONAL LOCAL TYPE I EXCLUSION

Overview. This part collects the preceding results and proves the local pointwise Type I exclusion conditional on the residual closure hypothesis. The argument is linear: local concentration, local endpoint weak-Serrin no-escape, admissible local Type I concentration sequence, raw Seregine extraction, passage to the raw generated state space with invariant nonvanishing, exhaustion into Liouville classes plus residual class, exclusion of the four non-residual classes, and residual closure.

33. FINAL ASSEMBLY

Theorem 33.1 (Local Type I regularity under residual closure). *Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$, and let $z_* = (x_*, T)$ be a terminal point satisfying the local pointwise Type I bound, for some $\rho > 0$,*

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} < \infty.$$

Assume the residual closure hypothesis [Hypothesis 16.1](#) holds for every raw generated descendant hull

$$\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

associated with invariant normalization data obtained from admissible local Type I concentration sequences at z_ . Then z_* is regular.*

Proof. Assume for contradiction that $z_* = (x_*, T)$ is singular.

Step 1: concentration. By [Corollary 5.8](#), there are $\eta_v > 0$ and $r_n \downarrow 0$ such that

$$C(z_*, r_n) \geq \eta_v.$$

Step 2: terminal no-escape. The local pointwise Type I bound gives

$$u \in L_t^{2,\infty} L_x^\infty(Q_\rho(z_*)).$$

By [Lemma 8.3](#), for each $0 < \theta < 1$ the terminal local A -quantity is uniformly bounded along the chosen small-scale sequence. Hence [Theorem 8.4](#) gives that the rescaled sequence satisfies

$$\limsup_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 = 0.$$

Step 3: nontrivial ancient extraction. [Corollary 8.5](#) promotes the sequence to an admissible local Type I concentration sequence. The concentration and no-escape properties give, by [Lemma 8.6](#), a fixed compact negative-time cylinder carrying positive L^3 mass. Then [Lemma 8.7](#) yields a raw extracted normalized Seregine ancient limit V satisfying

$$\iint_K |V|^3 > 0$$

for some compact $K \Subset \mathbb{R}^3 \times \mathbb{R}$. By [Lemma 13.12](#), V also satisfies the time-translation invariant local L^3 nonvanishing condition ([13.1](#)) for some data (R_0, η_*) .

Step 4: class decomposition. Let

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be the raw generated descendant hull formed with the fixed invariant normalization data obtained from the extracted profile V . By construction, the extracted profile V belongs to $\mathcal{S}$. By [Proposition 14.11](#),

$$V \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{sd}} \cup \mathcal{R}(\mathcal{S}).$$

Step 5: contradiction. The first four alternatives are excluded by [Proposition 15.10](#); each class definition includes exactly the hypotheses needed by its corresponding Liouville theorem. The residual alternative is excluded by [Hypothesis 16.1](#), applied to the raw generated descendant hull $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. No alternative is possible, contradicting the invariant nonvanishing normalization of [Corollary 13.19](#). This contradiction proves that z_* is regular. $\square$

Remark 33.2 (Conditional conclusion). Assuming [Hypothesis 16.1](#), [Theorem 33.1](#) gives the local pointwise Type I exclusion within the stated raw generated local pointwise Type I extraction class.

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